# Stochastic Analysis on Manifolds

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# Chapter 1

# SDE and Diffusion

Fix probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t\geq 0}$ of $\sigma$-fields contained in $\mathcal{F}$ with assumption that $\mathcal{F}_\infty = \mathcal{F}$. Also, $\mathbb{F}$ satisfies the usual condition, i.e., complete and right-continuous. Without specification, all semimartingales are assumed continuous. We use Einstein summation convention for convenience.

## 1.1 SDE on Euclidean Space

**Solution of SDE:** Let $Z = (Z_t)_{t\geq 0}$ be a continuous $\mathbb{R}^\ell$-valued $\mathbb{F}$-semimartingale, i.e.,

$$Z = M + A,$$

where $M$ is a local martingale and $A$ is an adapted process of local bounded variation such that $A_0 = 0$. Let $\sigma\colon \mathbb{R}^N \to \mathbb{R}^{N\times\ell}$ be smooth and locally Lipschitz, i.e. for all $R > 0$ there exists $R > 0$ and $C(R) > 0$ such that

$$\|\sigma(x) - \sigma(y)\|_{\mathrm{F}} \leq C(R)\,\|x - y\|\,, \quad \forall\, x, y \in B(R).$$

Let $X_0 \in \mathbb{R}^N$ be $\mathcal{F}_0$ measurable. For a stopping time $\tau$, consider a semimartingale $X = (X_{t\wedge\tau})_{t\geq 0}$ of the form

$$X_t = X_0 + \int_0^t \sigma(X_s)dZ_s, \quad 0 \leq t \leq \tau.$$

We say $X$ is a solution of a SDE driving by the semimartingale $Z$, and such equation denoted as $\mathrm{SDE}(\sigma, Z, X_0)$. Moreover, by Itô formula, for $f \in C^2(\mathbb{R}^N)$,

$$\begin{aligned}
f(X_t) &= f\left(X_0\right) + \int_0^t f_{x_i}\left(X_s\right)\sigma^i_\alpha\left(X_s\right)dZ^\alpha_s \\
&\quad + \frac{1}{2}\int_0^t f_{x_i x_j}\left(X_s\right)\sigma^i_\alpha\left(X_s\right)\sigma^j_\beta\left(X_s\right)d\left\langle Z^\alpha, Z^\beta\right\rangle_s \\
&= f\left(X_0\right) + \int_0^t f_{x_i}\left(X_s\right)\sigma^i_\alpha\left(X_s\right)dM^\alpha_s + \int_0^t f_{x_i}\left(X_s\right)\sigma^i_\alpha\left(X_s\right)dA^\alpha_s \\
&\quad + \frac{1}{2}\int_0^t f_{x_i x_j}\left(X_s\right)\sigma^i_\alpha\left(X_s\right)\sigma^j_\beta\left(X_s\right)d\left\langle M^\alpha, M^\beta\right\rangle_s
\end{aligned}$$

*Remark* 1.1.1. Note that if $X_0 \in L^2$ and $\sigma$ is globally Lipschitz continuous, then we don't need the stopping time $\tau$ and $\mathrm{SDE}(\sigma, Z, X_0)$ has a unique solution $X = (X_t)_{t\geq 0}$.

The problem why we need to consider a stopping time is because of the local Lipschitz property of $\sigma$. The local Lipschitz would make the usual solution exploded. To consider it rigorously, we first need the definition of explosion time.

Let $M$ be a locally compact metric space and $\widehat{M} = M \cup \{\partial_M\}$ be its one-point compactification. A continuous map $x\colon [0,\infty) \to \widehat{M}$ has an explosion time $e = e(x) > 0$ if $x_t \in M$ for $0 \le t < e$ and $x_t = \partial M$ for $t \ge e$ when $e < \infty$, i.e.

$$e\colon W(M) \to [0,\infty]$$

where $W(M) \subset C([0,\infty), \widehat{M})$ such that $e$ can be defined. Moreover, $W(M)$ can be equipped with the canonical filtration $\mathbb{F}_M = (\mathcal{F}_{M,t})_{t\ge 0}$, i.e., the natural filtration generated by coordinate processes. Then $e$ is a $\mathbb{F}_M$-stopping time.

**Definition 1.1.2.** Let $\tau$ be a stopping time. A continuous $X = (X_{t\wedge\tau})_{t\ge 0}$ is called a semimartingale up to $\tau$ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t\wedge\tau_n})$ is a semimartingale.

**Definition 1.1.3.** A semimartingale $X$ up to a stopping time $\tau$ is a solution of $\mathrm{SDE}(\sigma, Z, X_0)$ if there exists a sequence of stopping times $\tau_n \uparrow \tau$ such that $X^{\tau_n} = (X_{t\wedge\tau_n})$ is a semimartingale and

$$X_{t\wedge\tau_n} = X_0 + \int_0^{t\wedge\tau_n} \sigma\left(X_s\right) dZ_s, \quad t \ge 0.$$

**Theorem 1.1.4.** *If $\sigma$ is locally Lipschitz continuous, then there exists a unique $W(\mathbb{R}^N)$-valued random variable $X$ which is a solution of $\mathrm{SDE}(\sigma, Z, X_0)$ up to its explosion time $e(X)$. Moreover, if $X, Y$ are two solutions up to $\tau$ and $\eta$ respectively, then $X_{t\wedge\tau\wedge\eta} = Y_{t\wedge\tau\wedge\eta}$. In particular, if $X$ is a solution up to $e(X)$, then $\eta \le e(X)$ and $X_{t\wedge\tau} = Y_{t\wedge\tau}$.*

**Theorem 1.1.5.** *Suppose $\sigma$ and $b$ are locally Lipschitz. Then the weak uniqueness holds for*

$$X_t = X_0 + \int_0^t \sigma(X_s)dW_s + \int_0^t b(X_s)ds,$$

*where $W$ is a Brownian motion.*

**Stratonovich integral.** For two continuous semimartingales $H, Z$, the Stratonovich integral is defined as

$$\int_0^t H_s \circ dZ_s = \int_0^t H_s dZ_s + \frac{1}{2}\left\langle H, Z\right\rangle_t.$$

In such case, the Itô formula becomes

$$df(X_t) = \frac{\partial f}{\partial x^\alpha}(X_t) \circ dX_t^\alpha.$$

Suppose that $V_\alpha$, $\alpha = 1, 2, \cdots, \ell$ are smooth vector fields in $\mathbb{R}^N$, that is,

$$V_\alpha = (V_\alpha^1, \cdots, V_\alpha^N)\colon \mathbb{R}^N \to \mathbb{R}^N$$

and vector fields means that for any $f \in C^1(\mathbb{R}^N)$,

$$V_\alpha f = \sum_{i=1}^N V_\alpha^i \frac{\partial f}{\partial x^i}.$$

More generally, if

$$X_t = X_0 + \int_0^t V_\alpha\left(X_s\right) \circ dZ_s^\alpha,$$

then

$$X_t = X_0 + \int_0^t V_\alpha\left(X_s\right) dZ_s^\alpha + \frac{1}{2}\int_0^t \nabla_{V_\beta} V_\alpha\left(X_s\right) d\left\langle Z^\alpha, Z^\beta\right\rangle_s.$$

*Proof.* By definition,

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} d \left\langle V_\alpha(X_t), Z_t^\alpha \right\rangle .$$

or

$$dX_t^i = \sum_{\alpha=1}^{\ell} V_\alpha^i(X_t) dZ_t^\alpha + \frac{1}{2} \sum_{\alpha=1}^{\ell} d \left\langle V_\alpha^i(X_t), Z_t^\alpha \right\rangle .$$

For $d \left\langle V_\alpha^i(X_t), Z_t^\alpha \right\rangle$, first because $\sum_\alpha V_\alpha^i(X_t) dZ_t^\alpha$ is the local martingale part of $dX_t^i$,

$$d \left\langle X^i, Z^\alpha \right\rangle_t = \sum_{\beta=1}^{\ell} V_\beta^i(X_t) d \left\langle Z^\beta, Z^\alpha \right\rangle_t .$$

Then by Itô formula,

$$dV_\alpha^i(X_t) = \sum_{j=1}^{n} \frac{\partial V_\alpha^i}{\partial x^j} dX_t^j + \text{ bounded variation part } .$$

So

$$\begin{aligned} d \left\langle V_\alpha^i(X_t), Z_t^\alpha \right\rangle &= \sum_{j=1}^{n} \frac{\partial V_\alpha^i}{\partial x^j} d \left\langle X^j, Z^\alpha \right\rangle_t \\ &= \sum_{\beta=1}^{\ell} \sum_{j=1}^{n} V_\beta^j \frac{\partial V_\alpha^i}{\partial x^j} d \left\langle Z^\beta, Z^\alpha \right\rangle_t \\ &= \nabla_{V_\beta} V_\alpha^i d \left\langle Z^\beta, Z^\alpha \right\rangle_t . \end{aligned}$$

Therefore,

$$dX_t = V_\alpha(X_t) dZ_t^\alpha + \frac{1}{2} \nabla_{V_\beta} V_\alpha(X_t) d \left\langle Z^\beta, Z^\alpha \right\rangle_t . \qquad \square$$

*Remark* 1.1.6. Let $X, Y, Z, Y, H, K$ are be $\mathbb{R}$-semimartingales.

(1) Note that, if $dX = H \circ dW$ and $dY = K \circ dZ$, because

$$H \circ dW = H dW + \frac{1}{2} \left\langle H, W \right\rangle , \quad K \circ dZ = K dZ + \frac{1}{2} \left\langle K, Z \right\rangle$$

and $\langle H, W \rangle$ and $\langle K, Z \rangle$ are of bounded variation, then

$$d \left\langle X, Y \right\rangle = HK d \left\langle W, Z \right\rangle .$$

(2) In fact, we have

$$H \circ d \left( \int X \circ dW \right) = (HX) \circ dW,$$

because

$$d \left\langle HX, W \right\rangle = H d \left\langle X, W \right\rangle + X d \left\langle H, W \right\rangle ,$$

which is because of Itô formula, $d(HX) = H dX + X dH + d \langle H, X \rangle$.

In the sense of Stratonovich integral, if $X_t = X_0 + \int_0^t V_\alpha(X_s) \circ dZ_s^\alpha$ and $f \in C^2(\mathbb{R}^N)$, then

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha .$$

For a Itô SDE,

$$dX_t = b(X_t)dt + \sigma_n(X_t)dW_t^n,$$

where $W = (W^1, \cdots, W^\ell)$ is a $\ell$-dimensional Brownian motion. Let

$$Z_t^1 = t, \quad Z^2 = W^1\ , \ \cdots\ ,\ Z_{\ell+1} = W^\ell.$$

Clearly, they are semimartingales. Then by above, we can see if let

$$V_1(x) = b(x) - \frac{1}{2}\sigma_n^i \frac{\partial \sigma_n}{\partial x^i}, \quad V_2(x) = \sigma_1(x),\ \cdots, V_{\ell+1}(x) = \sigma_\ell(x),$$

the Itô SDE is equivalent to

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha = V_1(X_t)dt + V_\beta(X_t) \circ dZ_t^\beta.$$

## 1.2 SDE on Manifolds

**Definition 1.2.1.** Let $M$ be a smooth manifold. A continuous $M$-valued process $X$ is called a $M$-valued semi-martingale if $f(X)$ is a real-valued semi-martingale for all $f \in C^\infty(M)$.

*Remark* 1.2.2. We can also define it up to a stopping time $\tau$ as before. By Itô formula, it is obviously that $M = \mathbb{R}^N$ gives the usual definition of semimartingales. Moreover, the test function space can be chosen as $C_c^\infty(M)$.

Let $V_1, \cdots, V_\ell \in \Gamma(TM)$, $Z$ be an $\mathbb{R}^\ell$-valued semimartingale, and an $M$-valued random variable $X_0 \in \mathcal{F}_0$. Consider an equation symbolically written as

$$dX_t = V_\alpha \circ dZ_t^\alpha,$$

and denoted it as $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$.

**Definition 1.2.3.** An $M$-valued semimartingale $X$ defined up to a stopping time $\tau$ is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to $\tau$ if for all $f \in C^\infty(M)$,

$$f(X_t) = f(X_0) + \int_0^t (V_\alpha f)(X_s) \circ dZ_s^\alpha.$$

**Proposition 1.2.4.** *Suppose* $\Phi\colon M \to N$ *is a diffeomorphism and* $X$ *a solution of* $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$. *Then* $\Phi(X)$ *is a solution of* $\mathrm{SDE}(\Phi_* V_1, \cdots, \Phi_* V_\ell; Z, \Phi(X_0))$ *on* $N$.

*Proof.* Let $Y = \Phi(X)$. For any $f \in C^\infty(N)$, because $f \circ \Phi \in C^\infty(M)$ and $X$ is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$,

$$f \circ \Phi(X_t) = f \circ \Phi(X_0) + \int_0^t V_\alpha(f \circ \Phi)(X_s) \circ dZ_s^\alpha.$$

Note that $\Phi_*\colon \Gamma(TM) \to \Gamma(TN)$ such that for and $f \in C^\infty(N)$,

$$\Phi_* V(f)(\Phi(x)) = V(f \circ \Phi)(x).$$

It follows that

$$f(Y_t) = f(Y_0) + \int_0^t \Phi_* V_\alpha(f)(Y_s) \circ dZ_s^\alpha,$$

which means that $Y$ is a solution of $\mathrm{SDE}(\Phi_* V_1, \cdots, \Phi_* V_\ell; Z, \Phi(X_0))$. □

Note that by Whitney's embedding theorem, any smooth manifold $M$ can be properly embedded in a Euclidean space, i.e., $i\colon M \hookrightarrow \mathbb{R}^N$, where $i$ is a smooth proper map. Therefore, $i(M)$ is a closed submanifold in $\mathbb{R}^N$. Note that here closed submanifold means that a submanifold that is a closed subset. Therefore, we can always view $M \subset \mathbb{R}^N$ as an embedded submanifold that is closed. Moreover, we usually assume $M$ without boundary. When $M$ is non-compact, we can consider its one-point compactification $\widehat{M} = M \cup \{\partial_M\}$. So $\{x_n\}_n \subset M$ such that $x_n \to \partial_M$ in $\widehat{M}$ if and only if $\|x_n\|_{\mathbb{R}^N} \to \infty$.
In the following we always assumed that $M \subset \mathbb{R}^N$ is a submanifold that is closed and without boundary. Moreover, let
$$i = (f^1, \cdots, f^N)\colon M \hookrightarrow \mathbb{R}^N$$
and $f^i$ are coordinate functions that $f^i(x) = x^i$.

**Proposition 1.2.5.** *Suppose $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f^1, \cdots, f^N$ be coordinate functions. Let $X$ be an $M$-valued continuous process.*
*(1) $X$ is a semimartingale on $M$ if and only if it is an $\mathbb{R}^N$-valued semimartingale, i.e. $f^i(X)$ is a $\mathbb{R}$-valued semimartingale for each $i = 1, 2, \cdots, N$.*

*(2) $X$ is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to a stopping time $\sigma$ if and only if for each $i = 1, 2, \cdots, N$,*
$$f^i(X_t) = f^i(X_0) + \int_0^t (V_\alpha f^i)(X_s) \circ dZ_s^\alpha, \quad 0 \le t < \sigma.$$

*Proof.* (1) $\Rightarrow$: It is by the definition.
$\Leftarrow$: For any $f \in C^\infty(M)$, since $M$ is closed $\mathbb{R}^N$, it can be extended on $\tilde{f} \in C^\infty(\mathbb{R}^N)$ such that $f(X) = \tilde{f}(X)$. Because $X$ is $\mathbb{R}$-valued semimartingale, $f(X) = \tilde{f}(X)$ is also a semimartingale.

(2) $\Rightarrow$: It is by the definition.
$\Leftarrow$: For any $f \in C^\infty(M)$, let $\tilde{f} \in C^\infty(\mathbb{R}^N)$ be an extension. Then
$$f(X_t) = \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right).$$
Therefore,
$$\begin{aligned}
df(X_t) &= \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) \circ d(f^i(X_t)) \\
&= \tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) \circ V_\alpha f^i(X_t) \circ dZ_t^\alpha \\
&= \Big(\tilde{f}\left(f^1(X_t), \cdots, f^N(X_t)\right) V_\alpha f^i(X_t)\Big) \circ dZ_t^\alpha \\
&= V_\alpha \tilde{f}(X_t) \circ dZ_t^\alpha = V_\alpha f(X_t) \circ dZ_t^\alpha,
\end{aligned}$$
where the final equality is because $V_\alpha \in \Gamma TM$ and $\tilde{f}$ is an extension of $f$. $\square$

*Remark* 1.2.6. Note that for $f \in \mathbb{R}^k \to \mathbb{R}$ and $g\colon \mathbb{R}^n \to \mathbb{R}^k$
$$\begin{aligned}
d(f \circ g(M)) &= f_{x_i}(g(M)) \circ dg^i(M) \\
&= f_{x_i}(g(M)) \circ g^i_{x_j}(M) \circ dM^j \\
&= \nabla f(g(M))^\top Dg(M) \circ dM
\end{aligned}$$
under Stratonovich integral.

*Remark* 1.2.7. If $X$ is a solution of $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ up to its explosion time $e(X)$ and $X_0 \in M$, then $X_t \in M$ for $0 \le t < e(X)$. Moreover, $\mathrm{SDE}(V_1, \cdots, V_\ell; Z, X_0)$ is unique up its explosion time.

## 1.3 Diffusion Process

Let $L$ be a smooth second order elliptic, but not necessarily non degenerate, differential operator on a smooth manifold $M$, that is,

$$L\colon C^\infty(M) \to C^\infty(M)$$

such that on every chart $(U; x^1, \cdots, x^d)$

$$Lf(x) = a^{ij}(x)\frac{\partial^2}{\partial x^i \partial x^j}f(x) + b^i(x)\frac{\partial}{\partial x^i}f(x)$$

for some $a^{ij}, b^i \in C^\infty(U)$ and $\sum_{i,j} a^{ij}(x)\xi_i\xi_j \geq 0$.

*Remark* 1.3.1. If $M$ is equipped with a torsion-free connection $\nabla$, like a Riemannian manifold, then $L$ can be written as a coordinate-free formulation, that is,

$$Lf := \left\langle A, \nabla^2 f\right\rangle + \langle b, \nabla f\rangle,$$

for $b \in \Gamma(TM)$, and $A$, a symmetric $(2,0)$-tensor, where $\langle \cdot, \cdot\rangle$ is the natural dual operation.

For $f \in C^2(M)$ and $\omega \in W(M)$, let

$$M^f(\omega)_t = f(\omega_t) - f(\omega_0) - \int_0^t Lf(\omega_s)ds.$$

**Definition 1.3.2.** An $\mathbb{F}$-adapted process $X$ defined on a filtered $(\Omega, \mathbb{F}, \mathbb{P})$ and that is valued in $M$ is called a diffusion process generated by $L$ if $X$ is a $M$-valued semimartingale w.s.t. $\mathbb{F}$ up to $e(X)$ and

$$M^f(X)_t = f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds, \quad 0 \leq t < e(X),$$

is a $\mathbb{F}$-local martingale for all $f \in C^\infty(M)$.

A probability measure $\mu$ defined on $(W(M), \mathcal{F}_{M,\infty})$ is called a diffusion measure generated by $L$ if

$$M^f(\omega)_t = f\left(\omega_t\right) - f\left(\omega_0\right) - \int_0^t Lf\left(\omega_s\right)ds, \quad 0 \leq t < e(\omega),$$

is a local $\mathbb{F}_M$ local martingale for all $f \in C^\infty(M)$.

*Remark* 1.3.3. If $X$ is an $L$-diffusion, then $\mu^X = X_{\#}\mathbb{P}$ is an $L$-diffusion measure. Conversely, if $\mu$ is an $L$-diffusion measure, then the coordinate process is an $L$-diffusion process.

For given $L$, assume locally

$$L = \frac{1}{2}\sum_{i,j=1}^d a^{ij}(x)\frac{\partial^2}{\partial x^i \partial x^j} + \sum_{i=1}^d b^i(x)\frac{\partial}{\partial x^i},$$

where $(a^{ij}(x))$ is positive semi-definite. Define

$$\Gamma(f,g) = L(fg) - fLg - gLf, \quad f, g \in C^\infty(M).$$

So locally

$$\Gamma(f,g) = a^{ij}\frac{\partial f}{\partial x^i}\frac{\partial f}{\partial x^j}.$$

And the semi-elliptic means that for any $f \in C^\infty(M)$, $\Gamma(f,f) \geq 0$. Combining this with bilinearity, it implies that if $g^1, \cdots, g^n$, the matrix $(\Gamma(g^i, g^j))$ is positive semimartingale.
Here is another explanation for $\Gamma$. For any $f, g \in C^\infty(M)$, let $h = fg$. Let $Y = f(X)$ and $Z = g(X)$, so $h(X) = YZ$. First,

$$Y_t Z_t = h(X)_t = Y_0 Z_0 + M_t^h + \int_0^t L(fg)(X_s)ds.$$

On the other hand, because

$$\begin{aligned} Y_t &= Y_0 + M_t^f + \int_0^t Lf(X_s)ds \\ Z_t &= Z_0 + M_t^g + \int_0^t Lf(X_s)ds. \end{aligned}$$

and $M^f, M^g$ are local martingales, by Itô formula,

$$\begin{aligned} Y_t Z_t = Y_0 Z_0 &+ \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f \\ &+ \int_0^t (fLg + gLf)(X_s)ds \\ &+ \langle M^f, M^g \rangle_t . \end{aligned}$$

Therefore, we have

$$\int_0^t \Gamma(f,g)(X_s)ds - \langle M^f, M^g \rangle_t = \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f - M_t^h.$$

Note that the LHS is a process with bounded variation and the RHS is a continuous local martingale, which implies that the RHS is constant, i.e., 0. Then we have

$$M_t^h = \int_0^t f(X_s)dM_s^g + \int_0^t g(X_s)dM_s^f.$$

Moreover,

$$\langle Y, Z \rangle_t = \langle M^f, M^g \rangle_t = \int_0^t \Gamma(f,g)(X_s)ds.$$

**Existence of Diffusion Process.** Assume $M$ is a submanifold of $\mathbb{R}^N$, which is closed. Let $z = (z^1, \cdots, z^N)$ be coordinates in $\mathbb{R}^N$ and $f^\alpha$ ($\alpha = 1, 2, \cdots, N$) are coordinate maps $M \to \mathbb{R}^N$ by $f^\alpha(z) = z^\alpha$. Let

$$\tilde{a}^{\alpha\beta} = \Gamma(f^\alpha, f^\beta), \quad \tilde{b}^\alpha = Lf^\alpha.$$

Then they are in $C^\infty(M)$ and $(\tilde{a}^{\alpha\beta})$ is positive semi-definite. By the closedness of $M$, $\tilde{a}, \tilde{b}$ can be extended to $\mathbb{R}^N$ and let

$$\tilde{L} = \frac{1}{2} \sum_{\alpha,\beta=1}^N \tilde{a}^{\alpha\beta} \frac{\partial^2}{\partial z^\alpha \partial z^\beta} + \sum_{\alpha=1}^N \tilde{b}^\alpha \frac{\partial}{\partial z^\alpha}$$

be defined on $\mathbb{R}^N$. It is a true extension of $L$.

**Lemma 1.3.4.** *Suppose $f \in C^\infty(M)$. Then for any $\tilde{f} \in C^\infty(\mathbb{R}^N)$ which extends $f$ from $M$ to $\mathbb{R}^N$, $\tilde{L}\tilde{f} = Lf$ on $M$.*

*Proof.* Let $(x^1, \cdots, x^d)$ be a local coordinate on $M$. Note that

$$f(x) = \tilde{f}(f^1(x), \cdots, f^N(x)).$$

Then

$$\begin{aligned} Lf(x) &= L\tilde{f}(f^1(x), \cdots, f^N(x)) \\ &= \frac{1}{2}a^{ij}(x)\frac{\partial^2 \tilde{f}(f^1(x), \cdots, f^N(x))}{\partial x^i \partial x^j} + b^i(x)\frac{\partial \tilde{f}(f^1(x), \cdots, f^N(x))}{\partial x^i} \\ &= \tilde{L}\tilde{f}(z). \end{aligned}$$

$\square$

For $\tilde{L}$ on $\mathbb{R}^n$, if $\tilde{a} = \tilde{\sigma}\tilde{\sigma}^\top$, it is the generator of diffusion process that satisfies

$$X_t = X_0 + \int_0^t \tilde{\sigma}(X_s)dW_s + \int_0^t \tilde{b}(X_s)ds.$$

We can clearly choose $\tilde{\sigma} = \tilde{a}^{\frac{1}{2}}$ and $W$ a standard Brownian motion in $\mathbb{R}^N$ independent of $X_0$.

*Remark* 1.3.5. The existence of solution up to a stopping time needs the local Lipschitz of $\tilde{\sigma}$, which can be guaranteed by the local Lipschitz of $\tilde{a}$.

Moreover, for such diffusion process $X$ on $\mathbb{R}^N$, $\mu^X = X_\# \mathbb{P}$, the distribution on $W(\mathbb{R}^N)$, is concentrated on $W(M)$, i.e., $X$ is in fact a $L$-diffusion on $M$. In fact, we have the following theorem.

**Theorem 1.3.6.** *If $L$ is a smooth second order semi-elliptic operator a $C^\infty$-manifold $M$ and $\mu_0$ is a probability measure on $M$, then there exists a unique $L$-diffusion measure with initial distribution $\mu_0$.*

*Remark* 1.3.7. The existence and uniqueness of such $\mathbb{P}_{\mu_0}$ guarantee that the $L$ martingale problem is well-posed. So if $X$ is a $L$-diffusion process, then it is a strong Markov process w.s.t. $\mathbb{F}^X$ and its transition group $P_t f = \mathbb{E}^x[X_t]$ has the generator extending $L$. In this case, in fact, $X$ is a strong Markov process w.s.t. $\mathbb{F}$.

**Driving by Brownian Motion.** Let $M$ be a Riemannian manifold and consider all solutions of SDE

$$dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt,$$

where $V_\alpha, V_0 \in \Gamma(TM)$ and $W = (W^\alpha)$ is an $\ell$-dimensional Brownian motion. First, by Itô formula,

$$dV_\alpha f(X_t) = V_0(V_\alpha f(X_t))dt + V_\beta(V_\alpha f(X_t)) \circ dW_t^\beta,$$

which implies that

$$d\left\langle V_\alpha f(X_t), W_t^\alpha \right\rangle = V_\beta(V_\alpha f(X_t))d\left\langle W^\beta, W^\alpha \right\rangle_t = \sum_\alpha V_\alpha(V_\alpha f(X_t))dt$$

Therefore, we have

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t) \circ dW_t^\alpha \\ &= \int_0^t V_0 f(X_t)dt + \int_0^t V_\alpha f(X_t)dW_t^\alpha + \frac{1}{2}\left\langle V_\alpha f(X_t), W^\alpha \right\rangle_t \end{aligned}$$

$$= \int_0^t V_\alpha f(X_t) dW_t^\alpha + \int_0 \left( V_0 f(X_t) + \frac{1}{2} \sum_\alpha V_\alpha (V_\alpha f(X_t)) \right) dt$$

Denote $V_\alpha(V_\alpha f(x)) = V_\alpha^2 f(x)$ and let

$$L = \frac{1}{2} \sum_\alpha V_\alpha^2 + V_0,$$

which is obviously a semi-elliptic second order differential operator. From above,

$$f(X_t) - f(X_0) - \int_0^t Lf(X_t)dt = \int_0^t V_\alpha f(X_t) dW_t^\alpha.$$

So a solution to $dX_t = V_\alpha(X_t) \circ dW_t^\alpha + V_0(X_t)dt$ is a diffusion process generated by $L$. Moreover, assume that $V_\alpha, V_0$ satisfy

$$\sup_{t\in[0,T]} \sup_{x\in U} \mathbb{E}\left[ |D^\alpha f(X_t)| \mid X_0 = x \right] < \infty$$

for all $f \in C_c^\infty(M)$, $T > 0$, coordinate chart $U$ with $\overline{U}$ compact, and index $\alpha$, where $D^\alpha$ is the usual derivative.

*Remark* 1.3.8. In fact, assume $M \subset \mathbb{R}^n$ and $V_\alpha, V_0$ are the restriction of $\mathbb{R}^n$-vector fields $\tilde{V}_\alpha, \tilde{V}_0$ on $M$. If $\tilde{V}_\alpha = (a_\alpha^j), \tilde{V}_0 = (b^j)$ satisfy the condition of the existence and uniqueness of solution of SDE, then it satisfies the above condition.

Then define

$$u(t,x) = \mathbb{E}[f(X_t) \mid X_0 = x], \quad f \in C_c^\infty(M).$$

Because $f \in C_c^\infty(M)$,

$$M_t^f = f(X_t) - f(X_0) - \int_0^t Lf(X_s)ds$$

is a martingale, which implies that $\mathbb{E}[M_t^f \mid X_0] = M_0^f = 0$. Therefore,

$$u(t,x) = \mathbb{E}[f(X_t) \mid X_0 = x] = f(x) + \int_0^t \mathbb{E}\left[ Lf(X_s) \mid X_0 = x \right] ds,$$

called the Dynkin's formula. Moreover, for $u_t = u(t, \cdot)$,

$$\begin{cases} \dfrac{\partial u_t}{\partial t} = Lu_t, \\ u(0,x) = f(x). \end{cases}$$

This result can be extended to so-called Feynman-Kac formula. Let

$$v(t,x) = \mathbb{E}\left[ \exp\left( \int_0^t g(X_s)ds \right) f(X_t) \mid X_0 = x \right].$$

Then $v_t = v(t, \cdot)$ satisfies

$$\begin{cases} \dfrac{\partial v_t(x)}{\partial t} = Lv_t(x) + g(x)v_t(x), \\ v(0,x) = f(x). \end{cases}$$

# Chapter 2

# Stochastic Differential Geometry

## 2.1 Horizontal Lift

Let $M$ be $d$-dimensional smooth manifold equipped with an affine connection $\nabla$. Let $\Gamma_{ij}^k$ be the corresponding Christoffel symbol, i.e,

$$\nabla_{X_i} X_j = \Gamma_{ij}^k X_k,$$

where $X_i = \frac{\partial}{\partial x^i}$. $\nabla$ can induce concepts, such as, parallel moving ($\nabla_{\dot{c}(t)} V = 0$) and geodesic ($\nabla_{\dot{c}(t)} \dot{c}(t) = 0$).
For $x \in M$, a frame at $x$ is a $\mathbb{R}$-linear isomorphism $u \colon \mathbb{R}^d \to T_x M$. Let $F(M)_x$ be the set of all frameworks at $x$. $GL(d, \mathbb{R})$ acts on $F(M)_x$ as $u \mapsto ug = u \circ g$. Let $F(M) = \bigcup_{x \in M} F(M)_x$, called the frame bundle of dimension $d + d^2$, with the canonical projection $\pi \colon F(M) \to M$. Note that each $F(M)_x \cong GL(d, \mathbb{R})$. In fact, $(F(M), M, GL(d, \mathbb{R}))$ is a principal bundle $M \cong F(M)/GL(d, \mathbb{R})$. Let

$$F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d := F(M) \times \mathbb{R}^d / \sim,$$

where $(ug, v) \sim (u, gv)$. Then define

$$\Phi \colon F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d \to TM$$

by

$$\Phi([u, v]) = u(v) \in T_{\pi(u)} M.$$

$\Phi$ is a linear isomorphism and so $F(M) \times_{GL(d,\mathbb{R})} \mathbb{R}^d \cong TM$.
Consider the canonical projection $\pi \colon F(M) \to M$, because it is a smooth submersion and $\pi^{-1}(x) = F(M)_x \cong GL(d, \mathbb{R})$, for

$$d\pi_u \colon T_u F(M) \to T_{\pi(u)} M,$$

the vertical space

$$V_u F(M) := \ker(d\pi_u) = T_u F(M)_{\pi(u)} \cong \mathfrak{g}(d, \mathbb{R}).$$

Let $M$ be equipped with an affine connection $\nabla$. Let $c(t)$ be a smooth curve on $M$ and $u(t)$ be a lift of $c(t)$ on $F(M)$, i.e., $\pi(u(t)) = c(t)$. Because $u(t) \in F(M)_{c(t)}$, for any $e \in \mathbb{R}^d$, $u_e(t) = u(t)e \in T_{c(t)} M$, which means $u_e \in \Gamma(TM)$ is a vector field along $c(t)$. We say $u(t)$ is a horizontal lift if for any $e \in \mathbb{R}^d$, $u_e$ is parallel along $c$, i.e.,

$$\nabla_{\dot{c}(t)} u_e(t) \equiv 0.$$

**Lemma 2.1.1.** *For any smooth curve $c\colon (-\varepsilon, \varepsilon) \to M$, let $u_0 \in F(M)_{c(0)}$. Then there exists a unique $u\colon (-\varepsilon, \varepsilon) \to F(M)$ such that*

$$\pi \circ u = c, \quad u(0) = u_0, \quad \nabla_{\dot{c}(t)} \dot{u}_a(t) = 0, \ \forall\ a \in \mathbb{R}^d.$$

*Moreover, $u$ dependents smoothly on $(c, u_0)$, and if $u_1 = u_0 g$, the $t \mapsto u(t)g$ is the unique horizontal lift starting from $u_1$.*

*Proof.* WTLG, assume $c((-\varepsilon, \varepsilon))$ is contained in a chart $(U; x^1, \cdots, x^d)$ and let $\Gamma_{ij}^k$ be the corresponding Christoffel symbol for $\nabla$. Let $e_1, \cdots, e_d$ be the canonical basis of $\mathbb{R}^d$
Then a vector field $V(t) = V^i(t) \left.\frac{\partial}{\partial x^i}\right|_{c(t)}$ along $c(t)$ is parallel to $c(t)$ if and only if $V(t)$ satisfies

$$\dot{V}^i(t) + A_k^i(t) V^k(t) = 0, \quad \forall\ i,$$

where $A_k^i(t) = \Gamma_{jk}^i(c(t)) \dot{c}^j(t)$. For each $e_j$, above equation has a unique solution

$$V_{e_j}(t) = V_j^i(t) \left.\frac{\partial}{\partial x^i}\right|_{c(t)}$$

with initial value $V_{e_j}(0) = u_0 e_j = u_{0,j}^i \left.\frac{\partial}{\partial x^i}\right|_{c(0)}$. So the matrix $U(t) = (V_j^i(t))$ is the unique solution of ODE

$$\dot{U}(t) + A(t) U(t) = 0, \quad U(0) = U_0 := (u_{0,j}^i).$$

Moreover, because

$$\frac{d}{dt} \det U(t) = \operatorname{tr}\left(U(t)^{-1} \dot{U}(t)\right) \det U(t) = -\operatorname{tr}(A(t)) \det U(t),$$

and $u_0 \in F(M)_{c(0)}$ (i.e., $\det u_0 \neq 0$),

$$\det U(t) = \det U_0 \cdot \exp\left(-\int_{t_0}^t \operatorname{tr}(A(s)) ds\right) \neq 0,$$

it means $U(t) \in GL(d, \mathbb{R})$ for all $t$. Let

$$u(t)a := V_j^i(t) a^j \left.\frac{\partial}{\partial x^i}\right|_{c(t)}$$

for any $a = (a^j) \in \mathbb{R}^d$. Then $u(t) \in F(M)_{c(t)}$ and satisfies $u(0) = u_0$ and the parallel condition. The uniqueness of $u(t)$ is directly obtained by the uniqueness of solution of the ODE, which also implies that $t \mapsto u(t)g$ is the unique lift starting from $u_1$. Also, the smoothness on initial values is also by the theory of ODE. □

*Remark* 2.1.2. Moreover, give a curve $c(t)$ on $M$, let $u(t), v(t)$ be its horizontal lifts starting from $u_0, v_0$ respectively, because $u_0 = v_0(v_0^{-1} u_0)$, by above

$$u(t) = v(t)(v_0^{-1} u_0).$$

Moreover, for any $t_0, t_1$, it is obvious that

$$\mathcal{P}_{c,t_0,t_1} = u_{t_1} u_{t_0}^{-1} \colon T_{c(t_0)} M \to T_{c(t_1)} M$$

is the parallel moving along $c(t)$ by considering $V(t) = u_t(u_{t_0}^{-1}(X))$ and the uniqueness of parallel moving.

*Remark* 2.1.3. Moreover, if $\dot{c}(0) = X = X^i \left.\frac{\partial}{\partial x^i}\right|_{c(0)}$, then we can see

$$\dot{u}(0) = -\Gamma^i_{\ell k}(c(0)) X^\ell u^k_{0,j} \left.\frac{\partial}{\partial v^i_j}\right|_{u_0} + X^i \left.\frac{\partial}{\partial x^i}\right|_{u_0}$$

which implies $c(0)$, $\dot{c}(0)$, and $u_0$ uniquely determined $\dot{u}(0) \in T_{u_0}F(M)$.
For a given $X \in T_xM$, let $c$ be a smooth curve in $M$ with $c(0) = x$ and $\dot{c}(0) = X$. Let $u \in F(M)_x$. Let $u(t)$ be the unique horizontal lift of $c$ starting from $u$. Then define

$$X^H_u = \dot{u}(0) \in T_uF(M),$$

which is well-defined because it is independent of the choice of $c$, called a horizontal vector. Let

$$H_uF(M) := \left\{X^H_u \colon X \in T_xM\right\} \subset T_uF(M),$$

called the horizontal space.

**Lemma 2.1.4.** *For the canonical projection $\pi\colon F(M) \to M$ and any $u \in F(M)$ with $x = \pi(u)$,*

$$d\pi_u \colon H_uF(M) \to T_xM$$

*is an isomorphism. Therefore,*

$$T_uF(M) = V_uF(M) \oplus H_uF(M).$$

*Proof.* For any $X^H_u \in H_uF(M)$, $X^H_u = \dot{u}(0)$ for some $u(t)$ which is a horizontal lift of $c(t)$ with $c(0) = x$ and $\dot{c}(0) = X$ and starting from $u$ with $\pi(u) = x$. Then

$$d\pi_u(X^H_u) = \left.\frac{d}{dt}\right|_{t=0} \pi(u(t)) = \left.\frac{d}{dt}\right|_{t=0} c(t) = X.$$

And from the construction, we have seen $X \neq 0$ implies that $X^H_u \neq 0$. □

For any $e \in \mathbb{R}^d$ and $u \in F(M)$, $ue \in T_{\pi(u)}M$ and so we can consider

$$H_e(u) := (ue)^H_u = \dot{u}(0) \in H_uF(M),$$

where $u(t)$ is the horizontal lift of $c(t)$ for $c(0) = \pi(u)$ and $\dot{c}(0) = ue$ starting from $u$. By above, we know

$$dg_u H_e(u) = H_{g^{-1}e}(ug)$$

where $g\colon F(M) \to F(M)$ is $g(u) = u \circ g = ug$, because

$$dg_u H_e(u) = \left.\frac{d}{dt}\right|_{t=0} u(t)g.$$

*Remark* 2.1.5. Note that because

$$(d\pi_u)^{-1} \colon X \mapsto X^H_u$$

is $\mathbb{R}$-linear,

$$H_{\lambda e + \mu e'}(u) = \lambda H_e(u) + \mu H_{e'}(u), \quad \forall\ e, e' \in \mathbb{R}^d,\ u \in F(M),\ \lambda, \mu \in \mathbb{R}.$$

Let $e_1, \cdots, e_d$ be the canonical basis of $\mathbb{R}^d$. Then

$$H_j(u) = H_{e_j}(u) \in H_uF(M)$$

forms a basis of $H_uF(M)$ for $j = 1, \cdots, d$, because for any $v = (v^i) \in \mathbb{R}^d$, i.e., $v = v^i e_i$, then

$$H_v(u) = \sum_{i=1}^d v^i H_i(u).$$

$\{H_i\}_{i=1}^d$ are called the fundamental horizontal vector fields.

*Remark* 2.1.6. For any $X \in \Gamma(TM)$, its unique horizontal lift vector field $X^H \in \Gamma(TF(M))$ is defined as

$$X^H(u) := (X(\pi(u)))^H_u \in T_uF(M), \quad \forall\, u \in F(M).$$

Moreover, by definition, $d\pi_u(X^H(u)) = X(\pi(u))$, i.e,

$$\pi_* X^H = X.$$

This is the reason why $X^H$ is called a lift.

Let $c(t)$ be a smooth curve on $M$ and $u(t)$ be the horizontal lift of it starting from $u_0$. Then we can consider a curve $w(t)$ on $\mathbb{R}^d$, called the anti-development of $c(t)$ w.s.t. $u_0$, defined as

$$w_{u_0}(t) = \int_0^t u(s)^{-1}\dot{c}(s)ds,$$

which is because $\dot{c}(t) \in T_{c(t)}M$ and $u(t)^{-1} \colon T_{c(t)}M \to \mathbb{R}^d$. But note that $w_{u_0}(t)$ dependents on the choice of $u_0$. Let $v(t)$ be the horizontal lift of $c(t)$ starting from $v_0$. If $u_0 = v_0 g$, i.e., $v_0 = u_0 g^{-1}$, then $v(t) = u(t)g^{-1}$ and thus

$$w_{v_0}(t) = \int_0^t v(s)^{-1}\dot{c}(s)ds = g\int_0^t u(s)^{-1}\dot{c}(s)ds = g w_{u_0}(t).$$

Fix $u_0$ and let $w(t) = w_{u_0}(t)$. Because $\dot{w}(t) = u(t)^{-1}\dot{c}(t)$, i.e, $u(t)\dot{w}(t) = \dot{c}(t)$,

$$H_{\dot{w}(t)}(u(t)) = (u(t)\dot{w}(t))^H_{u(t)} = (\dot{c}(t))^H_{u(t)} = \dot{u}(t),$$

i.e.

$$\dot{u}(t) = \dot{w}^i(t)H_i(u(t)).$$

Note that from the definition of $H_i(u(t))$ this is linear in $u(t)$, when given $w(t)$ and $u_0$, this ODE has a unique solution $u(t)$ called the development of $w(t)$ in $F(M)$ and its projection $c(t) = \pi(u(t))$ is called the development of $w(t)$ in $M$.

**Orthogonal Frame.** When considering $(M, g)$ be a Riemannian manifold with the Levi-Civita connection $\nabla$. Instead of considering $F(M)$, we consider the orthogonal frame $O(M)$, that is, $O(M)_x$ consists of all $u \colon \mathbb{R}^d \to T_xM$ orthogonal map. The problem is if we can consider the horizontal lift on $O(M)$. To do that, let

$$u(t)a = V^i_j(t)a^j\frac{\partial}{\partial x^i}$$

be the parallel vector field by solving the ODE

$$\dot{U}(t) + A(t)U(t) = 0.$$

The problem is whether $u(t) \in O(M)_{c(t)}$, that is, whether

$$\langle u(t)e_i, u(t)e_j\rangle = \left\langle V^\alpha_i\frac{\partial}{\partial x^\alpha}, V^\beta_j\frac{\partial}{\partial x^\beta}\right\rangle = V^\alpha_i g_{\alpha\beta}V^\beta_j = (U(t)^\top G(t)U(t))_{ij} = \delta_{ij},$$

i.e., $U(t)^\top G(t)U(t) \equiv I$, where $U(t) = (V^i_j(t))$ and $G(t) = (g_{ij}(c(t)))$, the local expression of metric. Note that, by the consistency of metric

$$\dot{g}_{ij}(c(t)) = \dot{c}(t)\left\langle\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^i}\right\rangle$$

$$
\begin{aligned}
&= \left\langle \nabla_{\dot c(t)} \frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right\rangle + \left\langle \frac{\partial}{\partial x^i}, \nabla_{\dot c(t)} \frac{\partial}{\partial x^j} \right\rangle \\
&= \dot c^\alpha \Gamma^k_{\alpha i} g_{kj} + \dot c^\alpha \Gamma^k_{\alpha j} g_{ik} \\
&= A^\top G + GA,
\end{aligned}
$$

where $A = (A^i_k)$ for $A^i_k(t) = \Gamma^i_{jk}(c(t))\dot c^j(t)$. Moreover, because

$$
\dot U(t) + A(t)U(t) = 0,
$$

we have

$$
\begin{aligned}
\frac{d}{dt} U^\top G U &= \dot U^\top G U + U^\top \dot G U + U^\top G \dot U \\
&= -U^\top A^\top G U + U^\top (A^\top G + GA) U - U^\top G(AU) = 0.
\end{aligned}
$$

Therefore,

$$
U \top G U \equiv U^\top(0) G(0) U(0) = I,
$$

when $u(0) \in O(M)_{c(0)}$. Therefore, the horizontal lift lies on $O(M)$. For Riemannian manifolds, the following $F(M)$ can be replaced by $O(M)$.

## 2.2 Tensor Fields Scalarization

Let $\theta$ be a $(r, s)$-tensor field on $M$. For any $u \in F(M)$, let

$$
X_i = u e_i \in T_{\pi(u)} M
$$

for $i = 1, \cdot, d$, where $\{e_i\}$ is the canonical basis of $\mathbb{R}^d$. Then $\{X_i\}$ is a basis of $T_{\pi(u)}M$. Choose $\{X^i\}$ be the its dual basis of $T^*_{\pi(u)}M$. Then under these basis, we have

$$
\theta(\pi(u)) = \theta^{i_1 \cdots i_r}_{j_1 \cdots j_s} X_{i_1} \otimes \cdots \otimes X_{i_r} \otimes X^{j_1} \otimes \cdots \otimes X^{j_s}.
$$

and we define the scalarization

$$
\widetilde{\theta} \colon F(M) \to (\mathbb{R}^d)^{\otimes r} \otimes (\mathbb{R}^d)^{* \otimes s} =: \otimes^{(r,s)} \mathbb{R}^d,
$$

as

$$
\widetilde{\theta}(u) = \theta^{i_1 \cdots i_r}_{j_1 \cdots j_s} e_{i_1} \otimes \cdots \otimes e_{i_r} \otimes e^{j_1} \otimes \cdots \otimes e^{j_s},
$$

where $\{e^i\}$ is the dual basis of $\{e_i\}$ in $(\mathbb{R}^d)^*$. Note that $\widetilde{\theta}$ is $GL(d, \mathbb{R})$-equivariant, i.e.,

$$
\widetilde{\theta}(ug) = g\tilde{\theta}(u).
$$

If $\theta$ is $(0,0)$-tensor, i.e., $\theta = f \colon M \to \mathbb{R}$,

$$
\widetilde{f} = f \circ \pi \colon F(M) \to \mathbb{R}.
$$

Therefore, $\widetilde{\theta}$ can be considering as a lift of $\theta$ on $F(M)$.

*Remark* 2.2.1. For a finite dimensional vector space $V$, let $\{v_i\}$ be a basis of $V$ and $\{v_i^*\}$ be its dual basis in $V^*$. If

$$
w_j = g^i_j v_i
$$

and $\{w_j\}$ is another basis, i.e., $G = (g^i_j)_{i \times j} \in GL$, then for $w^*_\ell = \sum_k h^\ell_k (v_k)^*$, we have the matrix

$$
H = (h^\ell_k)_{\ell \times k} = G^{-1}.
$$

Moreover, for $u \in F(M)$, $u^{-1}\colon T_{\pi(u)}M \to \mathbb{R}^d$ and so we can consider $u^*\colon T^*_{\pi(u)}M \to (\mathbb{R}^d)^*$, which follows that we can defined $u^{-1}\colon \otimes^{(r,s)} T_{\pi(u)}M \to \otimes^{(r,s)}\mathbb{R}^d$ as

$$u^{-1} = \underbrace{u^{-1} \otimes \cdots \otimes u^{-1}}_{r} \times \underbrace{u^* \otimes \cdots \otimes u^*}_{s}.$$

Then it is obvious that for $\theta \in \Gamma(\otimes^{(r,s)}TM)$

$$\widetilde{\theta}(u) = u^{-1}\theta(\pi(u)).$$

On the other hand, for $u\colon \mathbb{R}^d \to T_xM$, consider $(u^{-1})^*\colon (\mathbb{R}^d)^* \to T^*_{\pi(u)}M$. Then it can extend $u$ to be defined on $\otimes^{(r,s)}(\mathbb{R}^d)^* \to \otimes^{(r,s)}T^*_{\pi(u)}M$ as

$$u = \underbrace{(u^{-1})^* \otimes \cdots \otimes (u^{-1})^*}_{r} \otimes \underbrace{u \otimes \cdots \otimes u}_{s},$$

where $\otimes^{(r,s)}(\mathbb{R}^d)^* = \otimes^r(\mathbb{R}^d)^* \otimes \otimes^s\mathbb{R}^d$ and $\otimes^{(r,s)}T^*_{\pi(u)}M = \otimes^r T^*_{\pi(u)}M \otimes \otimes^s T_{\pi(u)}M$. Then for any $\theta \in \Gamma(\otimes^{(r,s)}TM)$,

$$\theta_{\pi(u)}\colon \otimes^{(r,s)} T^*_{\pi(u)}M \to \mathbb{R}, \quad \widetilde{\theta}_u\colon \otimes^{(r,s)} (\mathbb{R}^d)^* \to \mathbb{R},$$

they have

$$\theta_{\pi(u)} \circ u = \widetilde{\theta}(u),$$

because $\{(u^{-1})^*e^j\}$ is the dual basis of $\{ue_i\}$ in $T^*_{\pi(u)}M$. For $f \in C^\infty(M)$, clearly $f_{\pi(u)} = \widetilde{f}(u)$. And so

$$\nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u).$$

**Proposition 2.2.2.** *Let $X \in \Gamma(TM)$ and $\theta \in \Gamma(\otimes^{(r,s)}TM)$. Then*

$$\widetilde{\nabla_X\theta} = X^H\widetilde{\theta},$$

*i.e., $\widetilde{\nabla_X\theta}(u) = X^H_u\widetilde{\theta}$ for all $u \in F(M)$, where $X^H_u\widetilde{\theta}$ acts on each component since $X^H_u \in T_uF(M)$.*

*Proof.* Fix $u \in F(M)$. Let $c(t)$ be a curve in $M$ with $c(0) = \pi(u)$ and $\dot{c}(0) = X_{\pi(u)}$. Let $u(t)$ be the unique horizontal lift of $c(t)$ starting from $u(0) = u$. Then as we know, $\tau_t = u(t)u^{-1}$ is the parallel moving along $c(t)$. So we have

$$\left.\frac{d}{dt}\right|_{t=0} \tau_t^{-1}\theta(c(t)) = \nabla_X\theta(\pi(u)).$$

On the other hand, because $\pi(u(t)) = c(t)$, we have

$$\widetilde{\theta}(u(t)) = u(t)^{-1}\theta(c(t)).$$

Moreover, because $X^H_u = \dot{u}(0)$,

$$\begin{aligned} X^H_u\widetilde{\theta} &= \left.\frac{d}{dt}\right|_{t=0} \widetilde{\theta}(u(t)) \\ &= \left.\frac{d}{dt}\right|_{t=0} u(t)^{-1}\theta(c(t)) \\ &= u^{-1} \left.\frac{d}{dt}\right|_{t=0} \tau_t^{-1}\theta(c(t)) \\ &= u^{-1}(\nabla_X\theta)(\pi(u)) = \widetilde{\nabla_X\theta}(u), \end{aligned}$$

where the first equality is because $\widetilde{\theta}$ is a tensor field on $\mathbb{R}^d$, i.e. the coordinate does not change along time. ☐

*Remark* 2.2.3. By the definition of $X_u^H\widetilde{\theta}$, because it is a tensor field on $\mathbb{R}^d$,

$$(X_u^H\widetilde{\theta})(v^1,\cdots,v^r,w_1,\cdots,w_s) = X_u^H\left(\widetilde{\theta}(u)(v^1,\cdots,v^r,w_1,\cdots,w_s)\right)$$

for any $v^i \in (\mathbb{R}^d)^*$ and $w_j \in \mathbb{R}^d$.

*Remark* 2.2.4. Above is also true for $f \in C^\infty(M)$.

$$\begin{aligned}
X_u^H\widetilde{f} &= \frac{d}{dt}\bigg|_{t=0}\widetilde{f}(u(t)) \\
&= \frac{d}{dt}\bigg|_{t=0} f(\pi(u(t))) = \frac{d}{dt}\bigg|_{t=0} f(c(t)) \\
&= \nabla_X f(c(0)) = \nabla_X f(\pi(u)) = \widetilde{\nabla_X f}(u).
\end{aligned}$$

**Example 2.2.5** (Hessian). For any $f \in C^\infty(M)$, consider the Hessian $\nabla^2 f \in \Gamma(\otimes^{0,2}TM)$,

$$\begin{aligned}
\nabla^2 f_{\pi(u)}(ue_i, ue_j) &= (\nabla_{ue_j}\nabla f)_{\pi(u)}(ue_i) = (\widetilde{\nabla_{ue_j}\nabla f})(u)(e_i) \\
&= \left((ue_j)_u^H\widetilde{\nabla f}\right)(e_i) = H_j\left(\widetilde{\nabla f}(u)(e_i)\right) \\
&= H_j\left(\nabla f_{\pi(u)}(ue_i)\right) = H_j\left((\nabla_{ue_i} f)_{\pi(u)}\right) \\
&= H_j\left(\widetilde{\nabla_{ue_i} f}(u)\right) = H_j H_i \widetilde{f}
\end{aligned}$$

where $\widetilde{f} = f \circ \pi$.

## 2.3 Stochastic Horizontal Lift

Given a SDE on $F(M)$:

$$dU_t = H(U_t) \circ dW_t = \sum_{i=1}^d H_i(U_t) \circ dW_t^i, \tag{2.1}$$

where $W$ is an $\mathbb{R}^d$-valued semimartingale and $\{H_i\}_{i=1}^d$ are the fundamental horizontal vector fields on $F(M)$.

**Definition 2.3.1.** (1) An $F(M)$-valued semimartingale is said to be horizontal if there exists an $\mathbb{R}^d$-valued semimartingale $W$ with $W_0 = 0$ such that SDE (2.1) holds. Then $W$ is called the anti-development of $U$ (or of its projection $X = \pi(U)$).

(2) Let $W$ be an $\mathbb{R}^d$-valued semimartingale with $W_0 = 0$ and $U_0$ is an $F(M)$-valued, $\mathcal{F}_0$-measurable random variable. The solution of SDE (2.1) with initial condition $U_0$ is called a stochastic development of $W$ in $F(M)$, so is its projection $X = \pi(U)$.

(3) Let $X$ be an $M$-valued semimartingale. An $F(M)$-valued horizontal semimartingale $U$ such that $\pi(U) = X$ is called a stochastic horizontal lift of $X$.

Note that $W \mapsto U$ is by considering the solution of SDE (2.1), and $U \mapsto X$ is just by projection, but we need $X \mapsto U$ and $U \mapsto W$.

Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. Let $f = (f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function and let lift it to

$$\widetilde{f}\colon F(M) \to M \subset \mathbb{R}^N$$

as $\widetilde{f} = f \circ \pi$, i.e. $\widetilde{f}(u) = f(\pi(u)) = \pi(u)$ as in $\mathbb{R}^N$.
For any $M$-valued semimartingale, we regard $X = (X^\alpha)_{\alpha=1}^N = (f^\alpha(X))_{\alpha=1}^N \in \mathbb{R}^N$ as an $\mathbb{R}^N$-valued semimartingale.
For any $x \in M$, let $P(x)\colon \mathbb{R}^N \to T_xM$ be the orthogonal projection by viewing $T_xM \subset \mathbb{R}^N$. Let $P_\alpha(x) = P(x)e_\alpha \in T_xM$ and so $P_\alpha \in \Gamma(TM)$.

*Remark* 2.3.2. For the smoothness of $P_\alpha$, first we can equip $M$ with the induced metric. Then on each local chart, there exists a smooth orthonormal frames $E_i$ on $TM$, which implies that

$$P_\alpha = \sum_i \langle e_\alpha, E_i \rangle E_i.$$

So $P_\alpha$ is smooth.
Define $P_\alpha^H \in \Gamma(TF(M))$ as

$$P_\alpha^H(u) = (P_\alpha(\pi(u)))_u^H \in T_uF(M), \quad \forall\ u \in F(M).$$

By using the fundamental horizontal lifts,

$$P_\alpha^H(u) = \left(u(u^{-1}P_\alpha(\pi(u)))\right)_u^H = (u^{-1}P_\alpha(\pi(u)))^i H_i(u).$$

**Lemma 2.3.3.** *For notations as above, when viewing $T_xM \subset \mathbb{R}^N$, we have*

$$P_\alpha^H(u)\widetilde{f} = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i(u)\widetilde{f^\alpha} = ue_i.$$

*Proof.* Let $u(t)$ be the horizontal lift from $u(0) = u$ of a curve $c(t)$ on $M$ with $c(0) = \pi(u)$ and $\dot{c}(0) = \widetilde{f}(u) = P_\alpha(\pi(u))$. It follows that $P_\alpha^H(u) = \dot{u}(0)$. Because $\widetilde{f}\colon F(M) \to \mathbb{R}^N$ and $P_\alpha^H(u) \in T_uF(M)$,

$$P_\alpha^H(u)\widetilde{f} = \left.\frac{d}{dt}\right|_{t=0} \widetilde{f}(u(t)) = \left.\frac{d}{dt}\right|_{t=0} \pi(u(t)) = \left.\frac{d}{dt}\right|_{t=0} c(t) = P_\alpha(\pi(u)).$$

Next, let $v(t)$ be the horizontal lift start from $v(0) = u$ of curve $x(t)$ with $x(0) = \pi(u)$ and $\dot{x}(0) = ue_i$. So

$$H_i(u)\widetilde{f} = \left.\frac{d}{dt}\right|_{t=0} \widetilde{f}(v(t)) = \left.\frac{d}{dt}\right|_{t=0} \pi(v(t)) = \left.\frac{d}{dt}\right|_{t=0} x(t) = ue_i,$$

which implies that $H_i(u)\widetilde{f} \in T_{\pi(u)}M$ and so

$$P(\pi(u))H_i(u)\widetilde{f} = H_i(u)\widetilde{f} \ \Rightarrow\ P_\alpha(\pi(u))H_i(u)\widetilde{f^\alpha} = H_i(u)\widetilde{f} = ue_i. \qquad \square$$

*Remark* 2.3.4. For a vector field $H \in \Gamma(TF(M))$, we usually denote $Hg(u) := H(u)g$ for any $u \in F(M)$ and $g\colon F(M) \to \mathbb{R}^n$. Under this notation,

$$P_\alpha^H\widetilde{f}(u) = P_\alpha(\pi(u)), \quad P_\alpha(\pi(u))H_i\widetilde{f^\alpha}(u) = ue_i.$$

*Remark* 2.3.5. Note that above two identities view $P_\alpha(\pi(u))$ as a vector in $\mathbb{R}^N$. More precisely, for any $g \in C^\infty(M)$, $g \circ \pi \in C^\infty(F(M))$. We similarly have

$$H_i(u)(g \circ \pi) = \left.\frac{d}{dt}\right|_{t=0} g(c(t)) = (ue_i)g.$$

and

$$P_\alpha^H(u)(g \circ \pi) = P_\alpha(\pi(u))g$$

or in other words,

$$d\pi_u(P_\alpha^H(u)) = P_\alpha(\pi(u)),$$

which is the definition of horizontal lift.

**Lemma 2.3.6.** *Assume $M \subset \mathbb{R}^N$ is a submanifold that is closed. If $X$ is an $M$-valued semimartingale, then*

$$X_t = X_0 + \int_0^t P(X_s) \circ dX_s = X_0 + \int_0^t P_\alpha(X_s) \circ dX_s^\alpha.$$

*Proof.* Let $Q_\alpha(x) = e_\alpha - P_\alpha(x)$, i.e., $Q_\alpha(x)$ normal to $T_xM$ and $Q_\alpha \in \Gamma(NM)$. Let

$$Y_t = X_0 + \int_0^t P_\alpha(X_s) \circ dX_t^\alpha$$

By the closedness of $M$, there exists $f \in C^\infty(\mathbb{R}^n)$ such that $f(x) \geq 0$ and $f^{-1}(0) = M$. Then by Itô formula,

$$f(Y_t) = f(X_0) + \int_0^t P_\alpha f(X_t) \circ dX_t^\alpha.$$

Because $P_\alpha \in \Gamma(TM)$ and $f|_M = 0$, $P_\alpha f(x) = 0$ for all $x \in M$. Since $X$ is $M$-valued, $f(Y_t) = 0$, i.e., $Y \in f^{-1}(0) = M$, and $Y$ is an $M$-valued semimartingale.
Next, consider a tubular neighborhood $U$ of $M$ and defined the natural projection $\tilde{h}\colon U \to M$, which can be extended a smooth $h\colon \mathbb{R}^N \to M$. Because $Q_\alpha \in \Gamma(NM)$, by choosing Fermi coordinates, we have

$$Q_\alpha h(x) = 0 \;\Rightarrow\; P_\alpha h(x) = \frac{\partial h}{\partial x^\alpha}.$$

So

$$\begin{aligned} Y_t = h(Y_t) &= h(X_0) + \int_0^t P_\alpha h(X_s) \circ dX_s^\alpha \\ &= X_0 + \int_0^t \frac{\partial h}{\partial x^\alpha} \circ dX_s^\alpha \\ &= h(X_t) = X_t. \end{aligned}$$

□

**Theorem 2.3.7.** *A horizontal semimartingale $U$ on $F(M)$ has a unique anti-development $W$. In fact,*

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha,$$

*for $X = \pi(U)$.*

*Proof.* By definition, $W$ should be

$$dU_t = H_i(U_t) \circ dW_t^i.$$

Let $\widetilde{f}$ be defined as above. Then

$$\widetilde{f}(U_t) = f(\pi(U_t)) = f(X_t) = X_t \in M \subset \mathbb{R}^N.$$

Then $X_t$ should satisfy

$$dX_t^\alpha = H_i \widetilde{f^\alpha}(U_t) \circ dW_t^i.$$

Therefore, by above formulas, we have

$$U_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha = U_t^{-1} P_\alpha(X_t) H_i \widetilde{f}^\alpha(U_t) \circ dW_t^i = e_i dW_t^i = dW_t$$

By the uniqueness of SDE, $W_t$ uniquely exists and

$$W_t = \int_0^t U_s^{-1} P_\alpha(X_s) \circ dX_s^\alpha.$$

□

*Remark* 2.3.8. Note that if $U_t^1$ and $U_t^2$ are two horizontal lifts of $X$ starting from $U_0^1$ and $U_0^2 = U_0^1 g$, then by above, we obviously have

$$W_t^1 = g^{-1} W_t^2.$$

*Remark* 2.3.9. On the other hand,

$$\begin{aligned} P_\alpha(X_t) \circ dX_t^\alpha &= P_\alpha(X_t) H_i \widetilde{f^\alpha}(U_t) \circ dW_t^i \\ &= U_t e_i \circ dW_t^i. \end{aligned}$$

Then by above,

$$X_t = X_0 + \int_0^s U_s e_i \circ dW_s^i.$$

In formally, we just denote

$$X_t = X_0 + \int_0^s U_s \circ dW_s.$$

Similarly, because $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$, we can informally write

$$W_t = \int_0^t U_s^{-1} \circ dX_s.$$

**Theorem 2.3.10.** *Suppose $X$ is an $M$-valued semimartingale (up to a stopping time $\tau$), and $U_0$ is an $F(M)$-valued $\mathcal{F}_0$-random variable such that $\pi(U_0) = X_0$. Then there exists a unique horizontal lift $U$ (up to $\tau$) of $X$ starting from $U_0$.*

*Proof.* Consider a SDE defined as

$$dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha.$$

Let $U_t$ be its unique solution, so $U_t$ is clearly a horizontal lift since $P_\alpha^H \in \Gamma F(M)$. It suffices to show $\pi(U_t) = X_t$. Let

$$Y_t = \widetilde{f}(U_t) = f(\pi(U_t)) = \pi(U_t) \in M \subset \mathbb{R}^N.$$

By above, we have

$$dY_t = P_\alpha^H \widetilde{f}(U_t) \circ dX_t^\alpha = P_\alpha(Y_t) \circ dX_t^\alpha.$$

By above lemma, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. By the uniqueness of SDE, $X = Y$.
Next, assume there exists another horizontal lift $\Pi_t$ of $X_t$. Then there exists a semimartingale $W$ such that

$$d\Pi_t = H_i(\Pi_t) \circ dW_t^i.$$

and by above theorem

$$dW_t = \Pi_t^{-1} P_\alpha(X_t) \circ dX_t^\alpha.$$

So

$$\begin{aligned} d\Pi_t &= \left(\Pi_t^{-1} P_\alpha(X_t)\right)^i H_i(\Pi_t) \circ dX_t^\alpha \\ &= P_\alpha^H(\Pi_t) \circ dX_t^\alpha. \end{aligned}$$

So $\Pi = U$ by the uniqueness of solution of SDE. □

*Remark* 2.3.11. When considering explosion time, there is a fact, if $X$ on $M$ is a semimartingale up to $\tau$, then its horizontal lift $H$ on $F(M)$ is also defined up to $\tau$.

Similarly as the deterministic case, given a semimartingale $X$ on $M$, the horizontal lift $U_t$ provide the stochastic parallel moving along $X_t$,

$$\tau^X_{t_1 t_2} = U_{t_2} U_{t_1}^{-1} \colon T_{X_{t_1}} M \to T_{X_{t_2}} M.$$

**Proposition 2.3.12.** *Let a semimartingale $X$ on a manifold $M$ be the solution of* $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$ *and $V_\alpha^H$ be the horizontal lift of $V_\alpha$. Then the horizontal lift $U$ of $X$ starting from $U_0$ satisfy* $\mathrm{SDE}(V_1^H, \cdots, V_N^H; Z, U_0)$*, and the anti-development of $X$ is given by*

$$W_t = \int_0^t U_s^{-1} V_\alpha(X_s) \circ dZ_s^\alpha.$$

*Proof.* Assume $M \subset \mathbb{R}^N$ that is closed and use above notations. First, we similar have

$$V_\alpha^H \widetilde{f}(u) = V_\alpha(\pi(u)).$$

Let $\Pi$ be the solution of $\mathrm{SDE}(V_1^H, \cdots, V_N^H; Z, U_0)$ and $Y_t = \widetilde{f}(\Pi_t) = \pi(\Pi_t)$.

$$dY_t = V_\alpha^H \widetilde{f}(\Pi_t) \circ dZ_t^\alpha = V_\alpha(Y_t) \circ dZ_t^\alpha,$$

which implies that $Y = X$ by the uniqueness of solution of $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$. And by the uniqueness of horizontal lift, $U = \Pi$.
Next, the anti-development of $X$ is given by

$$dW_t = U_t^{-1} P_\beta(X_t) \circ dX_t^\beta.$$

Because $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$, i.e., $dX_t^\beta = V_\alpha^\beta(X_t) \circ dZ_t^\alpha$,

$$dW_t = U_t^{-1} P_\beta(X_t) V_\alpha^\beta(X_t) \circ dZ_t^\alpha.$$

Because $V_\alpha(X_t) \in T_{X_t} M$,

$$P_\beta(X_t) V_\alpha^\beta(X_t) = P(X_t) V_\alpha(X_t) = V_\alpha(X_t).$$

So

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

□

## 2.4 Stochastic Line Integral

**Line Integral.** For any 1-form $\omega = f(t)dt$ on $[a, b]$ of $\mathbb{R}$, define

$$\int_{[a,b]} \omega = \int_a^b f(t) dt.$$

Let $M$ be a smooth manifold and $\omega \in \Gamma(T^*M)$. Let $\gamma \colon [a, b] \to M$ smooth. Define the line integral of $\omega$ along $\gamma$ as

$$\int_\gamma \omega = \int_{[a,b]} \gamma^* \omega.$$

Clearly, it satisfies the linearity in $\omega$ and additivity w.s.t. $\gamma$. Moreover, for any smooth $F \colon M \to N$ and $\eta \in \Gamma(T^*N)$ and $\gamma$ in $M$,

$$\int_\gamma F^* \eta = \int_{F \circ \gamma} \eta.$$

**Proposition 2.4.1.** *Let $\omega \in \Gamma(T^*M)$ and $\gamma\colon [a,b] \to M$ smooth.*

$$\int_\gamma \omega = \int_a^b \omega_{\gamma(t)}(\dot\gamma(t))dt$$

*In particular, if $\omega = df$ for some $f \in C^\infty(M)$,*

$$\int_\gamma df = \int_a^b (f\circ\gamma)'(t)dt = f(\gamma(b)) - f(\gamma(a)).$$

*Proof.* Let $\gamma(t) = (\gamma^1(t), \cdots, \gamma^d(t))$ and $\omega = \omega_i dx^i$ be the coordinate expression. Then

$$(\gamma^*\omega)_t = \omega_i(\gamma(t))dx^i(\gamma(t)) = \omega_i(\gamma(t))d\gamma^i(t) = \omega_i(\gamma(t))(\gamma^i)'(t)dt$$

On the other hand,

$$\omega_{\gamma(t)}(\dot\gamma(t)) = \omega_i(\gamma(t))dx^i\left((\gamma^j)'(t)\frac{\partial}{\partial x^j}\right) = \omega_i(\gamma(t))(\gamma^i)'(t).$$

Therefore, we get the desired result. $\square$

**Stochastic Line Integral.** Let $x\colon [0,t] \to M$ and $u(t)$ be its horizontal lift on $F(M)$ starting from $u_0$. Let $w(t)$ be the anti-development of $x(t)$, i.e., $\dot w_t = u_t^{-1}\dot x_t$ and $w_0 = 0$. Let $\{e_i\}$ be the canonical basis of $\mathbb{R}^d$ and so $\{u_t e_i\}$ is a basis of $T_{x_t}M$. Let

$$w_t = w_t^i e_i \;\Rightarrow\; \dot x_t = \dot w_t^i\left(u_t e_i\right).$$

Let $\theta \in \Gamma(t^*M)$. Then

$$\int_x \theta = \int_0^t \theta_{x_s}(\dot x_s)ds = \int_0^t \dot w_s^i \theta_{x_s}(u_s e_i)ds.$$

**Definition 2.4.2.** Let $\theta \in \Gamma(t^*M)$ and $X$ be an $M$-valued semimartingale. Let $U$ be a horizontal lift of $X$ and $W$ be its anti-development. The stochastic line integral of $\theta$ along $X$ is defined by

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta_{X_s}(U_s e_i)\circ dW_s^i.$$

*Remark* 2.4.3. For such $\theta$, let $\widetilde\theta$ be the scalarization of 1-form $\theta$. Then by the definition, $\widetilde\theta\colon F(M) \to \mathbb{R}^d$ by viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$ and

$$\widetilde\theta(u) = \left(\theta_{\pi(u)}(ue_1), \cdots, \theta_{\pi(u)}(ue_d)\right)^\top \in \mathbb{R}^d.$$

So

$$\int_{X_{[0,t]}} \theta = \int_0^t \widetilde\theta(U_s)_i \circ dW_s^i = \int_0^t \widetilde\theta(U_s)^\top \circ dW_s.$$

Note that $\int_{X_{[0,t]}}\theta$ seems depend on the choice of horizontal lift, and so the connection on $M$. Actually, it is independent of the above choices by the following proposition and the fact that any semimartingale on $M$ is a solution of some SDE shown in above section.

**Proposition 2.4.4.** *Let $\theta \in \Gamma(T^*M)$ and $X$ be the solution of $dX_t = V_\alpha(X_t)\circ dZ_t^\alpha$. Then*

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_\alpha)(X_s)\circ dZ_s^\alpha.$$

*Proof.* Choose a horizontal lift $U$ and corresponding anti-development $W$ of $X$. Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

Therefore,

$$\begin{aligned}
\widetilde{\theta}(U_s)_i \circ dW_s^i &= \widetilde{\theta}(U_s)_i (U_t^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\
&= \theta_{X_s}(U_s e_i)(U_t^{-1} V_\alpha(X_s))^i \circ dZ_s^\alpha \\
&= \theta_{X_s}\left(U_s\left(e_i (U_t^{-1} V_\alpha(X_s))^i\right)\right) \circ dZ_s^\alpha \\
&= \theta_{X_s}(V_\alpha(X_s)) \circ dZ_s^\alpha.
\end{aligned}$$

Therefore, we have

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_\alpha)(X_s) \circ dZ_s^\alpha. \qquad \square$$

*Remark* 2.4.5. Note that the stochastic line integral of 1-form $\theta$ along a given $M$-valued semimartingale $X$ is independent of the choice of horizontal lift and also the connection. Symbolically,

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta \circ dX_t$$

**Example 2.4.6.** (1) For all $f \in C^\infty(M)$, assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$.

$$\int_{X_{[0,t]}} df = \int_0^t V_\alpha f(X_s) \circ dZ_s^\alpha = f(X_t) - f(X_0)$$

(2) Let $(U, x^1, \cdots, x^d)$ be a local chart of $M$. Let $\theta = \theta_i dx^i$. For any $X$ a semimartingale on $M$, let $X^i = x^i(X)$ that is a semimartingale on $\mathbb{R}$.

$$dX_t = V_i(X_t) \circ dX_t^i, \quad V_i = \frac{\partial}{\partial x^i},$$

because $df(X_t) = \frac{\partial f}{\partial x^i}(X_t) \circ dX_t^i$. Then

$$\int_{X_{[0,t]}} \theta = \int_0^t \theta(V_i(X_s)) \circ dX_s^i = \int_0^t \theta_i(X_s) \circ dX_s^i,$$

which is the local expression of stochastic line integral.

**Definition 2.4.7.** Let $\theta\colon \Gamma(TF(M)) \to C^\infty(F(M), \mathbb{R}^d)$, i.e., a $\mathbb{R}^d$-valued 1-form on $F(M)$. Define

$$\theta(Z)(u) = \theta_u(Z_u) := u^{-1}(d\pi_u(Z_u)),$$

for any $Z \in \Gamma(TF(M))$ and $u \in F(M)$, where $\pi\colon F(M) \to M$ is the natural projection. Then $\theta$ is called the solder form.

**Proposition 2.4.8.** *Let $U$ be a horizontal semimartingale on $F(M)$. Then its the corresponding anti-development is given by*

$$W_t = \int_{U_{[0,t]}} \theta,$$

*where $\theta$ is the solder form.*

*Proof.* Let $X_t = \pi(U_t)$. Then we have

$$dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha.$$

Note that $\pi_* P_\alpha^H = P_\alpha$. So we get

$$\begin{aligned}
\int_{U_{[0,t]}} \theta &= \int_0^t \theta(P_\alpha^H)(U_s) \circ dX_s^\alpha \\
&= \int_0^t U_s^{-1} d\pi_{U_s}(P_\alpha^H(U_s)) \circ dX_s^\alpha \\
&= \int_0^t U_s^{-1} P_\alpha(\pi(U_s)) \circ dX_s^\alpha = W_t.
\end{aligned}$$

□

**Quadratic Variation.** Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and $\widetilde{h}$ be its scalarization. So

$$\widetilde{h}(u)(e, e') = h_{\pi(u)}(ue, ue'), \quad \forall\ e, e' \in \mathbb{R}^d,\ u \in F(M).$$

Let

$$h^{\mathrm{sym}}(e, e') = \frac{h(e, e') + h(e', e)}{2}.$$

**Definition 2.4.9.** Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and $X$ be an $M$-valued semimartingale. Let $U$ be a horizontal lift of $X$ and $W$ be its anti-development. Note that

$$dX_t = U_t e_i \circ dW_t^i,$$

Then the $h$-quadratic variation of $X$ is defined as

$$\begin{aligned}
\int_0^t h(dX_s, dX_s) &:= \int_0^t h_{X_s}(U_s e_i, U_s e_j) d\left\langle W^i, W^j \right\rangle_s \\
&= \int_0^t \widetilde{h}(U_s)(e_i, e_j) d\left\langle W^i, W^j \right\rangle_s.
\end{aligned}$$

By viewing $dW_s^i dW_s^j = d\left\langle W^i, W^j \right\rangle_s$, then

$$\int_0^t h(dX_s, dX_s) = \int_0^t \widetilde{h}(U_s)(dW_s, dW_s).$$

Obviously,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h^{\mathrm{sym}}(dX_s, dX_s)$$

and if $h$ is anti-symmetric, $\int_0^t h(dX_s, dX_s) = 0$.

*Remark* 2.4.10. Note that this definition is independent the choice of $U$, i.e., the choice of $U_0$, because the anti-development of $U_t g$ is $g^{-1}W_t$ and

$$\begin{aligned}
h(U_t g e_i, U_t g e_j) d\left\langle (g^{-1}W)^i, (g^{-1}W)^j \right\rangle_t &= \widetilde{h}(U_t)(g e_i, g e_j) d\left\langle (g^{-1}W)^i, (g^{-1}W)^j \right\rangle_t \\
&= g_i^\alpha g_j^\beta \widetilde{h}(U_t)(e_\alpha, e_\beta) \tilde{g}_k^i \tilde{g}_\ell^j d\left\langle W^k, W^\ell \right\rangle_t \\
&= \widetilde{h}(U_t)(e_k, e_\ell) d\left\langle W^k, W^\ell \right\rangle_t,
\end{aligned}$$

where $g e_i = g_i^j$ and $(\tilde{g}_i^j) = (g_i^j)^{-1}$. So $h$-quadratic variation is well-defined.

**Proposition 2.4.11.** *Let $h \in \Gamma(\otimes^{(0,2)}TM)$ and $X$ be the solution of* $\mathrm{SDE}(V_1, \cdots, V_N; Z, X_0)$. *Then*

$$\int_0^t h(dX_s, dX_s) = \int_0^t h(V_\alpha, V_\beta)(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

*Proof.* Note that

$$dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$$

Let $U$ be a horizontal lift of $X$ and $W$ be its anti-development. Then

$$dW_t = U_t^{-1} V_\alpha(X_t) \circ dZ_t^\alpha.$$

and so

$$d\left\langle W^i, W^j \right\rangle_t = \left(U_t^{-1} V_\alpha(X_t)\right)^i \left(U_t^{-1} V_\beta(X_t)\right)^j d\left\langle Z^\alpha, Z^\beta \right\rangle_t.$$

It implies that

$$\begin{aligned} h(U_t e_i, U_t e_j) d\left\langle W^i, W^j \right\rangle_t &= h(U_t e_i, U_t e_j) \left(U_t^{-1} V_\alpha(X_t)\right)^i \left(U_t^{-1} V_\beta(X_t)\right)^j d\left\langle Z^\alpha, Z^\beta \right\rangle_t \\ &= h(V_\alpha, V_\beta)(X_t) d\left\langle Z^\alpha, Z^\beta \right\rangle_t. \end{aligned}$$

□

**Example 2.4.12.** (1) Let $(U, x^1, \cdots, x^d)$ be a coordinate and $h \in \Gamma(\otimes^{(0,2)}TM)$ locally written as

$$h = h_{ij} dx^i \otimes dx^j.$$

For $X = (X^i)$ on $M$,

$$\int_0^t h(dX_s, dX_s) = \int_0^t h_{ij}(X_s) d\left\langle X^i, X^j \right\rangle_s,$$

by $dX_t = \frac{\partial}{\partial x^i}(X_t) \circ dX_t^i$.

(2) For $f \in C^\infty(M)$, $\nabla^2 f \in \Gamma(\otimes^{(0,2)}TM)$ and so

$$\begin{aligned} \int_0^t \nabla^2 f(dX_s, dX_s) &= \int_0^t \nabla^2 f_{X_s}(U_s e_i, U_s e_j) d\left\langle W^i, W^j \right\rangle_s \\ &= \int_0^t H_j H_i \widetilde{f}(U_s) d\left\langle W^i, W^j \right\rangle_s. \end{aligned}$$

(3) If $f, g \in C^\infty(M)$,

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \langle f(X), g(X) \rangle_t.$$

*Proof.* For $X$ an $M$-valued semimartingale, WTLG assume $dX_t = V_\alpha(X_t) \circ dZ_t^\alpha$. Then

$$\int_0^t (df \otimes dg)(dX_s, dX_s) = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

On the other hand, because

$$df(X_t) = V_\alpha f(X_t) \circ dZ_t^\alpha, \quad dg(X_t) = V_\beta g(X_t) \circ dZ_t^\beta,$$

we have

$$\langle f(X), g(X) \rangle_t = \int_0^t V_\alpha f(X_s) V_\beta f(X_s) d\left\langle Z^\alpha, Z^\beta \right\rangle_s.$$

□

**Proposition 2.4.13.** *If $\theta \in \Gamma(\otimes^{(0,2)}TM)$ is positive semi-definite, then the $\theta$-quadratic variation of a semimartingale $X$ is nondecreasing.*

*Proof.* First,

$$\int_0^t h\left(dX_s, dX_s\right) = \int_0^t h\left(U_s e_i, U_s e_j\right) d\left\langle W^i, W^j\right\rangle_s .$$

Because $h$ is positive semi-positive, the matrix $(h\left(U_s e_i, U_s e_j\right))$ is positive semi-definite and let $(m_i^k(s))$ be its squared root matrix. Let

$$J_t^k = \int_0^t m_i^k(s) dW_s^i.$$

Therefore,

$$\int_0^t h\left(dX_s, dX_s\right) = \sum_k \int_0^t m_i^k(s) m_j^k(s) d\left\langle W^i, W^j\right\rangle_s = \sum_k \left\langle J^k, J^k\right\rangle_t . \qquad \square$$

## 2.5 Martingale on Manifold

**Definition 2.5.1.** Suppose $M$ is a $C^\infty$ manifold with a connection $\nabla$. An $M$-valued semimartingale $X$ is called a $\nabla$-martingale if its anti-development $W$ w.s.t $\nabla$ is an $\mathbb{R}^d$-valued local martingale.

*Remark* 2.5.2. It is well defined because if $W$ and $W'$ are two anti-development of $X$, then there exists a $g \in GL(d, \mathbb{R})$ such that $W' = gW$.

*Remark* 2.5.3. Note that when $M = \mathbb{R}^N$ is equipped with the canonical connection. Then for any $c(t)$ on $M$, its horizontal lift starting from $u_0$ is

$$u(t)a = (u_0)^i_j a^j \left.\frac{\partial}{\partial x^i}\right|_{c(t)} .$$

When viewing $T_x M \subset \mathbb{R}^N$, $u(t)a \equiv u_0 a$. Define $f\colon F(M) \to \mathbb{R}^N$ as $f(u) = ue_i$. Then

$$P_\alpha^H(u) f = \left.\frac{d}{dt}\right|_{t=0} f(u_t) = \left.\frac{d}{dt}\right|_{t=0} u_t e_i = 0.$$

Let $X$ be an $M$-valued semimartingale. Because $dU_t = P_\alpha^H(U_t) \circ dX_t^\alpha$,

$$df(U_t) = P_\alpha^H f(U_t) \circ dX_t^\alpha = 0 \ \Rightarrow \ f(U_t) = U_t e_i = U_0 e_i.$$

Therefore,

$$\begin{aligned} X_t &= X_0 + \int_0^t U_s e_i \circ dW_s^i \\ &= X_0 + \int_0^t U_0 e_i dW_s^i \\ &= X_0 + U_0 W_t. \end{aligned}$$

It follows that when $M = \mathbb{R}^N$, if $X$ is an $M$-martingale, then it is a local martingale in the usual sense.

**Proposition 2.5.4.** *An $M$-valued semimartingale $X$ is a $\nabla$-martingale if an only if*

$$N^f(X)_t := f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \nabla^2 f(dX_s, dX_s),$$

*is an $\mathbb{R}$-valued local martingale for every $f \in C^\infty(M)$.*

*Remark* 2.5.5. For Euclidean case, a continuous semimartingale $X$ is a local martingale if and only if $N^f(X)$ is a local martingale for all $f \in C^\infty$ (or $f \in C_c^\infty$) by Itô formula.

*Proof.* $\Rightarrow$: Let $U$ be a horizontal lift of $X$ and $W$ be its anti-development. Then

$$dU_t = H_i(U_t) \circ dW_t^i.$$

For $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi \in C^\infty(F(M))$. So

$$\begin{aligned} f(X_t) - f(X_0) &= \widetilde{f}(U_t) - \widetilde{f}(U_0) \\ &= \int_0^t H_i\widetilde{f}(U_s) \circ dW_s^i \\ &= \int_0^t H_i\widetilde{f}(U_s) dW_s^i + \frac{1}{2}\left\langle H_i\widetilde{f}(U), W^i \right\rangle_t. \end{aligned}$$

Note that

$$dH_i\widetilde{f}(U_t) = H_jH_i\widetilde{f}(U_t) \circ dW_t^j,$$

which implies that

$$\begin{aligned} \left\langle H_i\widetilde{f}(U), W^i \right\rangle_t &= \int_0^t H_jH_i\widetilde{f}(U_s) d\left\langle W^i, W^j \right\rangle_s \\ &= \int_0^t \nabla^2 f(dX_s, dX_s). \end{aligned}$$

Therefore,

$$N^f(X)_t = \int_0^t H_i\widetilde{f}(U_t) dW_t^i.$$

When $X$ is an $M$-local semimartingale, i.e., $W$ is an $\mathbb{R}^d$-local martingale, then $N^f(X)$ is obviously a local martingale.
$\Leftarrow$: Assume $M \subset \mathbb{R}^N$ properly embedded. Let $f\colon M \to \mathbb{R}^N$ be the coordinate function $f(x) = x$ and $\widetilde{f} = f \circ \pi$ be its lift on $F(M)$. When viewing $T_xM \subset \mathbb{R}^N$, we already have $H_i\widetilde{f}(u) = ue_i$. So similarly as above calculation,

$$\begin{aligned} N^f(X)_t &:= N^{f^\alpha}(X)_t e_\alpha \\ &= e_\alpha \int_0^t H_i\widetilde{f^\alpha}(U_s) dW_s^i \\ &= e_\alpha \int_0^t (U_s e_i)^\alpha dW_t^i \\ &= \int_0^t U_s e_i dW_t^i = \int_0^t U_s dW_s, \end{aligned}$$

by viewing $U_s\colon \mathbb{R}^d \to \mathbb{R}^N$ where $U_s\colon \mathbb{R}^d \to T_xM$ is an isomorphism. Define $V_s\colon \mathbb{R}^N = T_xM \oplus N_xM \to \mathbb{R}^d$ as

$$V_s\xi = \begin{cases} U_s^{-1}\xi, & \xi \in T_{X_s}M, \\ 0, & \xi \perp T_{X_s}M \end{cases}$$

Then $V_sU_se = e$ for all $\mathbb{R}^d$. Therefore,

$$\int_0^t V_s dN^f(X)_s = \int_0^t V_sU_s dW_s = W_t.$$

Because $N^f(X)$ is a local martingale, $W_t$ is a local martingale. It follows that $X$ is an $M$-martingale. $\square$

*Remark* 2.5.6. From the proof, we can see that when viewing $M \subset \mathbb{R}^N$ and let $f = (f^1, \cdots, f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function, if $N^{f^\alpha}$ for $\alpha = 1, \cdots, N$ are local martingales, then $X$ is an $M$-local martingale.

**Proposition 2.5.7.** *Suppose $(U, x^1, \cdots, x^d)$ is a local chart on $M$ and $X = (X^i)$, i.e. $X^i = x^i(X)$, a semimartingale on $M$. Then $X$ is an $M$-martingale if and only if*

$$N_t^i = X_t^i - X_0^i + \frac{1}{2}\int_0^t \Gamma_{jk}^i(X_s) d\left\langle X^j, X^k\right\rangle_s$$

*is a local martingale.*

*Proof.* $\Rightarrow$: Locally, we know

$$\nabla^2 x^i = -\Gamma_{jk}^i dx^j \otimes dx^k,$$

which implies that

$$\int_0^t \nabla^2 x^i(dX_s, dX_s) = -\int_0^t \Gamma_{jk}^i(X_s) d\left\langle X^j, X^k\right\rangle_s.$$

Therefore, this direction can be directly obtained by applying above theorem to each $f = x^i$.
$\Leftarrow$: Choose a local coordinate for $F(M)$ as $(\tilde{x}^i, e_j^i)$, where

$$\tilde{x}^i(u) = x^i(\pi(u)), \quad e_j^i(u) \text{ for } ue_j = e_j^i(u) \left.\frac{\partial}{\partial x^j}\right|_{\pi(u)}.$$

Let $U$ be a horizontal lift with the anti-development $W$ of $X$. Then

$$dU_t = H_j(U_t) \circ dW_t^j.$$

Because $X_t = \pi(U_t)$ and $U_te_j = e_j^i(t) \left.\frac{\partial}{\partial x^j}\right|_{X_t}$, where $e_j^i(t) = e_j^i(U_t)$, we have

$$\begin{aligned} dX_t^i = d\tilde{x}^i(U_t) &= H_j\tilde{x}^i(U_t) \circ dW_t^j \\ &= (U_te_j)(x^i) \circ dW_t^j \\ &= e_j^i(t) \circ dW_t^j. \end{aligned}$$

Moreover, we also know

$$dU_t = P_k^H(U_t) \circ dX_t^k.$$

Note that in the local chart, $M$ can be viewed as $\mathbb{R}^d$ and so $P(x)\colon \mathbb{R}^d \to T_xM$ the orthogonal projection is the identity, which implies that

$$P_k(X_t) = P(X_t)e_k = e_k = \delta_k^\ell \left.\frac{\partial}{\partial x^\ell}\right|_{X_t}.$$

Let $v(s)$ be the horizontal lift starting from $U_t$ of a curve $c(s)$ with $c(0) = U_t$ and $\dot{c}(0) = P_k(X_t) = \delta_k^m \frac{\partial}{\partial x^m}\big|_{X_t}$. Then by using the coordinates, we have

$$\dot{v}(0) = P_k^H(U_t) = -\Gamma^i_{m\ell}(X_t)\delta_k^m e_j^\ell(t) \left.\frac{\partial}{\partial e_j^i}\right|_{U_t} + \delta_k^m \left.\frac{\partial}{\partial \tilde{x}^m}\right|_{U_t}.$$

So

$$P_k^H e_j^i(U_t) = -\Gamma^i_{k\ell}(X_t)e_j^\ell(t).$$

It follows that

$$de_j^i(t) = -\Gamma^i_{k\ell}(X_t)e_j^\ell(t) \circ dX_t^k.$$

Let $(f_j^i) = (e_j^j)^{-1}$. Then we get

$$dW_t^j = f_k^j(t) \circ dX_t^k.$$

So

$$\begin{aligned}
dX_t^i &= e_j^i(t)dW_t^i + \frac{1}{2}d\left\langle e_j^i, W^j\right\rangle_t \\
&= e_j^i(t)dW_t^i - \frac{1}{2}\Gamma^i_{k\ell}\left(X_t\right)e_j^\ell(t)f_m^j(t)d\left\langle X^k, X^m\right\rangle_t \\
&= e_j^i(t)dW_t^i - \frac{1}{2}\Gamma^i_{k\ell}\left(X_t\right)d\left\langle X^k, X^\ell\right\rangle,
\end{aligned}$$

which implies that

$$dN_t^i = e_j^i(t)dW_t^i \;\Rightarrow\; dW_t^i = f_j^i(t)dN_t^i.$$

Therefore, if $N_t^i$ are all local martingales, so is $W_t$ and $X$ is an $M$-valued martingale. $\square$

## 2.6 Martingales on Submanifolds

Let $M, N$ be two manifolds equipped with connection $\nabla^M, \nabla^N$ respectively, and $M \subset N$ be a submanifold. Consider the second fundamental form $\mathrm{II}\colon \Gamma(TM) \times \Gamma(TM) \to \Gamma(TN|_M)$ defined as

$$\mathrm{II}(V, W) = \nabla_V^N W - \nabla_V^M W,$$

which is obviously bilinear over $C^\infty(M)$, so II is a $TN|_M$-valued $(0, 2)$-tensor field on $M$.

*Remark* 2.6.1. A direct result of this is, $\mathrm{II}(V, W)(p)$ is fully determined by $V_p$ and $W_p$

Moreover, for a connection $\nabla$, let

$$T(V, W) = \nabla_V W - \nabla_W V - [V, W].$$

We have

$$\mathrm{II}(V, W) - \mathrm{II}(W, V) = T^N(V, W) - T^M(V, W),$$

so II is symmetric if $\nabla^M, \nabla^N$ are torsion-free. If $\mathrm{II} \equiv 0$, then $M$ is called totally geodesic.

**Proposition 2.6.2.** *If $M$ is totally geodesic, then any $\nabla^M$-geodesic in $M$ is also a $\nabla^N$-geodesic in $N$. When $\nabla^M, \nabla^N$ are torsion-free, the converse is also true.*

**Lemma 2.6.3.** *Let $M \subset \mathbb{R}^N$ be a submanifold and $f\colon M \to \mathbb{R}^N$ be the coordinate function on $M$ with lift $\widetilde{f} = f \circ \pi$ to the frame bundle $F(M)$. Then*

$$H_i H_j \widetilde{f}(u) = \mathrm{II}(ue_i, ue_j)$$

*Proof.* Let $u_t$ be the horizontal lift starting from $u_0 = u$ of a curve $x_t$ with $x_0 = \pi(u)$ and $\dot{x}_0 = ue_i$, i.e., $\dot{u}(0) = H_i(u)$, and $V(x_t) = u_t e_j$ is a $\nabla^M$-parallel along $x_t$. So

$$\nabla^M_{\dot{x}_t} V = 0 \ \Rightarrow \ \nabla^M_{ue_i} V = 0.$$

It implies that

$$\mathrm{II}(ue_i, ue_j) = \nabla^{\mathbb{R}^N}_{ue_i} V.$$

On the other hand, recall when viewing $T_xM \subset \mathbb{R}^N$,

$$H_j \widetilde{f}(u_t) = u_t e_j = V(x_t).$$

Therefore,

$$H_i H_j \widetilde{f}(u) = \left.\frac{d}{dt}\right|_{t=0} V(x_t) = \nabla^{\mathbb{R}^N}_{ue_i} V = \mathrm{II}(ue_i, ue_j)$$

for viewing $V(x_t) = (V^1(x_t), \cdots, V^N(x_t)) \in \mathbb{R}^N$. □

**Theorem 2.6.4.** *Using notations as above and assume* $\dim M = l$ *and* $\dim N = d$. *Suppose that* $X$ *is a semimartingale on* $M \subset N$. *Let* $U^M$ *be a horizontal lift of* $X$ *in* $F(M)$ *with the corresponding* $\mathbb{R}^d$*-valued anti-development* $W^M$, *and* $U^N$ *be a horizontal lift of* $X$ *in* $F(N)$ *with the corresponding* $\mathbb{R}^l$*-valued anti-development* $W^N$. *Then*

$$W^N_t = \int_0^t (U^N_s)^{-1} U^M_s dW^M_s + \frac{1}{2} \int_0^t (U^N_s)^{-1} \mathrm{II}(dX_s, dX_s).$$

*Proof.* First,

$$\mathrm{II}(dX_s, dX_s) = \mathrm{II}(U^M_s e_i, U^M_s e_j) d \left\langle (W^M)^i, (W^M)^j \right\rangle_s.$$

So we need to show

$$W^N_t = \int_0^t (U^N_s)^{-1} U^M_s e_i d(W^M)_s + \int_0^t (U^N_s)^{-1} \mathrm{II}(U^M_s e_i, U^M_s e_j) d \left\langle (W^M)^i, (W^M)^j \right\rangle_s.$$

First, assume $N = \mathbb{R}^l$. WTLG, we can choose $U^N_0 = I$. In such case, we have

$$W^N = X - X_0, \quad U^N \equiv I.$$

For simplicity, denote $U = U^M$ and $W = W^M$. Let $f \colon M \to \mathbb{R}^l$ be the coordinate map and $\widetilde{f} = f \circ \pi$. Then we have been shown

$$X_t = X_0 + \int_0^t H_i \widetilde{f}(U_s) dW^i_s + \frac{1}{2} \int_0^t H_i H_j \widetilde{f}(U_s) d \left\langle W^i, W^j \right\rangle_s.$$

Note that

$$H_i \widetilde{f}(u) = ue_i, \quad H_i H_j \widetilde{f}(u) = \mathrm{II}(ue_i, ue_j).$$

Therefore,

$$W^N_t = X_t - X_0 = \int_0^t U_s dW_s + \frac{1}{2} \int_0^t \mathrm{II}\left(dX_s, dX_s\right).$$

Next, for general $N$, assume $N \subset \mathbb{R}^N$ embedded manifold. Let $\mathrm{II}^M$ and $\mathrm{II}^N$ be the second fundamental forms of $M$ and $N$ in $\mathbb{R}^N$ respectively. Then

$$X_t = X_0 + \int_0^t U^M_s dW^M_s + \frac{1}{2} \int_0^t \mathrm{II}^M\left(dX_s, dX_s\right)$$

$$X_t = X_0 + \int_0^t U_s^N dW_s^N + \frac{1}{2}\int_0^t \mathrm{II}^N\left(dX_s, dX_s\right),$$

where the two equations are viewed in $\mathbb{R}^N$, $T_xM = \mathbb{R}^l$ and $T_xN = \mathbb{R}^d$. Note that

$$\mathrm{II}(X,Y) = \mathrm{II}^M(X,Y) - \mathrm{II}^N(X,Y).$$

By above, $dX_t = U_s^N dW_s^N + \frac{1}{2}\mathrm{II}^N\left(dX_s, dX_s\right)$,

$$\begin{aligned} W_t^N &= \int_0^t (U_s^N)^{-1} dX_t - \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^N\left(dX_s, dX_s\right) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^M\left(dX_s, dX_s\right) \\ &\quad - \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}^N\left(dX_s, dX_s\right) \\ &= \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \frac{1}{2}\int_0^t (U_s^N)^{-1}\mathrm{II}(dX_s, dX_s). \end{aligned}$$

□

**Corollary 2.6.5.** *Suppose that $M \subset N$ is a totally geodesic submanifold of $N$. Then every $\nabla^M$-martingale on $M$ is also a $\nabla^N$-martingale on $N$.*

For $x \in M$, define

$$\operatorname{ran}\mathrm{II}_x := \{\mathrm{II}(X,Y)(x) \colon X, Y \in \Gamma(TM)\} \subset T_xN.$$

II is called independent of $TM$ if $\operatorname{ran}\mathrm{II}_x \cap T_xM = \{0\}$ for every $x \in M$. Note that for Riemannian submanifold, II is obviously independent of $TM$.

**Corollary 2.6.6.** *Suppose that $M \subset N$ such that* II *is independent of $TM$. Then a $\nabla^N$-martingale on $N$ which lives on $M$ is also a $\nabla^M$-martingale on $M$.*

*Proof.* First, $W^M$ is clearly a continuous semi-martingale, so it can be uniquely decomposed into

$$W_t = M_t + A_t,$$

where $M$ is a local martingale and $A$ is a process of bounded variation with $A_0 = 0$. It suffices to prove $A = 0$. By above,

$$W_t^N = \int_0^t (U_s^N)^{-1} U_s^M dW_s^M + \int_0^t (U_s^N)^{-1}\left(U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s)\right).$$

Because $W^N$ is a local martingale,

$$\int_0^t (U_s^N)^{-1}\left(U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s)\right) = 0$$

and so

$$U_s^M dA_s + \frac{1}{2}\mathrm{II}(dX_s, dX_s) = 0.$$

However,

$$U_s^M dA_s \in T_{\pi(U_s^M)}M, \quad \mathrm{II}(dX_s, dX_s) \in \operatorname{ran}\mathrm{II}_{\pi(U_s^M)}.$$

The independence implies that $dA_s = 0$, and so $A = 0$. □

**Example 2.6.7.** (1) Let $M \subset N$ be a Riemannian hypersurface. It is called (strictly) convex if the scalar fundamental form is (strictly) positive definite at each point. Assume $M$ is strictly convex. Because II is independent of $TM$, any $\nabla^N$-martingale on $N$ which lies on $M$ is also a $\nabla^M$-martingale. It implies that

$$\mathrm{II}(dX_s, dX_s) = 0.$$

However, because of the strict positivity, $dX_s = 0$, and so $X$ is constant.

(2) Let $M$ be a Riemannian manifold of dimension $\mathbb{R}^d$. A semimartingale $X$ on $M$ is called a Brownian motion if its anti-development $W$ is a $d$-dimensional standard Brownian motion. Let $M \subset \mathbb{R}^N$ be a minimal submanifold, i.e., the mean curvature field $H = \mathrm{tr}_g \mathrm{II} = 0$. Then a Brownian motion on $M$ is a local martingale on $\mathbb{R}^N$, because

$$\mathrm{II}\left(U_t e_i, U_t e_j\right) d\left\langle W^i, W^j\right\rangle_t = \sum_i \mathrm{II}\left(U_t e_i, U_t e_i\right) dt = H\left(X_t\right) dt = 0.$$

Note that for Riemannian manifold, we consider $O(M)$, and so $\{U_t e_i\}$ is orthonormal in $T_{\pi(U_t)}M$.

**Proposition 2.6.8.** *Suppose that $M \subset N$ be totally geodesic. For any $x \in M$, there exists a neighborhood $O$ of $p$ in $N$ such that if $X$ is an $\nabla^N$-martingale in $O$ on $[0,T]$ with $X_T \in M$, then $X_t \in M$ for all $t \in [0,T]$.*

**Theorem 2.6.9.** *Let $M$ be a manifold with a connection. For any $x \in M$, there exists a neighborhood $O$ of $x$ such that: if $(X_t)_{t\in[0,T]}$ and $(Y_t)_{t\in[0,T]}$ are two martingales on $V$ such that $X_T = Y_T$, then $X_t = Y_t$ for all $t \in [0,T]$.*

# Chapter 3

# Brownian Motion on Manifolds

## 3.1 Laplace-Beltrami operator

Let $(M, g)$ be a Riemannian manifold equipped with the Levi-Civita connection $\nabla$ and the standard Riemannian volume measure $m$. For $f, g \in C_c(M)$,

$$(f, g) := \int_M f(x)g(x)m(dx)$$

and for $X, Y \in \Gamma_c(TM)$, i.e., vector fields with compact support,

$$(X, Y) = \int_M \langle X, Y \rangle_x m(dx),$$

and for $\theta, \psi \in \Gamma_c(T^*M)$, i,e, 1-forms with compact support,

$$(\theta, \psi) = \int_M \langle \theta, \psi \rangle_x m(dx).$$

Let $\Delta_M$ be the Laplace-Beltrami operator. Because locally

$$\Delta_M f = \mathrm{tr}_g(\nabla^2 f) = g^{ij} \frac{\partial^2}{\partial x^i \partial x^j} f - g^{k\ell} \Gamma^i_{k\ell} \frac{\partial}{\partial x^i} f,$$

$\Delta_M$ is a nondegenerate second order elliptic operator. Moreover, by the local expression, for $L = \frac{1}{2}\Delta_M$, the corresponding

$$\Gamma(f, g) = g^{ij} \frac{\partial f}{\partial x^i} \frac{\partial g}{\partial x^j} = \langle \nabla f, \nabla g \rangle.$$

Let $\pi \colon O(M) \to M$ be the orthogonal frame bundle. Define a second order differential operator on $O(M)$ as

$$\Delta_{O(M)} := \sum_{i=1}^{d} H_i^2,$$

called Bochner's horizontal Laplacian.

**Proposition 3.1.1.** *For any $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi \in C^\infty(O(M))$. Then for any $u \in O(M)$,*

$$\Delta_M f(x) = \Delta_{O(M)} \widetilde{f}(u),$$

*where $x = \pi(u)$.*

*Proof.* Fix a $f \in C^\infty(M)$. Note that for any 1-form, when viewing $(\mathbb{R}^d)^* = \mathbb{R}^d$, the scalarization $\widetilde{\theta}\colon O(M) \to \mathbb{R}^d$ satisfies

$$\widetilde{\theta}(u) = (\theta(ue_1), \cdots, \theta(ue_d))^\top \in \mathbb{R}^d.$$

Therefore,

$$(\widetilde{df})_i = df(ue_i) = ue_i(f) = H_i \widetilde{f}(u),$$

and

$$\widetilde{df} = \left(H_1 \widetilde{f}, H_2 \widetilde{f}, \cdots, H_d \widetilde{f}\right)^\top.$$

Because $(ue_i)$ is a normal basis in $T_{\pi(u)}M$, we have

$$\sharp(ue_i)^* = ue_i.$$

It follows that, first

$$\widetilde{\operatorname{grad} f} = \left(H_1 \widetilde{f}, H_2 \widetilde{f}, \cdots, H_d \widetilde{f}\right)^\top.$$

because $\operatorname{grad} f((ue_i)^*) = \langle \operatorname{grad} f, \sharp(ue_i)^* \rangle = \langle \operatorname{grad} f, ue_i \rangle = df(ue_i)$. Second, for any $X \in \Gamma(TM)$,

$$\operatorname{div} X(\pi(u)) = \sum_i \nabla X((ue_i)^*, ue_i) = \sum_i \langle \nabla_{ue_i} X, \sharp(ue_i)^* \rangle = \sum_i \langle \nabla_{ue_i} X, ue_i \rangle.$$

Moreover, since $u \in O(M)$, orthogonal map,

$$\begin{aligned}
\operatorname{div} X\,(\pi(u)) &= \sum_i \left\langle u^{-1} \nabla_{ue_i} X(\pi(u)), e_i \right\rangle \\
&= \sum_i \left\langle \widetilde{\nabla_{ue_i} X}(u), e_i \right\rangle \\
&= \sum_i \left\langle H_i \widetilde{X}(u), e_i \right\rangle = \sum_i \left(H_i \widetilde{X}\right)^i (u).
\end{aligned}$$

It follows that

$$\Delta_M f(\pi(u)) = \operatorname{div}(\operatorname{grad} f)(\pi(u)) = \sum_i \left(H_i (\widetilde{\operatorname{grad} f})\right)^i (u) = \sum_i H_i(H_i \widetilde{f}) = \Delta_{O(M)} \widetilde{f}(u). \quad \square$$

**Theorem 3.1.2** (Nash's Embedding Theorem)**.** *Every Riemannian manifold can be isometrically embedded in some Euclidean space with the standard metric.*

*Remark* 3.1.3. Note that if the Riemannian manifold is complete, then we can find an Euclidean space such that the image is closed, i.e., a complete Riemannian manifold can be viewed as a Riemannian submanifold of an Euclidean space, which is also a closed subset.

In the following, we assume a $M \subset \mathbb{R}^N$ be a Riemannian submanifold that is closed. So all above theories can be applied to such $M$. Similarly as above, let $P(x)\colon \mathbb{R}^N \to T_xM$ be the orthogonal projection and $P_\alpha = Pe_\alpha \in \Gamma(TM)$. Note that if $X \in T_xM$, we have

$$\begin{aligned}
X = P(x)X = P(x) \sum_{\alpha=1}^N \langle X, e_\alpha \rangle_{\mathbb{R}^N} e_\alpha &= \sum_{\alpha=1}^N \langle X, e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x) \\
= \sum_{\alpha=1}^N \langle P(x)X, e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x) &= \sum_{\alpha=1}^N \langle X, P(x)e_\alpha \rangle_{\mathbb{R}^N} P_\alpha(x)
\end{aligned}$$

$$= \sum_{\alpha=1}^{N} \langle X, P_\alpha(x) \rangle_{T_xM} P_\alpha(x).$$

Let $\widetilde{\nabla}$ be the standard Euclidean connection. The induced Levi-Civita connection on $M$ is

$$\nabla_X Y = (\widetilde{\nabla}_X Y)^\top = P(\widetilde{\nabla}_X Y), \quad X, Y \in \Gamma(TM).$$

Moreover, for any $X \in \Gamma(TM)$ and any $f \in C^\infty(M)$,

$$X(f) = \langle \operatorname{grad} f, X \rangle_{TM} = \left\langle \overline{\operatorname{grad} f}, X \right\rangle_{\mathbb{R}^N}, \ \Rightarrow \ \operatorname{grad} f = P\overline{\operatorname{grad} f},$$

where $\overline{\operatorname{grad} f}$ is the usual gradient on $\mathbb{R}^N$.

**Theorem 3.1.4.** *Using notations as above,*

$$\Delta_M = \sum_{\alpha=1}^{N} P_\alpha^2.$$

*Proof.* Let $f \in C^\infty(M)$.

$$\operatorname{grad} f = \sum_{\alpha=1}^{N} \langle \operatorname{grad} f, P_\alpha \rangle P_\alpha = \sum_{\alpha=1}^{N} P_\alpha(f) P_\alpha.$$

Note that for any $X \in \Gamma(TM)$ and $h \in C^\infty(M)$, it is not difficult such that

$$\operatorname{div}(hX) = X(h) + h \operatorname{div}(X).$$

It follows that

$$\Delta_M f = \sum_{\alpha=1}^{N} P_\alpha^2(f) + \sum_{\alpha=1}^{N} P_\alpha(f) \operatorname{div}(P_\alpha).$$

For the second term, fix $x \in M$. First,

$$\operatorname{div}(P_\alpha)(x) = \sum_{i=1}^{d} \left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \left.\frac{\partial}{\partial x^i}\right|_x \right\rangle,$$

where $(x^1, \cdots, x^d)$ is a normal coordinate at $x$, because $\left\{ \frac{\partial}{\partial x^i}\big|_x \right\}$ is an orthonormal basis of $T_xM$. Moreover, because $\Gamma_{ij}^k(x) = 0$,

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.$$

So

$$\begin{aligned}
\left\langle \nabla_{\frac{\partial}{\partial x^i}} P_\alpha(x), \left.\frac{\partial}{\partial x^i}\right|_x \right\rangle_{T_xM} &= \frac{\partial}{\partial x^i} \left\langle P_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) = \frac{\partial}{\partial x^i} \left\langle Pe_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{TM} (x) \\
&= \frac{\partial}{\partial x^i} \left\langle e_\alpha, P\frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) = \frac{\partial}{\partial x^i} \left\langle e_\alpha, \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} (x) \\
&= \left\langle \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} e_\alpha(x), \frac{\partial}{\partial x^i} \right\rangle_{\mathbb{R}^N} + \left\langle e_\alpha(x), \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} \\
&= \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N}.
\end{aligned}$$

It follows that

$$\begin{aligned}
\sum_{\alpha=1}^{N} \operatorname{div}(P_\alpha)(x) P_\alpha(x) &= \sum_{\alpha=1}^{N}\sum_{i=1}^{d} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P_\alpha(x) \\
&= \sum_{\alpha=1}^{N}\sum_{i=1}^{d} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} P(x) e_\alpha \\
&= \sum_{i=1}^{d} P(x) \sum_{\alpha=1}^{N} \left\langle e_\alpha, \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) \right\rangle_{\mathbb{R}^N} e_\alpha \\
&= \sum_{i=1}^{d} P(x) \widetilde{\nabla}_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = \sum_{i=1}^{d} \nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^i}(x) = 0.
\end{aligned}$$

Therefore, $\sum_{\alpha=1}^{N} P_\alpha(f) \operatorname{div}(P_\alpha) = 0$ for any $f \in C^\infty(M)$. Then

$$\Delta_M f = \sum_\alpha P_\alpha^2(f).$$

□

**Proposition 3.1.5.** *Using same notations as above,*

$$\Delta_M f = \sum_\alpha \nabla^2 f(P_\alpha, P_\alpha).$$

*Proof.* Fix $x \in M$. Let $\{X_i\}$ be an orthonormal basis of $T_x M$ and consider decomposition

$$X_i = \sum_{\alpha=1}^{N} \langle X_i, P_\alpha(x) \rangle P_\alpha(x).$$

Then

$$\begin{aligned}
\Delta_M f(x) &= \sum_{i=1}^{d} \nabla^2 f(x)(X_i, X_i) \\
&= \sum_{i=1}^{d} \sum_{\alpha,\beta=1}^{N} \nabla^2 f(P_\alpha, P_\beta)(x) \langle X_i, P_\alpha(x) \rangle \langle X_i, P_\beta(x) \rangle \\
&= \sum_{\alpha=1}^{N} \nabla^2 f \left( P_\alpha, \sum_{i=1}^{d}\sum_{\beta=1}^{N} \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta \right)(x).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\sum_{i=1}^{d}\sum_{\beta=1}^{N} \langle X_i, P_\alpha \rangle \langle X_i, P_\beta \rangle P_\beta &= \sum_{\beta=1}^{N} \left\langle \sum_{i=1}^{d} \langle X_i, P_\beta \rangle X_i, P_\alpha \right\rangle P_\beta \\
&= \sum_{\beta=1}^{N} \langle P_\beta, P_\alpha \rangle P_\beta = P P_\alpha = P_\alpha.
\end{aligned}$$

□

## 3.2 Brownian Motion on Manifolds

**Theorem 3.2.1.** *Let $X : \Omega \to W(M)$ be a measurable map defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\mu = (X_0)_\# \mathbb{P}$ be the initial distribution on $M$. TFAE.*

*(1) $X$ is a $\frac{1}{2}\Delta_M$-diffusion process w.s.t. $\mathbb{F}^X$, i.e., $X$ is a $\mathbb{F}^X$-semimartingale, and*

$$M^f(X)_t = f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \Delta_M f(X_s)ds, \quad 0 \leq t < e(X),$$

*is an $\mathbb{F}^X$-local martingale for all $f \in C^\infty(M)$.*

*(2) $X$ is a $\mathbb{F}^X$-semimartingale on $M$ whose anti-development is a standard Euclidean $\mathbb{F}^X$-Brownian motion.*

*Remark* 3.2.2. An $M$-valued process $X$ satisfies above condition is called a Riemannian Brownian motion on $M$. Moreover, $\mathbb{P}_\mu = X_\# \mathbb{P}$ is called a Wiener measure on $W(M)$. Note that a Wiener measure is uniquely determined by $\Delta_M/2$ and an initial measure $\mu$.

*Proof.* $\Leftarrow$: Let $U$ be a horizontal lift of $X$ and $W$ is its anti-development. Then we have

$$dU_t = H_i(U_t) \circ dW_t.$$

For any $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi$ be its lift on $O(M)$. Then by above mention,

$$f(X_t) = f(X_0) + \int_0^t H_i \widetilde{f}(U_s)\, dW_s^i + \frac{1}{2}\int_0^t H_i H_j \widetilde{f}(U_s)\, d\left\langle W^i, W^j \right\rangle_s.$$

Because $W$ is a Brownian motion, $\langle W^i, W^j \rangle_t = \delta^{ij} t$. Then

$$\begin{aligned}
\int_0^t H_i H_j \widetilde{f}(U_s)\, d\left\langle W^i, W^j \right\rangle_s &= \int_0^t \sum_i H_i^2 \widetilde{f}(U_s) ds \\
&= \int_0^t \Delta_{O(M)} \widetilde{f}(U_s) ds \\
&= \int_0^t \Delta_M f(X_s) ds.
\end{aligned}$$

Therefore,

$$M^f(X)_t = \int_0^t H_i \widetilde{f}(U_s) dW_s^i,$$

which is clearly a $\mathbb{F}^X$-local martingale.
$\Rightarrow$: Let $f = (f^\alpha)\colon M \to \mathbb{R}^N$ be the coordinate function $f(x) = x$. Let $\widehat{f^\alpha} = f^\alpha \circ \pi$ be the lift on $O(M)$. By assumption, for $X_t^\alpha = f^\alpha(X)_t$,

$$X_t^\alpha = X_0^\alpha + M_t^\alpha + \frac{1}{2}\int_0^t \Delta_M f^\alpha(X_s) ds,$$

where $M_t^\alpha$ is a local martingale, $\alpha = 1, 2, \cdots, N$. On the other hand, let $H$ be the horizontal lift of $X$ and $W$ be its anti-development. Then

$$X_t^\alpha = X_0^\alpha + \int_0^t H_i \widetilde{f^\alpha}(U_s) dW_s^i + \frac{1}{2}\int_0^t \nabla^2 f^\alpha(dX_s, dX_s).$$

**Claim:** For any $f \in C^\infty(M)$, we can have

$$\int_0^t \nabla^2 f(dX_s, dX_s) = \int_0^t \Delta_M f(X_s) ds.$$

First, we know $dX_t = P_\alpha(X_t) \circ dX_t^\alpha$. Then we get

$$\int_0^t \nabla^2 f\left(dX_s, dX_s\right) = \int_0^t \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s) d\left\langle X^\alpha, X^\beta\right\rangle_s.$$

By the assumption, because $X$ is a $\Delta_M/2$-diffusion process,

$$d\left\langle X^\alpha, X^\beta\right\rangle_s = d\left\langle M^\alpha, M^\beta\right\rangle_s = \Gamma(f^\alpha, f^\beta)(X_s)ds,$$

where

$$\Gamma(f^\alpha, f^\beta) = \left\langle \nabla f^\alpha, \nabla f^\beta\right\rangle.$$

Because

$$\operatorname{grad} f^\alpha = P(\overline{\operatorname{grad}} f^\alpha) = Pe_\alpha = P_\alpha,$$

we have

$$\Gamma(f^\alpha, f^\beta) = \left\langle \operatorname{grad} f^\alpha, \operatorname{grad} f^\beta\right\rangle = \left\langle P_\alpha, P_\beta\right\rangle.$$

Moreover,

$$\sum_{\alpha,\beta} \nabla^2 f(P_\alpha, P_\beta)\left\langle P_\alpha, P_\beta\right\rangle = \sum_\alpha \nabla^2 f(P_\alpha, \sum_\beta \left\langle P_\alpha, P_\beta\right\rangle P_\beta) = \sum_\alpha \nabla^2 f(P_\alpha, P_\alpha) = \Delta_M f$$

Then

$$\begin{aligned}
\int_0^t \nabla^2 f\left(dX_s, dX_s\right) &= \int_0^t \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s)\Gamma(f^\alpha, f^\beta)(X_s)ds \\
&= \int_0^t \sum_{\alpha,\beta} \nabla^2 f\left(P_\alpha, P_\beta\right)(X_s)\left\langle P_\alpha, P_\beta\right\rangle ds \\
&= \int_0^t \Delta_M f(X_s)ds.
\end{aligned}$$

So the claim has been proved.
Then we have

$$M_t^\alpha = \int_0^t H_i\widetilde{f^\alpha}(U_s)dW_s^i.$$

Next, we will use the Lévy's theorem to check $W$ is a Brownian motion. First, because $\widetilde{f^\alpha} = f^\alpha \circ \pi$,

$$\begin{aligned}
H_i\widetilde{f^\alpha}(u) &= (ue_i)(f^\alpha) \\
&= \left\langle ue_i, \operatorname{grad} f^\alpha\right\rangle_{TM} \\
&= \left\langle ue_i, Pe_\alpha\right\rangle_{TM} = \left\langle ue_i, Pe_\alpha\right\rangle_{\mathbb{R}^N} \\
&= \left\langle Pue_i, e_\alpha\right\rangle_{\mathbb{R}^N} = \left\langle ue_i, e_\alpha\right\rangle_{\mathbb{R}^N}
\end{aligned}$$

Moreover,

$$\begin{aligned}
\sum_{\alpha=1}^N \left\langle ue_i, e_\alpha\right\rangle\left\langle ue_j, e_\alpha\right\rangle &= \left\langle ue_i, \sum_{\alpha=1}^N \left\langle ue_j, e_\alpha\right\rangle e_\alpha\right\rangle \\
&= \left\langle ue_i, ue_j\right\rangle = \left\langle e_i, e_j\right\rangle = \delta_{ij}.
\end{aligned}$$

Therefore, we have

$$dW_t^j = \left\langle e_\alpha, U_t e_j\right\rangle dM_t^\alpha,$$

which implies that $W$ is a continuous $\mathbb{F}^X$-local martingale since $M^\alpha$ are $\mathbb{F}^X$-local martingales. Again, because

$$d\left\langle M^\alpha, M^\beta\right\rangle_s = \langle P_\alpha, P_\beta\rangle\, ds,$$

we have

$$\begin{aligned}
\left\langle W_t^i, W_t^j\right\rangle &= \sum_{\alpha,\beta}\int_0^t \langle e_\alpha, U_s e_j\rangle \langle e_\beta, U_s e_j\rangle \langle P_\alpha, P_\beta\rangle\, ds \\
&= \sum_\alpha \int_0^t \langle e_\alpha, U_s e_j\rangle \left\langle \sum_\beta \langle P_\alpha, P_\beta\rangle\, e_\beta, U_s e_j\right\rangle ds \\
&= \sum_\alpha \int_0^t \langle e_\alpha, U_s e_j\rangle \left\langle \sum_\beta \langle P P_\alpha, e_\beta\rangle\, e_\beta, U_s e_j\right\rangle ds \\
&= \sum_\alpha \int_0^t \langle e_\alpha, U_s e_j\rangle \langle P_\alpha, U_s e_j\rangle\, ds \\
&= \int_0^t \sum_\alpha \langle e_\alpha, U_s e_j\rangle \langle e_\alpha, U_s e_j\rangle\, ds = \delta^{ij} t.
\end{aligned}$$

Therefore, by the Lévy's theorem, $W$ is a $\mathbb{F}^X$-Brownian motion. $\square$

*Remark* 3.2.3. We can make $\mathbb{F}^X$ larger. $X$ is called a $\mathbb{F}$-Brownian motion if it is defined as above w.s.t. $\mathbb{F}$ and it is a strong Markov process w.s.t. $\mathbb{F}$.

If $X$ be a Brownian motion on $M$, then its horizontal lift $U$ is called a horizontal Brownian motion. Since

$$dU_t = H_i(U_t) \circ dW_t^i,$$

where $W$ is a Brownian motion, then $U$ is a diffusion process generated by the operator

$$L = \frac{1}{2}\sum_i H_i^2 = \frac{1}{2}\Delta_{O(M)}.$$

**Proposition 3.2.4.** *Let $U\colon \Omega \to W(O(M))$ be a measurable map defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. TFAE.*

*(1) $U$ is a horizontal $\mathbb{F}^U$-semimartingale whose projection $X = \pi(U)$ is a Brownian motion on $M$.*

*(2) $U$ is a horizontal $\mathbb{F}^U$-semimartingale whose anti-development is a standard Euclidean Brownian motion.*

*(3) $U$ is a $\Delta_{O(M)}/2$-diffusion w.s.t. $\mathbb{F}^U$.*

*Remark* 3.2.5. An $O(M)$-valued process $U$ satisfies any of above is called a horizontal Brownian motion on $O(M)$.

*Proof.* (1) $\Rightarrow$ (2) : Since $X$ is a Brownian motion, by above its anti-development $W$ is a standard Euclidean Brownian motion.
(2) $\Rightarrow$ (3) : Then by

$$dU_t = H_i(U_t) \circ dW_t^i,$$

it is a diffusion process of $\Delta_{O(M)}/2$.
(3) $\Rightarrow$ (1) : For any $f \in C^\infty(M)$, let $\widetilde{f} = f \circ \pi \in C^\infty(O(M))$. Because

$$\Delta_M f(\pi(u)) = \Delta_{O(M)} \widetilde{f}(u),$$

we have

$$f(X_t) - f(X_0) - \frac{1}{2}\int_0^t \Delta_M f(X_s)ds = \widetilde{f}(U_t) - \widetilde{f}(U_0) - \frac{1}{2}\int_0^t \Delta_{O(M)}\widetilde{f}(U_s)ds.$$

The RHS is a $\mathbb{F}^U$-local martingale, which means that the LHS is also a $\mathbb{F}^U$-local martingale. Moreover, $X = \pi(U)$ implies $\mathbb{F}^X \subset \mathbb{F}^U$, so the LHS is a $\mathbb{F}^X$-local martingale. □

*Remark* 3.2.6. Note that let $M \subset \mathbb{R}^N$ and $B$ is a Brownian motion on $\mathbb{R}^N$. Then if

$$dX_t = P_\alpha(X_t) \circ dB_t^\alpha,$$

then $X_t$ is a diffusion process generated by

$$L = \frac{1}{2}\sum_\alpha P_\alpha^2 = \frac{1}{2}\Delta_M.$$

Therefore, $X$ is a Brownian motion on $M$.

## 3.3 Examples of Brownian Motion

**Riemannian Structure of Sphere.** First, consider the standard Riemannian structure on $\mathbb{S}^d$.

1. $d = 1$ : Choosing a chart $(0, 2\pi) \to \mathbb{S}^1$ and consider the embedding

$$\iota\colon \mathbb{S}^1 \hookrightarrow \mathbb{R}^2$$

given by $\theta \mapsto (x, y)$, where $x = \cos\theta$ and $y = \sin\theta$. Note that any $V_{(x,y)} \in T_{(x,y)}\mathbb{R}^2$ is equivalent to $V_{(x,y)} \in \mathbb{R}^2$ by setting

$$V_{(x,y)} = a(x,y)\frac{\partial}{\partial x} + b(x,y)\frac{\partial}{\partial y} = (a(x,y), b(x,y))^\top.$$

and for any $f(x, y) \in C^\infty(\mathbb{R}^2)$,

$$V_{(x,y)}f = a(x,y)\frac{\partial f}{\partial x}(x,y) + b(x,y)\frac{\partial f}{\partial y}(x,y).$$

So for $\iota_*\colon T\mathbb{S}^1 \to \mathbb{R}^2$, at $\theta_0$

$$\iota_{*,\theta_0}\left(\left.\frac{d}{d\theta}\right|_{\theta_0}\right) = -\sin\theta_0\frac{\partial}{\partial x} + \cos\theta_0\frac{\partial}{\partial y}.$$

which is because for any $f(x, y) \in C^\infty(M)$,

$$\iota_{*,\theta_0}\left(\left.\frac{d}{d\theta}\right|_{\theta_0}\right) f = \left.\frac{d}{d\theta}\right|_{\theta_0} f\circ\iota = \left.\frac{d}{d\theta}\right|_{\theta_0} f(\cos\theta, \sin\theta) = -\sin\theta_0\frac{\partial f}{\partial x}(\theta_0) + \cos\theta_0\frac{\partial f}{\partial y}(\theta_0).$$

And note that $\iota_*$ is $C^\infty(\mathbb{S}^1)$-linear. Then the induce metric on $\mathbb{S}^1$ is given by

$$g_{\theta\theta}(\theta_0) = \left\langle \left.\frac{d}{d\theta}\right|_{\theta_0}, \left.\frac{d}{d\theta}\right|_{\theta_0} \right\rangle = \left\langle \iota_{*,\theta_0}\left.\frac{d}{d\theta}\right|_{\theta_0}, \iota_{*,\theta_0}\left.\frac{d}{d\theta}\right|_{\theta_0} \right\rangle = \langle(-\sin\theta_0, \cos\theta_0), (-\sin\theta_0, \cos\theta_0)\rangle = 1$$

Therefore, $g = d\theta \otimes d\theta$, which implies that the corresponding Christoffel coefficients

$$\Gamma^{\theta}_{\theta\theta} = \frac{1}{2} g^{\theta\theta} \frac{d}{d\theta} g_{\theta\theta} = 0,$$

It follows that that for any $X = a(\theta)\frac{d}{d\theta}, Y = b(\theta)\frac{d}{d\theta}$, we have

$$\nabla_X Y = a(\theta)\dot{b}(\theta)\frac{d}{d\theta},$$

because $\nabla_{\frac{d}{d\theta}} \frac{d}{d\theta} = 0$. Moreover, we also have

$$\Delta_{\mathbb{S}^1} h(\theta) = \frac{d^2}{d\theta^2} h \;\Rightarrow\; \Delta_{\mathbb{S}^1} = \frac{d^2}{d\theta^2}.$$

On the other hand, consider $\mathbb{S}^1 \subset \mathbb{R}^2$ embedded, let $T = \iota_*(d/d\theta)$. Then

$$\widetilde{X} = \iota_*(X) = a(\theta)T, \quad \widetilde{Y} = \iota_*(Y) = b(\theta)T.$$

Moreover, they can be extended to vector fields on $\mathbb{R}^2$. Let $\widetilde{T} = (-y, x)$, i.e., $T = \widetilde{T}|_{\mathbb{S}^1}$. Then

$$\widetilde{X}(x, y) = \widetilde{a}(x, y)\widetilde{T}, \quad \widetilde{Y}(x, y) = \widetilde{b}(x, y)\widetilde{T},$$

where $\widetilde{a}(\cos\theta, \sin\theta) = a(\theta)$ and $\widetilde{b}(\cos\theta, \sin\theta) = b(\theta)$. Note that

$$\widetilde{T}(\widetilde{a})|_{\mathbb{S}^1}(\theta) = -y\frac{\partial \widetilde{a}}{\partial x} + x\frac{\partial \widetilde{a}}{\partial y} = \dot{a}(\theta)$$
$$\widetilde{T}(\widetilde{b})|_{\mathbb{S}^1}(\theta) = -y\frac{\partial \widetilde{b}}{\partial x} + x\frac{\partial \widetilde{b}}{\partial y} = \dot{b}(\theta)$$

and so we have

$$\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}|_{\mathbb{S}^1}(\theta) = -\widetilde{T}(y)\frac{\partial}{\partial x} + \widetilde{T}(x)\frac{\partial}{\partial y} = -\cos\theta\frac{\partial}{\partial x} - \sin\theta\frac{\partial}{\partial y},$$

which mean that $\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}|_{\mathbb{S}^1} = -\vec{n} \in N_\theta\mathbb{S}^1$. So

$$\nabla_T T = (\widetilde{\nabla}_{\widetilde{T}}\widetilde{T})^\top = 0,$$

and by viewing $T = d/d\theta$, $\Gamma^\theta_{\theta\theta} = 0$ as same calculation as above. Moreover,

$$\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y}|_{\mathbb{S}^1}(\theta) = a(\theta)\left(\widetilde{T}(\widetilde{b})\widetilde{T} + \widetilde{b}\widetilde{\nabla}_{\widetilde{T}}\widetilde{T}\right) = a(\theta)\dot{b}(\theta)T - a(\theta)b(\theta)\vec{n}(\theta).$$

It follows that

$$\nabla_X Y(\theta) = (\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})^\top = a(\theta)\dot{b}(\theta)\frac{d}{d\theta},$$

and

$$\mathrm{II}(X, Y) = (\widetilde{\nabla}_{\widetilde{X}}\widetilde{Y})^\perp = -a(\theta)b(\theta)\vec{n}(\theta) = -\langle X, Y\rangle\,\vec{n}.$$

2. In general case, by embedding $\mathbb{S}^d \subset \mathbb{R}^{d+1}$,

$$\nabla_X Y = \widetilde{\nabla}_X Y + \langle X, Y\rangle\,\vec{n}$$

where $\vec{n} \in \Gamma(N\mathbb{S}^d)$ is the outer unit normal vector. Then let $\{X_i\}_{i=1}^d$ be an orthonormal frame of $T\mathbb{S}^d$ and so $\{X_i, \vec{n}\}$ is an orthonormal frame of $\mathbb{R}^{d+1}$. For any $f \in C^\infty(\mathbb{S}^d)$, let $F \in C^\infty(\mathbb{R}^{d+1})$ be an extension of $f$. As we know $\operatorname{grad} f = (\overline{\operatorname{grad}} f)^\top$, $\widetilde{\nabla}_{X_i}\vec{n} = X_i$, and

$$\Delta_{\mathbb{R}^{d+1}} F = \sum_{i=1}^d \widetilde{\nabla}^2 F(X_i, X_i) + \widetilde{\nabla}^2 F(\vec{n}, \vec{n}),$$

we have

$$\begin{aligned}
\Delta_{\mathbb{S}^d} f &= \sum_{i=1}^d \nabla^2 f(X_i, X_i) = \sum_{i=1}^d \langle \nabla_{X_i} \operatorname{grad} f, X_i \rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \operatorname{grad} F + \langle \operatorname{grad} F, X_i \rangle \vec{n}, X_i \right\rangle = \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \operatorname{grad} F, X_i \right\rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} (\overline{\operatorname{grad}} F)^\top, X_i \right\rangle = \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} (\overline{\operatorname{grad}} F - \langle \overline{\operatorname{grad}} F, \vec{n} \rangle \vec{n}), X_i \right\rangle \\
&= \sum_{i=1}^d \left\langle \widetilde{\nabla}_{X_i} \overline{\operatorname{grad}} F, X_i \right\rangle - d \langle \overline{\operatorname{grad}} F, \vec{n} \rangle = \sum_{i=1}^d \widetilde{\nabla}^2 F(X_i, X_i) - d \widetilde{\nabla}_{\vec{n}} F \\
&= \Delta_{\mathbb{R}^{d+1}} F - \widetilde{\nabla}^2 F(\vec{n}, \vec{n}) - d \widetilde{\nabla}_{\vec{n}} F
\end{aligned}$$

On the other hand, choose a polar coordinate chart for $\mathbb{R}^{d+1} \backslash \{0\}$, $(r, \theta = (\theta^1, \cdots, \theta^d))$. Then for Euclidean coordinate $(x^1, \cdots, x^{d+1})$,

$$x^i = r\iota^i(\theta),$$

where $\iota \colon \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ is the inclusion. Then we have

$$\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r}\frac{\partial}{\partial x^i} = \iota^i \frac{\partial}{\partial x^i}$$

and

$$\frac{\partial}{\partial \theta^\alpha} = \frac{\partial x^i}{\partial \theta^\alpha}\frac{\partial}{\partial x^i} = r\frac{\partial \iota^i}{\partial \theta^\alpha}\frac{\partial}{\partial x^i}$$

Then we have, first

$$g_{rr} = \left\langle \frac{\partial}{\partial r}, \frac{\partial}{\partial r} \right\rangle = \delta_{ij}\iota^i\iota^j = 1,$$

which also implies that

$$0 = \frac{\partial}{\partial \theta^\alpha}\delta_{ij}\iota^i\iota^j = 2\delta_{ij}\iota^i\frac{\partial \iota_j}{\partial \theta^\alpha} = 0.$$

So, second

$$g_{\alpha r} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial r} \right\rangle = r\delta_{ij}\iota^i\frac{\partial \iota^i}{\partial \theta^\alpha} = 0.$$

Finally,

$$g_{\alpha\beta} = \left\langle \frac{\partial}{\partial \theta^\alpha}, \frac{\partial}{\partial \theta^\beta} \right\rangle = r^2\delta_{ij}\frac{\partial \iota^i}{\partial \theta^\alpha}\frac{\partial \iota^i}{\partial \theta^\beta}.$$

For $g_{\alpha\beta}$, note that $\iota$ is the isometric inclusion, so

$$g_{\mathbb{S}^d}\left( \left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d}, \left.\frac{\partial}{\partial \theta^\beta}\right|_{\mathbb{S}^d} \right) = g\left( \iota_* \left.\frac{\partial}{\partial \theta^\alpha}\right|_{\mathbb{S}^d}, \iota_* \left.\frac{\partial}{\partial \theta^\beta}\right|_{\mathbb{S}^d} \right) = \delta_{ij}\frac{\partial \iota^i}{\partial \theta^\alpha}\frac{\partial \iota^i}{\partial \theta^\beta},$$

because
$$\iota_* \frac{\partial}{\partial \theta^\alpha}\bigg|_{\mathbb{S}^d} = \frac{\partial \iota^i}{\partial \theta^\alpha}\frac{\partial}{\partial x^i}.$$
It follows that
$$g_{\alpha\beta} = r^2 h_{\alpha\beta}, \quad h_{\alpha\beta} := g_{\mathbb{S}^d}\left( \frac{\partial}{\partial\theta^\alpha}\bigg|_{\mathbb{S}^d}, \frac{\partial}{\partial\theta^\beta}\bigg|_{\mathbb{S}^d} \right).$$
So we have
$$g = dr\otimes dr + r^2 h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta.$$
It also implies that $(r,\theta)$ is a semi-geodesic coordinate of $\mathbb{R}^{d+1}$ w.s.t. $\mathbb{S}^d$ and distance function $r$. Moreover, based on this coordinate system, we have
$$g = (g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 h_{\alpha\beta} \end{pmatrix}, \quad g^{-1} = (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} h^{\alpha\beta} \end{pmatrix},$$
which implies that
$$\sqrt{|g|} = r^d \sqrt{|h|}.$$
Let $(y^1, \cdots y^{d+1}) = (r, \theta^1, \cdots, \theta^d)$. For any $F = F(r,\theta) \in C^\infty(\mathbb{R}^{d+1})$, note that
$$g^{ij}\frac{\partial F}{\partial y^j} = \begin{cases} \frac{\partial F}{\partial r}, & i = 1 \\ \frac{h^{i\beta}}{r^2}\frac{\partial F}{\partial \theta^\beta}, & i \neq 1 \end{cases}$$
Then by the definition of Laplace-Beltrami operator,
$$\begin{aligned}\Delta_{\mathbb{R}^{d+1}} F &= \frac{1}{|g|}\frac{\partial}{\partial y^i}\left(\sqrt{|g|} g^{ij}\frac{\partial F}{\partial y^j}\right) \\ &= \frac{1}{|g|}\frac{\partial}{\partial r}\left(\sqrt{|g|}\frac{\partial F}{\partial r}\right) + \frac{1}{|g|}\frac{\partial}{\partial \theta^\alpha}\left(\sqrt{|g|}\frac{h^{\alpha\beta}}{r^2}\frac{\partial F}{\partial\theta^\beta}\right) = \mathrm{I} + \mathrm{II}.\end{aligned}$$
For I,
$$\begin{aligned}\mathrm{I} &= \frac{1}{|g|}\frac{\partial}{\partial r}\left(\sqrt{|g|}\frac{\partial F}{\partial r}\right) = \frac{1}{r^d\sqrt{|h|}}\frac{\partial}{\partial r}\left(r^d\sqrt{|h|}\frac{\partial F}{\partial r}\right) \\ &= \frac{d}{r}\frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2}.\end{aligned}$$
For II,
$$\begin{aligned}\mathrm{II} &= \frac{1}{|g|}\frac{\partial}{\partial\theta^\alpha}\left(\sqrt{|g|}\frac{h^{\alpha\beta}}{r^2}\frac{\partial F}{\partial\theta^\beta}\right) \\ &= \frac{1}{r^d\sqrt{|h|}}\frac{\partial}{\partial\theta^\alpha}\left(r^{d-2}\sqrt{|h|}h^{\alpha\beta}\frac{\partial F}{\partial\theta^\beta}\right) \\ &= \frac{1}{r^2\sqrt{|h|}}\frac{\partial}{\partial\theta^\alpha}\left(\sqrt{|h|}h^{\alpha\beta}\frac{\partial F}{\partial\theta^\beta}\right) \\ &= \frac{1}{r^2}\Delta_{\mathbb{S}^d}\hat{F}_r,\end{aligned}$$
where $\hat{F}_r(\theta) = F(r,\theta) \in C^\infty(\mathbb{S}^d)$. So we have
$$\Delta_{\mathbb{R}^{d+1}} F = \frac{d}{r}\frac{\partial F}{\partial r} + \frac{\partial^2 F}{\partial r^2} + \frac{1}{r^2}\Delta_{\mathbb{S}^d}\hat{F}_r.$$

Then on $\mathbb{S}^d$, i.e., $r = 1$, for any $f \in C^\infty(\mathbb{S}^d)$, choose any extension $F$, then

$$\Delta_{\mathbb{S}^d} f = \left( \Delta_{\mathbb{R}^{d+1}} - d\frac{\partial}{\partial r} - \frac{\partial^2}{\partial r^2} \right) F|_{r=1}.$$

In particular, if we choose $F(r, \theta) = f(\theta)$ for any $r$, then

$$\Delta_{\mathbb{S}^d} f = \Delta_{\mathbb{R}^{d+1}} F.$$

*Remark* 3.3.1. If we embedded $\iota \colon \mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$ by

$$\begin{aligned}
x_1 &= \cos\theta_1 \\
x_2 &= \sin\theta_1 \cos\theta_2 \\
x_3 &= \sin\theta_1 \sin\theta_2 \cos\theta_3 \\
&\vdots \\
x_d &= \sin\theta_1 \cdots \sin\theta_{d-1} \cos\theta_d \\
x_{d+1} &= \sin\theta_1 \cdots \sin\theta_{d-1} \sin\theta_d
\end{aligned}$$

then the induced metric metric is given by

$$g_{\mathbb{S}^d} = d\theta_1 \otimes d\theta_1 + \sin^2\theta_1 d\theta_2 \otimes d\theta_2 + \sin^2\theta_1 \sin^2\theta_2 d\theta_3 \otimes d\theta_3 + \cdots + \left( \prod_{i=1}^{d-1} \sin^2\theta_i \right) d\theta_d \otimes d\theta_d.$$

Note that it implies that $\left\{ \frac{\partial}{\partial\theta^\alpha} \right\}$ is a orthogonal frame of $\mathbb{S}^d$ but not normal.

**Radially Symmetric Space.** For a Riemannian manifold $(M, g)$ of dimension $d$, consider a normal coordinate system, i.e., $\exp_p \colon U \to M$, and using the polar coordinate $(r, \theta) \in (0, \infty) \times \mathbb{S}^{d-1}$, then we have

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta.$$

**Definition 3.3.2.** A Riemannian manifold $(M, g)$ is called radially symmetric if on any normal polar chart, for

$$g = dr \otimes dr + g_{\alpha\beta}(r, \theta) d\theta^\alpha \otimes d\theta^\beta,$$

it has

$$g_{\alpha\beta}(r, \theta) = G(r)^2 h_{\alpha\beta}(\theta),$$

where

$$G(0) = 0, \quad G'(0) = 1,$$

and $h_{\alpha\beta} d\theta^\alpha \otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$.

**Proposition 3.3.3.** *Let $(M, g)$ be a Riemannian manifold with constant sectional curvature $\kappa$. Then $M$ is radially symmetric, i.e.,*

$$g = dr \otimes dr + G(r)^2 h_{\alpha\beta}(\theta) d\theta^\alpha \otimes d\theta^\beta,$$

*where $G(0) = 0, \quad G'(0) = 1$. Moreover,*

$$G''(r) + \kappa G(r) = 0.$$

*Proof.* Fix $p \in M$, and on $T_pM$, choosing an orthonormal basis such that $T_pM = \mathbb{R}^d$. For the $\Phi \colon (0, \delta) \times \mathbb{S}^{d-1} \to M$, $\Phi(r, \theta) = \exp_p(r\theta)$ is a geodesic, which also provides the normal polar coordination $(r, \theta)$. Fix a $\theta_0 \in \mathbb{S}^{d-1}$. Consider a normal radial geodesic $\gamma(r)$ with $\gamma(0) = p$ and $\dot\gamma(0) = \theta_0 \in \mathbb{S}^{d-1}$. Let $\theta_\alpha \colon (-\varepsilon, \varepsilon) \to \mathbb{S}^{d-1}$ be a smooth map with $\theta(0) = \theta_0$ and $\dot\theta(0) = \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1}, \theta_0}$. Note that:

(i) $W_\alpha = \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1},\theta_0} = \iota_* \frac{\partial}{\partial\theta^\alpha}|_{\mathbb{S}^{d-1},\theta_0} \in \mathbb{R}^d$, i.e., an vector in $T_pM$,

(ii) because $|\theta(s)| \equiv 1$, $W_\alpha \perp \theta_0$.

Consider the radial variation

$$F_\alpha(s,r) = \exp_p(r\theta_\alpha(s))$$

and its two vector fields

$$V_\alpha(r) = \frac{\partial F_\alpha}{\partial s}(r,0) = (d\exp_p)_{r\theta_0}(rW_\alpha), \quad T(r) = \frac{\partial F_\alpha}{\partial r}(r,0) = \dot\gamma(r).$$

Then $V_\alpha(r)$ is a Jacobian field. Moreover, we have

$$V_\alpha(0) = 0, \quad \nabla_T V_\alpha(0) = W_\alpha.$$

Because $W_\alpha \perp T(0) = \theta_0$ and $V_\alpha(0) \perp T(0)$, $V_\alpha \perp T$. It implies that

$$R(V_\alpha, T)T = \kappa V_\alpha,$$

since $M$ has constant curvature, i.e.

$$\langle R(V_\alpha, T)T, V_\alpha\rangle = \kappa\left(|V_\alpha|^2 |T|^2 - \langle V_\alpha, T\rangle^2\right) = \kappa |V_\alpha|^2.$$

So the Jacobian equation is

$$\frac{D^2}{dt^2}V_\alpha(t) + \kappa V_\alpha = 0.$$

So when choosing an orthogonal frame $\{E_i(r)\}$ with $E_1(r) = T(r)$ along $\gamma(r)$ and $E_\alpha(r)$ are obtained by parallel moving $W_\alpha$ along $\gamma(r)$, which are orthogonal because $\langle W_\alpha, W_\beta\rangle = 0$. Using similar result, because

$$\langle V_\alpha(0), E_\beta(0)\rangle = \langle \nabla_T V_\alpha(0), E_\beta(0)\rangle = \langle W_\alpha, W_\beta\rangle = 0, \ \forall\ \beta \neq \alpha,$$

$V_\alpha \perp E_\beta$. So

$$V_\alpha(r) = G_\alpha(r) E_\alpha(r)$$

and $G_\alpha(r)$ should satisfy

$$f''(r) + \kappa f(r) = 0, \quad f(0) = 0,\ f'(0) = 1$$

By the uniqueness of ODE, for all $\alpha$

$$G(r) = G_\alpha(r) = \begin{cases} \frac{1}{\sqrt{\kappa}}\sin(\sqrt{\kappa}r), & \kappa > 0, \\ r, & \kappa = 0, \\ \frac{1}{\sqrt{-\kappa}}\sinh(\sqrt{-\kappa}r), & \kappa < 0, \end{cases}$$

Next, the the coordinate tangent vectors on $M$ are given by

$$\left.\frac{\partial}{\partial\theta^\alpha}\right|_M = \Phi_*\left(\left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1}}\right)$$

So by definition

$$d\Phi_{(r,\theta_0)}\left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1},\theta_0} = \left.\frac{d}{ds}\right|_{s=0}\Phi(r,\theta_\alpha(s)) = \left.\frac{d}{ds}\right|_{s=0}\exp_p(r\theta_\alpha(s)) = V_\alpha(r),$$

i.e.,

$$\left.\frac{\partial}{\partial\theta^\alpha}\right|_{M,(r,\theta_0)} = V_\alpha(r).$$

Therefore,

$$g_{\alpha\beta}(r,\theta_0) = \langle V_\alpha(r), V_\beta(r)\rangle = G(r)^2\,\langle E_\alpha(r), E_\beta(r)\rangle = G(r)^2\,\langle E_\alpha(0), E_\beta(0)\rangle = G(r)^2\,\langle W_\alpha, W_\beta\rangle$$

because $E_\alpha(r)$ are obtained by parallel moving. Let

$$h_{\alpha\beta}(\theta_0) = \langle W_\alpha, W_\beta\rangle = \left\langle \left.\frac{\partial}{\partial\theta^\alpha}\right|_{\mathbb{S}^{d-1},\theta_0}, \left.\frac{\partial}{\partial\theta^\beta}\right|_{\mathbb{S}^{d-1},\theta_0} \right\rangle,$$

i.e., $h_{\alpha\beta}d\theta^\alpha \otimes d\theta^\beta$ is the canonical Riemannian metric on $\mathbb{S}^{d-1}$. So we have

$$g = dr\otimes dr + G(r)^2 h_{\alpha\beta}(\theta)d\theta^\alpha\otimes d\theta^\beta. \qquad \square$$

For a radially symmetric space $(M,g)$, using similar calculation as $\mathbb{S}^d$, we have

$$\Delta_M = L_r + \frac{1}{G(r)^2}\Delta_{\mathbb{S}^{d-1}},$$

where

$$L_r = \frac{\partial^2}{\partial r^2} + (d-1)\frac{G'(r)}{G(r)}\frac{\partial}{\partial r},$$

is called the radial Laplacian.

**Examples.**

**Example 3.3.4** (Circle)**.** Consider

$$\mathbb{S}^1 = \left\{e^{i\theta} \colon 0\le\theta<2\pi\right\} \subset \mathbb{R}^2.$$

Let $B$ be a Brownian motion. Define

$$X_t = e^{iB_t} = (\cos B_t, \sin B_t).$$

Then $X$ is a Brownian motion on $\mathbb{S}^1$.

*Proof by Laplacian Operator.* First, When viewing $\mathbb{S}^1\subset\mathbb{R}^2$, $X_t = (\cos B_t, \sin B_t)$, by Itô formula,

$$dX_t = (-\sin B_t, \cos B_t)\circ dB_t = \iota_*\frac{d}{d\theta}(B_t)\circ dB_t.$$

Therefore,

$$V = \iota_*\frac{d}{d\theta} = \frac{d}{d\theta}.$$

Then $X_t$ is a diffusion process with

$$L = \frac{1}{2}\frac{d^2}{d\theta^2} = \frac{1}{2}\Delta_{\mathbb{S}^1}.$$

So $X$ is a Brownian motion on $\mathbb{S}^1$. $\square$

*Proof by Anti-development.* First, note that at any $\theta \in \mathbb{S}^{\not K}$, $O(\mathbb{S}^1)_\theta = \{-1, 1\}$ and so the orthonormal frame bundle $O(M) = \bigcup_{\theta\in\mathbb{S}^1}\{-1,1\}$. By $\Gamma^\theta_{\theta\theta} = 0$, any horizontal vector $\dot{u}(0)$ vanishes on the vertical part. Then by $dU_t = P^H_\alpha \circ dX^\alpha_t$, the horizontal lift of $X$ starting from $U_0 = 1$, $U_t \equiv 1$.
On the other hand, consider $P(\theta)\colon \mathbb{R}^2 \to T_\theta\mathbb{S}^1$. Note that $n = (-\sin\theta, \cos\theta) \in N_\theta\mathbb{S}^1$. So

$$P(\theta) = \begin{pmatrix} \sin^2\theta & -\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \cos\theta^2 \end{pmatrix}.$$

Therefore,

$$P_1(\theta) = (\sin^2\theta, -\sin\theta\cos\theta)^\top = \iota_*\left(-\sin\theta\frac{d}{d\theta}\right)$$
$$P_2(\theta) = (-\sin\theta\cos\theta, \cos^2\theta)^\top = \iota_*\left(\cos\theta\frac{d}{d\theta}\right).$$

Then by $dW^t = U_t^{-1}P_\alpha(X_t) \circ dX^\alpha_t$,

$$\begin{aligned} dW_t &= -\sin B_t \circ dX^1_t + \cos B_t \circ dX^2_t \\ &= \sin^2 B_t \circ dB_t + \cos^2 B_t \circ dB_t \\ &= dB_t, \end{aligned}$$

and $W_0 = 0$, so $W_t = B_t$. $\square$

**Example 3.3.5** (Sphere)**.** Consider $\mathbb{S}^d \subset \mathbb{R}^{d+1}$. Let $\vec{n}$ be the outer unit normal vector. Then the orthogonal projection

$$P(\theta)\colon \mathbb{R}^{d+1} \to T_\theta\mathbb{S}^d$$

is given by $P(\theta)(x) = x - \langle x, \vec{n}(\theta)\rangle\,\vec{n}(\theta)$. The matrix expression of $P(\theta)$ is

$$P(\theta)_{ij} = \delta_{ij} - n(\theta)_i n(\theta)_j$$

Or when embedding $\mathbb{S}^d \hookrightarrow \mathbb{R}^{d+1}$, $P(x)_{ij} = \delta_{ij} - x^i x^j$. Therefore, by $\Delta_M = \sum_\alpha P^2_\alpha$, $X$ is a Brownian motion on $\mathbb{S}^d$, when it satisfies

$$X^i_t = X^i_0 + \sum_{j=1}^{d+1}\int_0^t (\delta_{ij} - X^i_s X^j_s) \circ dW^j_t,$$

where $W$ is a standard Brownian motion on $\mathbb{R}^{d+1}$.

**Example 3.3.6** (Radially Symmetric Space)**.** Let $(M, g)$ be a radially symmetric space and $X_t = (r_t, \theta_t)$ be a Brownian motion on $M$ expressed as polar coordinate. Consider its radial process and angular process separately.
(1) Radial process: Choose $f(r,\theta) = r$ to the martingale problem,

$$\begin{aligned} \beta_t &= r_t - r_0 - \frac{1}{2}\int_0^t \Delta_M f(X_s)ds \\ &= r_t - r_0 - \frac{d-1}{2}\int_0^t \frac{G'(r_s)}{G(r_s)}ds \end{aligned}$$

is a local martingale. Moreover, the quadratic variation is given by

$$\langle\beta,\beta\rangle_t = \int_0^t \Gamma(r,r)(X_s)ds,$$

where we have

$$\Gamma(r,r) = \langle \nabla r, \nabla r \rangle = g^{ij} \frac{\partial r}{\partial x^i} \frac{\partial r}{\partial x^j} = |\nabla r|^2 = 1.$$

So

$$\langle \beta, \beta \rangle_t = t,$$

which implies that $\beta$ is a 1-dimensional Brownian motion. On the other hand, we can see the radial process $r_t$ is generated by the radial operator $L_r$.

(2) Angular process: Let $f \in C^\infty(\mathbb{S}^{d-1})$, which can be extended to $f(r,\theta) = f(\theta) \in C^\infty(M)$. Then

$$f(X_t) = f(\theta_t), \quad \Delta_M f(r,\theta) = \frac{1}{G(r)^2} \Delta_{\mathbb{S}^{d-1}}.$$

It implies that

$$M_t^f = f(\theta_t) - f(\theta_0) - \frac{1}{2} \int_0^t \frac{1}{G(r_s)^2} \Delta_{\mathbb{S}^{d-1}} f(\theta_s) ds$$

Let

$$T_{t'} = \int_0^{t'} \frac{1}{G(r_s)^2} ds,$$

which is a nondecreasing process. So we can consider

$$\tau_t = \inf \{t' \geq 0 \colon T_{t'} \geq t\} = T^{-1}(t),$$

where the final equality is because $T$ is strictly increasing in $t$. Then by considering the time-change, let $z_t = \theta_{\tau_t}$, we have

$$M_{\tau_t}^f = f(z_t) - f(z_0) - \frac{1}{2} \int_0^t \Delta_{\mathbb{S}^{d-1}} f(z_s) ds.$$

Since $M_t$ is a local martingale, $t \mapsto M_{\tau_t}^f$ is also a local martingale w.s.t. $\mathcal{F}_{\tau_t}$. Therefore, $t \mapsto z_t = \theta_{\tau_t}$ is a Brownian motion on $\mathbb{S}^{d-1}$

**Claim:** $r$ and $z$ are independent.

*Proof.* I. First, we embed $\mathbb{S}^{d-1} \hookrightarrow \mathbb{R}^d$ with $\theta \mapsto y = (y^\alpha)$ and choose $f(\theta) = f(y) = f^\alpha(y) = y^\alpha$, the coordinate function on $\mathbb{R}^\alpha$. Let $U_t$ be a horizontal lift of $\theta_t = Y_t \in \mathbb{R}^d$ and $W_t$ be its anti-development. Then

$$W_t = \int_0^t U_s^{-1} P_\alpha(Y_s) \circ dY_s^\alpha.$$

Because $\tau_t$ is continuous, for any continuous semimartingales $H, K$,

$$\int_0^{\tau_t} H_s dK_s = \int_0^t H_{\tau_s} dK_{\tau_s}, \quad \langle H \circ \tau, K \circ \tau \rangle_t = \langle H, K \rangle_{\tau_t}.$$

So $U_{\tau_t}$ is a horizontal lift of $\theta_{\tau_t}$ and $W_{\tau_t}$ is the anti-development of $\theta_{\tau_t}$. Because $\theta_{\tau_t}$ is a Brownian motion on $\mathbb{S}^{d-1}$, $\{W_{\tau_t}\}_{t\geq 0}$ is an Euclidean Brownian motion.

II. Next, we need to check

$$\langle W \circ \tau, \beta \rangle = 0$$

For any $f \in C^\infty(\mathbb{S}^{d-1})$,

$$\langle M^f, \beta \rangle_t = \int_0^t \Gamma(f,r)(X_s) ds = \int_0^t \langle \nabla f, \nabla r \rangle (X_s) ds.$$

Note that $f$ is viewing as $f(r,\theta)=f(\theta)\in C^\infty(M)$. By

$$\langle \nabla f^\alpha, \nabla r\rangle = g^{ij}\frac{\partial f^\alpha}{\partial x^i}\frac{\partial r}{\partial x^j}=0,$$

we have

$$\left\langle M^{f^\alpha},\beta\right\rangle = \langle Y^\alpha,\beta\rangle \equiv 0, \quad \forall\ \alpha=1,2,\cdots,d,$$

which implies that

$$\langle W,\beta\rangle_t=\int_0^t U_s^{-1}P_\alpha(Y_s)d\,\langle Y^\alpha,\beta\rangle=0$$

By the following lemma, we have

$$\langle W\circ\tau,\beta\rangle_t=0.$$

III. Because $\{W_{\tau_t}\}_{t\geq 0}$ and $\{\beta_t\}_{t\geq 0}$ are two Brownian motions, they are independent. Moreover, since

$$d\theta_{\tau_t}=U_{\tau_t}^{-1}e_i\circ dW_{\tau_t}^i$$

and

$$dr_t=d\beta_t+\frac{d-1}{2}\frac{G'(r_t)}{G(r_t)}dt,$$

$r_t$ and $z_t=\theta_{\tau_t}$ are independent. □

**Lemma 3.3.7.** *Let $W$ and $M$ be two continuous local martingale and $\tau$ be a continuous time-change. If $\langle W,M\rangle\equiv 0$, then $\langle W\circ\tau,M\rangle\equiv 0$.*

**Example 3.3.8** (In Local Coordinates)**.** Let $(x^1,\cdots,x^d)$ be a local chart of $M$ and $(x^i, i=1,\cdots,d; e_j^i, 1\leq i\leq j\leq d)$ be a chart of $O(M)$. The equation of a horizontal Brownian motion is

$$dU_t=H_i(U_t)\circ dW_t^i,$$

where $W$ is a $d$-dimensional Brownian motion. We have already shown that

$$H_i(u)=e_j^i\frac{\partial}{\partial x^j}-e_i^je_m^\ell\Gamma_{j\ell}^k(x)\frac{\partial}{\partial e_m^k}.$$

So for $U_t=(X_t^i,e_j^i(t))$,

$$\begin{aligned}dX_t^i&=e_j^i(t)\circ dW_t^j,\\ de_j^i(t)&=-\Gamma_{k\ell}^i(X_t)e_j^\ell(t)e_m^k(t)\circ dW_t^m.\end{aligned}$$

For the first equation,

$$dX_t^t=e_j^i(t)dW_t^j+\frac{1}{2}d\left\langle e_j^i,dW^j\right\rangle_t.$$

Let

$$dM_t^i=e_j^i(t)dW_t^j\ \Rightarrow\ d\left\langle M^i,M^j\right\rangle_t=\sum_{k=1}^d e_k^i(t)e_k^j(t)dt.$$

Note that because $u\in O(M)$ and $ue_\ell=e_\ell^i\frac{\partial}{\partial x^i}$,

$$\delta_{\ell m}=\langle ue_\ell,ue_m\rangle=e_\ell^ig_{ij}e_m^j,$$

i.e., $ege^\top = I$ for $e = (e^i_j)$. Moreover, because $e$ is invertible,

$$e^\top = g^{-1}, \quad i.e. \sum_{k=1}^{d} e^i_k e^j_k = g^{ij}.$$

Therefore,

$$d\left\langle M^i, M^j \right\rangle_t = g^{ij}\left(X_t\right)dt.$$

Then let $\sigma = g^{-\frac{1}{2}}$ and define

$$B_t = \int_0^t \sigma\left(X_s\right)^{-1} dM_s,$$

Because

$$d\left\langle B^i, B^j \right\rangle_t = \delta^{ij} dt,$$

$B$ is an Euclidean Brownian motion. Also,

$$dM_t = \sigma(X_t)dB_t.$$

On the other hand, by the second equation,

$$d\left\langle e^i_j, dW^j \right\rangle_t = -\Gamma^i_{k\ell}\left(X_t\right)e^\ell_m(t)e^k_m(t) = -g^{\ell k}\left(X_t\right)\Gamma^i_{k\ell}\left(X_t\right).$$

Therefore, $X_t$ satisfies

$$dX^i_t = \sigma^i_j\left(X_t\right)dB^j_t - \frac{1}{2}g^{\ell k}\left(X_t\right)\Gamma^i_{k\ell}\left(X_t\right)dt,$$

where $B$ is a $d$-dimensional Brownian motion. Note that $X_t$ is a diffusion process generated by

$$L = -\frac{1}{2}g^{\ell k}\Gamma^i_{k\ell}\frac{\partial f}{\partial x^i} + \frac{1}{2}g^{ij}\frac{\partial^2 f}{\partial x^i \partial x^j}.$$

Also, because

$$\Delta_M f = g^{ij}\frac{\partial^2 f}{\partial x^i \partial x^j} - g^{k\ell}\Gamma^i_{k\ell}\frac{\partial f}{\partial x^i} = 2L,$$

$X$ is a $\Delta_M/2$-diffusion process, and so $X$ is a Brownian motion on $M$.

## 3.4 Radial Process

Let $(M, g)$ be a complete Riemannian manifold of dimension $d$. Fix $o \in M$ and let $C_o$ be the cut locus and $E_o = M\backslash C_o$. Let

$$r(x) = d(x, o).$$

Then $r$ is smooth on $M\backslash C_o \cup \{o\}$ and $|\nabla r| \equiv 1$.
Let $X$ be a Brownian motion on $M$ starting within $E_o$. Before it hitting $C_o$, i.e., $t < T_{C_o}$, where $T_{C_o}$ is the hitting time of $C_o$,

$$\beta_t = r(X_t) - r(X_0) - \frac{1}{2}\int_0^t \Delta_M r(X_s)ds.$$

is a local martingale. Moreover, because

$$\langle r, r\rangle_t = \langle \beta, \beta\rangle_t = \int_0^t \Gamma\left(r, r\right)(X_s)ds = \int_0^t \left|\nabla r(X_s)\right|^2 ds = t,$$

$r(X)$ is a Brownian motion on $\mathbb{R}$.

**Model Manifolds.** Choose the polar coordinate $(r,\theta)$ of $\mathbb{R}^n$, then equip $\mathbb{R}^n$ with Riemannian metric

$$g = dr\otimes dr + G(r)^2 g_{\mathbb{S}^{n-1}}.$$

where $G\colon [0,\infty)\to[0,\infty)$ is smooth with $G(0)=0$, $G'(0)=1$, and $G(r)>0$ for $r\neq 0$. Then any Riemannian manifold $(\mathbb{M},g)$ isometric to $(\mathbb{R}^n,g)$ is called a model manifold, and denote $0\in\mathbb{M}$ be a fixed point corresponding to $0\in\mathbb{R}^n$. For the Levi-Civita connection $\nabla$, we have the Koszul formula

$$\begin{aligned} 2\left\langle \nabla_U V, W\right\rangle &= U\left\langle V,W\right\rangle + V\left\langle W,U\right\rangle - W\left\langle U,V\right\rangle \\ &\quad + \left\langle [U,V],W\right\rangle - \left\langle [V,W],U\right\rangle - \left\langle [U,W],V\right\rangle. \end{aligned}$$

So, let $X_r = \frac{\partial}{\partial r}$ and $Y,Z\in\Gamma(T\mathbb{S}^{n-1})$, because

$$\left\langle X_r,X_r\right\rangle = 1,\quad \left\langle Y,Z\right\rangle = G(r)^2\left\langle Y,Z\right\rangle_{\mathbb{S}^{n-1}},\quad [X_r,Y]=[X_r,Z]=0,$$

we have the following calculations.

(1) First, because $\left\langle X_r,X_r\right\rangle = 1$,

$$0 = X_r\left\langle X_r,X_r\right\rangle = 2\left\langle \nabla_{X_r}X_r, X_r\right\rangle = 0.$$

So $\nabla_{X_r}X_r\in\Gamma(T\mathbb{S}^{n-1})$. On the other hand, because $Y\perp X_r$, by Koszul formula,

$$2\left\langle \nabla_{X_r}X_r, Y\right\rangle = -Y\left\langle X_r,X_r\right\rangle = 0.$$

Therefore,

$$\nabla_{X_r}X_r = 0.$$

(2) By Koszul formula, $\left\langle \nabla_{X_r}Y, X_r\right\rangle = 0$, which implies that $\nabla_{X_r}Y\in\Gamma(T\mathbb{S}^{n-1})$. Similarly,

$$2\left\langle \nabla_{X_r}Y, Z\right\rangle = X_r\left(\left\langle Y,Z\right\rangle\right) = 2G(r)G'(r)\left\langle Y,Z\right\rangle_{\mathbb{S}^{n-1}} = 2\frac{G'(r)}{G(r)}\left(\left\langle Y,Z\right\rangle\right).$$

Therefore,

$$\nabla_{X_r}Y = \frac{G'(r)}{G(r)}Y.$$

By torsion-free,

$$\nabla_Y X_r = [X_r,Y] + \nabla_{X_r}Y = \nabla_{X_r}Y = \frac{G'(r)}{G(r)}Y.$$

(3) Also, by Koszul formula,

$$\left\langle \nabla_Y Z, X_r\right\rangle = -\frac{G'(r)}{G(r)}\left\langle Y,Z\right\rangle.$$

Therefore,

$$\nabla_Y Z = \nabla_Y^{\mathbb{S}^{n-1}} Z - \frac{G'(r)}{G(r)}\left\langle Y,Z\right\rangle X_r,$$

where $\nabla^{\mathbb{S}^{n-1}}$ is the induced Levi-Civita connection.

The radical curvature $\kappa_{\text{rad}}$ is defined as the sectional curvature restricted to radical section, that is a $\Pi_x\subset T_xM$, where $\Pi_x = \operatorname{span}\{X_r, Y\}$ for some unit $Y\in\Gamma(T\mathbb{S}^{n-1})$. To calculate the radical curvature, by above, we have

$$\nabla_Y Y = \nabla_Y^{\mathbb{S}^{n-1}} Y - \frac{G'(r)}{G(r)}X_r,$$

which implies that

$$\nabla_{X_r}\nabla_Y Y = \frac{G'}{G}\nabla_Y^{\mathbb{S}^{n-1}}Y - \frac{G''G-(G')^2}{G^2}X_r,$$

and

$$\begin{aligned}\nabla_Y\nabla_{X_r}Y &= \nabla_Y\left(\frac{G'}{G}Y\right) = \frac{G'}{G}\nabla_Y Y\\ &= \frac{G'}{G}\nabla_Y^{\mathbb{S}^{n-1}}Y - \left(\frac{G'}{G}\right)^2 X_r.\end{aligned}$$

Therefore,

$$\begin{aligned}R(X_r,Y)Y &= \nabla_{X_r}\nabla_Y Y - \nabla_Y\nabla_{X_r}Y - \nabla_{[X_r,Y]}Y\\ &= \nabla_{X_r}\nabla_Y Y - \nabla_Y\nabla_{X_r}Y\\ &= -\frac{G''}{G}X_r.\end{aligned}$$

It follows that

$$K(\Pi_x) = \frac{\langle R(X_r,Y)Y, X_r\rangle}{|X_r|^2|Y|^2 - \langle X_r, Y\rangle^2} = -\frac{G''(r)}{G(r)},$$

where $r = d(0,x)$. Note that the radial curvature is independent of the choice of $Y$, i.e.

$$\kappa_{\text{rad}}(r) = -\frac{G''(r)}{G(r)}.$$

i.e., $G$ is the unique solution of ODE

$$G''(r) + \kappa_{\text{rad}}(r)G(r) = 0, \quad G(0)=0,\ G'(0)=1.$$

Moreover,

$$\text{Ric}(X_r,X_r) = -(n-1)\frac{G''(r)}{G(r)}$$

Then let $\gamma(t) = (t,\theta_0)$ be a normal radial geodesic with $\dot\gamma(t) = X_r(t)$. Let $E_1 = X_r, E_2, \cdots, E_n$ be a orthonormal frame along $\gamma$. If

$$J(t) = f(t)E(t)$$

is an orthogonal Jacobian field, then $f$ satisfies

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0.$$

Therefore, for any $X \in T_{\gamma(t_0)}\mathbb{S}^{n-1}$, let $J(t)$ be the orthogonal Jacobian field with $J(0)=0$ and $J(t_0)=X$. Note that for such $J(t) = f(t)E(t)$,

$$f''(t) + \kappa_{\text{rad}}(t)f(t) = 0, \quad f(0)=0,$$

which implies that $f(t) = cG(t)$ for some constant $c$. Then

$$\nabla^2 r(X,X) = \langle\nabla_{X_r}J(t_0), J(t_0)\rangle = \frac{f'(t_0)}{f(t_0)}|X|^2 = \frac{G'(t_0)}{G(t_0)}|X|^2.$$

Therefore,

$$\Delta r(t) = \sum_{i=2}^{\infty}\nabla^2 r(E_i,E_i) = (n-1)\frac{G'(t_0)}{G(t_0)}.$$

**Generalize Comparison Theorem.** Let

$$\begin{aligned} K_M(x) &= \{K(\Pi)\colon \Pi \subset T_xM \text{ 2-dimensional section}\}, \\ \mathrm{Ric}_M(x) &= \{\mathrm{Ric}(X,X)\colon X \in T_xM,\ |X| = 1\}. \end{aligned}$$

Assume there are two model manifolds with radial curvature $\kappa_1$ and $\kappa_2$ such that

$$\begin{aligned} &\kappa_1(r) \geq \sup\{K_M(x)\colon r(x) = r\} \\ &\kappa_2(r) \leq \inf\{\mathrm{Ric}_M(x)\colon r(x) = r\}\,(d-1)^{-1}. \end{aligned}$$

Then by replacing the space form with the model manifold $(\mathbb{M}, g)$ in the proof of Hessian comparison theorem and Laplacian comparison theorem, we have the following comparison,

$$(d-1)\frac{G_1'(r)}{G_1(r)} \leq \Delta_M r \leq (d-1)\frac{G_2'(r)}{G_2(r)},$$

where

$$G''(r) + \kappa(r)G(r) = 0, \quad G(0) = 0, \quad G'(0) = 1.$$

# Chapter 4

# Riemannian Langevin Dynamics

Let $(M, g)$ be a complete Riemannian manifold with Levi-Civita connection $\nabla$ and the canonical Riemannian volume measure $dm$. In the following, we denote $\operatorname{grad} f = \nabla f$ for convenience. Let $B^M$ is a Brownian motion on $M$. Let $U \in C^\infty(M)$. Consider the following SDE

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}dB_t^M, \tag{4.1}$$

which is called a Riemannian Langevin dynamics.

## 4.1 Diffusion Operator

Because $B^M$ is a Brownian motion on $M$, its anti-development $W$ is a standard Euclidean Brownian motion. So

$$dB_t^M = V_\alpha \circ dW_t^\alpha,$$

with

$$\Delta_M = \sum_\alpha V_\alpha^2.$$

Therefore,

$$dX_t = -\nabla U(X_t)dt + \sqrt{2}V_\alpha \circ dW_t^\alpha,$$

which implies that the diffusion operator of $X$ is

$$\begin{aligned} Lf &= \frac{1}{2}\sum_\alpha (\sqrt{2}V_\alpha)^2 f - \nabla U(f) \\ &= \Delta_M f - \langle \nabla U, \nabla f \rangle, \end{aligned}$$

i.e., $L = \Delta_M - \langle \nabla U, \nabla \cdot \rangle$. For such $L$, in note *Stochastic Analysis*, we have shown that

$$L^* p = \Delta_M p + \operatorname{div}(p\nabla U)$$

in the sense of on $L^2(dm)$. Also,

$$d\mu = e^{-U}dm,$$

is the invariant measure of such diffusion process, which is also symmetric for $L$. Note that $L$ is essentially self-adjoint on $C_c^\infty(M)$ w.s.t. $\mu$. Therefore, it has a unique self-adjoint extension $L$ on $L^2(\mu, M)$. Let

$$P_t = e^{-tL}$$

be the functional calculus of $L$, defined on $L^2(\mu, M)$. On the other hand, define

$$Q_t f(x) = \mathbb{E}\left[f(X_t) \mid X_0 = x\right], \quad \forall\, f \in C_c^\infty(M),$$

By note *Markov Operator*, $Q_t$ can be extended to $L^2(\mu)$ and has the generator $A$ that is self-adjoint on $L^2(\mu)$. By Dynkin's formula,

$$Af = Lf, \quad \forall\, f \in C_c^\infty(M).$$

Therefore, $L \subset A$. By the essential self-adjointness of $L$, $L = A$, which implies that

$$P_t f(x) = Q_t f(x) = \mathbb{E}\left[f(X_t) \mid X_0 = x\right], \quad \forall\, f \in L^2(\mu).$$

*Remark* 4.1.1. Note that if $\nabla U$ is globally Lipschitz continuous (in particular, $M$ is compact), i.e., $|\nabla^2 U| \le L$, then it can show that $(Q_t)_{t\ge 0}$ is a Feller-Dynkin semigroup so that it can be defined on $C_0(M)$.

*Remark* 4.1.2. For $P_t$, note that

$$\frac{\partial}{\partial t} P_t = L P_t = P_t L,$$

called the backward Kolmogorov equation. Moreover, it can define the Markov transition kernel

$$p_t(x, A) = P_t \mathbb{I}_A(x) = \mathbb{E}\left[\mathbb{I}_A(X_t) \mid X_0 = x\right] = \mathbb{P}_x(X_t \in A).$$

Then

$$P_t f(x) = \int_M f(y) p_t(x, dy).$$

It follows that we can define

$$\int_M f d(P_t^* \nu) = \int_M P_t f d\nu.$$

Then we have

$$\frac{\partial}{\partial t} P_t^* = L^* P_t^* = P_t^* L^*,$$

called the forward Kolmogorov equation/Fokker-Planck equation, where $L^*$ acts on $\nu$ by

$$\int_M f d(L^* \nu) = \int_M L f d\nu, \quad \forall\, f \in \mathcal{D}(L).$$

Note that $L^*\nu$ may not be a measure. Note that $\mu$ is an invariant measure if and only if $L^*\mu = 0$, i.e.,

$$\int_M L f d\mu = 0, \quad \forall\, f \in C_c^\infty(M).$$

## 4.2 Deterministic Flow

Let $V(t, x)$ be a time-dependent $C^\infty$-vector field. Consider ODE on $M$

$$\begin{cases} \dfrac{dx_t}{dt} = V(t, x_t) \\ x_0 = x \end{cases} \tag{4.2}$$

and denote $\varphi_t(x)$ be its flow.

**Theorem 4.2.1.** *Let $\{\varphi_t\}_{t\in[0,T)}$ be a locally Lipschitz family of diffeomorphisms of $M$ with $\varphi_0 = id$. Let $V(t, x)$ be the time-dependent vector fields associated with $\{\varphi_t\}_{t\in[0,T)}$. Let $\nu$ be a probability measure on $M$ and $\nu_t = (\varphi_t)_\# \nu$. Then $\nu_t$ is the unique solution of the following PDE*

$$\begin{cases} \dfrac{\partial \nu_t}{\partial t} + \operatorname{div}(\nu_t V_t) = 0, \quad t \in [0, T) \\ \nu_0 = \nu \end{cases} \tag{4.3}$$

*in $C([0, T), \mathcal{P}(M))$, where $\mathcal{P}(M)$ is equipped with the weak topology.*

*Remark* 4.2.2. For the measure PDE, it means that for any $f \in C_c^\infty(M)$,

$$\langle \partial_t \nu_t, f \rangle + \langle \operatorname{div}(\nu_t V_t), f \rangle = 0,$$

where

$$\langle \partial_t \nu_t, f \rangle = \frac{\partial}{\partial t} \langle \nu_t, f \rangle = \frac{\partial}{\partial t} \int_M f d\nu_t,$$

$$\langle \operatorname{div}(\nu_t V_t), f \rangle = -\int_M f d(\operatorname{div}(\nu_t V_t)) = -\int_M \operatorname{div}(f V_t) d\nu_t = \int_M \langle \nabla f, V_t \rangle \, d\nu_t.$$

Moreover, if $d\nu_0 = pdm$ for $p \in C^\infty$, which implies there are $p_t \in C^\infty$ such that $d\nu_t = p_t dm$, then the PDE is equivalent to

$$\begin{cases} \dfrac{\partial p_t}{\partial t} + \operatorname{div}(p_t V_t) = 0, \quad t \in [0, T) \\ p_0 = p. \end{cases}$$

The theorem implies that if there is a family of probability measures $\{\nu_t\}_{t \in [0,T)}$ satisfies above PDE (4.3), then for a random process $\widetilde{X}_t$ on $M$ satisfying the ODE (4.2) with $\widetilde{X}_0 \sim \nu$, we have $\widetilde{X}_t \sim \nu_t$.

Consider the Riemannian Langevin dynamics (4.1) with $X_0 \sim \nu$, where $d\nu = pdm$ for a smooth $p$. Then $X_t \sim \nu_t$, for $d\nu_t = d(P_t^* \nu) = p_t dm$. By the Fokker-Planck equation, we have

$$\frac{\partial \nu_t}{\partial t} = L^* \nu_t,$$

which is equivalent to

$$\frac{\partial p_t}{\partial t} = \Delta_M p_t + \operatorname{div}(p_t \nabla U) = \operatorname{div}(p_t(\nabla U + \nabla \log p_t)).$$

Therefore, if $\widetilde{X}_t$ satisfies

$$\begin{cases} \dfrac{d\widetilde{X}_t}{dt} = -\nabla U(\widetilde{X}_t) - \nabla \log p_t(\widetilde{X}_t), \\ \widetilde{X}_0 \sim \nu, \end{cases}$$

then $\widetilde{X}_t \sim \nu_t$, i.e., $\widetilde{X}_t \overset{\mathrm{d}}{=} X_t$.

## 4.3 Convergence to Invariant Measure

**KL Divergence.** If we have

$$\operatorname{Ric}_U = \operatorname{Ric} + \nabla^2 U \geq \rho g,$$

for some $\rho > 0$, then above process satisfies the $\mathrm{CD}(\rho, \infty)$ and so $\mathrm{LSI}(1/\rho)$, which implies that

$$\operatorname{End}_\mu(P_t g) \leq e^{-2\rho t} \operatorname{Ent}_\mu(g), \quad \forall\, g \in L^1(\mu).$$

(see note *Stochastic Analysis*). If $X_0 \sim \nu_0$ with $d\nu_0 = f d\mu$ (assume $f \in L^1(\mu)$) and $\mathrm{KL}(\nu_0 \parallel \mu) < \infty$, then

$$X_t \sim \nu_t = P_t^* \nu_0,$$

where $d(P_t^* \nu_0) = P_t f d\mu$. Then we can get

$$\mathrm{KL}(\nu_t \parallel \mu) \leq e^{-2\rho t} \mathrm{KL}(\nu_0 \parallel \mu).$$

Therefore,

$$\mathrm{KL}(\nu_t \parallel \mu) \leq \mathcal{O}(e^{-t}).$$

On the other hand, by Pinsker inequality $d_{\mathrm{TV}}(\nu_t, \mu) \leq \frac{1}{2} \mathrm{KL}(\nu_t \parallel \mu)$,

$$d_{\mathrm{TV}}(\nu_t, \mu) \leq \mathcal{O}(e^{-t}).$$

**Wasserstein Distance.** Under the same condition,

$$\mathrm{Ric}_U = \mathrm{Ric} + \nabla^2 U \geq \rho g,$$

we can have

$$W_2(\nu_t, \mu) \leq e^{-\rho t} W_2(\nu_0, \mu).$$

*Remark* 4.3.1. It needs techniques from the optimal transport theory, like $W_2$-geodesic convexity, et. al., or it can be obtained by directly calculating from the SDE.

## 4.4 Time-reversal Process

This section is based on results in De Bortoli et al. 2022. Let $(M, g)$ be a compact Riemannian manifold (without boundary) with Levi-Civita connection $\nabla$ and canonical volume measure $dm$ and the corresponding probability measure $dp_{\mathrm{ref}} = dm/m(M)$. In be following, we consider a more general equation.

$$dX_t = b(X_t)dt + dB_t^M, \quad t \in [0, T]$$

where $b \in \Gamma(TM)$, $B^M$ is a Brownian motion on $M$. The main goal of this section is to consider its time-reversal process $Y_t = X_{T-t}$. It can show that

$$dY_t = (-b(Y_t) + \nabla \log p_{T-t}(Y_t))dt + dB_t^M,$$

where $X_t \sim \mu_t$ with $d\mu_t = p_t dp_{\mathrm{ref}}$. It is equivalent to prove that for any $f \in C^\infty(M)$,

$$M_t^{Y,f} := f(Y_t) - f(Y_0) - \int_0^t \langle -b(Y_s) + \nabla \log p_{T-s}(Y_s), \nabla f(Y_s)\rangle + \frac{1}{2}\Delta_M f(Y_s) ds$$

is a martingale, by the weak uniqueness of Martingale problem. Furthermore, it suffices to show that for any $g \in C^\infty(M)$ and any $t \geq s \in [0, T]$,

$$\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] = \mathbb{E}\left[g(Y_s)\int_s^t \tilde{\mathcal{A}}_s f(Y_s) ds\right],$$

where

$$\tilde{\mathcal{A}}_t f = \langle -b + \nabla \log p_{T-t}, \nabla f\rangle + \frac{1}{2}\Delta_M f.$$

Note that for $t \geq s \in [0, T]$

$$\begin{aligned}
\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] &= \mathbb{E}\left[g(X_{T-s})(f(X_{T-t}) - f(X_{T-s}))\right] \\
&= \mathbb{E}\left[\mathbb{E}[g(X_{T-s}) \mid X_{T-t}] f(X_{T-t})\right] - \mathbb{E}[g(X_{T-s}) f(X_{T-s})] \\
&= \mathbb{E}\left[v(T-t, X_{T-t}) f(X_{T-t})\right] - \mathbb{E}\left[v(T-s, X_{T-s}) f(X_{T-s})\right],
\end{aligned}$$

where $v\colon [0, T-s] \times M \to \mathbb{R}$ is given by

$$v(u, x) = \mathbb{E}[g(X_{T-s}) \mid X_u = x] = P_{T-s-u} g(x),$$

where $P_t$ is the semigroup associated with $X_t$. Assume $v$ is regular enough. So it satisfies the backward Kolmogorov equation

$$\partial_u v(u, x) = -\mathcal{A} v(u, x).$$

Let $h\colon [0, T-s] \times M \to \mathbb{R}$ be defined as

$$h(u, x) = v(u, x) f(x).$$

Note that by Itô's formula, because $\mathcal{A}$ is the generator for $X$,

$$M_t^{X,h} = h(t, X_t) - h(0, X_0) - \int_0^t (\partial_u + \mathcal{A})h(u, X_u)du$$

is martingale. It implies that

$$\mathbb{E}\left[v(T-t, X_{T-t})f(X_{T-t})\right] - \mathbb{E}\left[v(T-s, X_{T-s})f(X_{T-s})\right] = \int_{T-s}^{T-t} \mathbb{E}\left[(\partial_u + \mathcal{A})h(u, X_u)\right] du.$$

To calculate it, first, we have

$$\begin{aligned}\partial_u h(u,x) + \mathcal{A}h(u,x) &= f(x)\partial_u v(u,x) + f(x)\mathcal{A}v(u,x) + v(u,x)\mathcal{A}f(x) + \langle \nabla f(x), \nabla v(u,x)\rangle \\ &= v(u,x)\mathcal{A}f(x) + \langle \nabla f(x), \nabla v(u,x)\rangle,\end{aligned}$$

because $\Delta_M(fg) = f\Delta_M g + g\Delta_M f + 2\langle \nabla f, \nabla g\rangle$ by $\operatorname{div}(fX) = \langle \nabla f, X\rangle + f \operatorname{div} X$. By the divergence theorem,

$$\mathbb{E}\left[\langle \nabla f(X_u), \nabla v(u, X_u)\rangle\right] = -\mathbb{E}\left[v\left(u, X_u\right)\Delta_M f\left(X_u\right)\right] - \mathbb{E}\left[v\left(u, X_u\right)\langle \nabla f\left(X_u\right), \nabla \log p_u\left(X_u\right)\rangle\right].$$

Therefore, it implies that

$$\begin{aligned}\mathbb{E}\left[(\partial_u + \mathcal{A})h\left(u, X_u\right)\right] &= \mathbb{E}\left[v\left(u, X_u\right)\left(\langle b\left(X_u\right) - \nabla \log p_u\left(X_u\right), \nabla f\left(X_u\right)\rangle - \frac{1}{2}\Delta_M f\left(X_u\right)\right)\right] \\ &= -\mathbb{E}\left[v\left(u, X_u\right)\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right].\end{aligned}$$

Therefore, we have

$$\begin{aligned}\mathbb{E}\left[g(Y_s)(f(Y_t) - f(Y_s))\right] &= \int_{T-t}^{T-s} \mathbb{E}\left[v\left(u, X_u\right)\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[\mathbb{E}\left[g(X_{T-s}) \mid X_u\right]\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[\mathbb{E}\left[g(X_{T-s})\tilde{\mathcal{A}}_{T-u} f\left(X_u\right) \mid X_u\right]\right] du \\ &= \int_{T-t}^{T-s} \mathbb{E}\left[g(X_{T-s})\tilde{\mathcal{A}}_{T-u} f\left(X_u\right)\right] du \\ &= \mathbb{E}\left[g(Y_s)\int_s^t \tilde{\mathcal{A}}_u f\left(Y_u\right) du\right].\end{aligned}$$

**Integration by Parts.** Let $(\mathsf{X}, \mathcal{X})$ be a polished space. Let $\mathbb{P}$ be a probability measure on $\Omega = C([0,T], \mathsf{X})$ equipping with canonical measurable sets. Let $X$ be the canonical process valued on $\mathcal{X}$ defined on $\Omega$, i.e., $X\colon C([0,T], \mathsf{X}) \to \mathsf{X}$,

$$X_t(\omega) := \omega_t,$$

and $\mathbb{F}^X = (\mathcal{F}_t)_{t\in[0,T]}$ with $\mathcal{F}_t = \sigma(X_s\colon s \in [0,t])$. For any $t \in [0,T]$, $\mathbb{P}_t := (X_t)_\#\mathbb{P}$ and more general, $\mathbb{P}_\mathcal{T} = (X_\mathcal{T})_\#\mathbb{P}$.
Let $M = (M_t)_{t\in[0,T]}$ be a $\mathsf{X}$-valued and $\mathbb{F}^X$-adapted process on $\Omega$, which is called a $\mathbb{P}$-local martingale if it is a local martingale w.s.t. $\mathbb{F}^X$.
For such $\mathbb{P}$, it can define the extended generator $\bar{\mathcal{A}}_\mathbb{P}$ (No Markov in advance). For a function,

$$u\colon [0,T] \times \mathsf{X} \to \mathbb{R}$$

if there exists a $\mathsf{X}$-valued process $A = (A_t)_{t\in[0,T]}$ on $\Omega$ such that

(a) $A$ is $\mathbb{F}^X$-adapted,

(b) $\int_0^T |A_s|\, ds < \infty$ $\mathbb{P}$-a.e.,

(c) $M = (M_t)_{t\in[0,T]}$ defined as

$$M_t = u(t, X_t) - u(0, X_0) - \int_0^t A_s ds,$$

is a $\mathbb{P}$-local martingale.

Note that if $A$ exists, then it is unique, because

$$\int_0^t A_s - B_s ds \text{ a local martingale} \quad \Rightarrow \quad A_t = B_t$$

$dt\otimes\mathbb{P}$-a.e.. Moreover, because $A$ is $\left(\sigma\left(X_{[0,t]}\right); 0 \le t \le T\right)$-adapted, i.e., $A_t$ is $\sigma(Y)$-measurable, for $Y = X_{[0,t]}$, $A_t = A_t(Y) = A_t(X_{[0,t]})$. Then we say $u \in \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}})$ and

$$A_t =: \bar{\mathcal{A}}_{\mathbb{P}} u(t, X_{[0,t]}).$$

Let $(u, v)$ be $u, v\colon [0, T] \times \mathsf{X} \to \mathbb{R}$ such that $u, v, uv \in \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}})$. For $(u, v)$, define the carré du champ $\bar{\Upsilon}_{\mathbb{P}}$

$$\bar{\Upsilon}_{\mathbb{P}} = \bar{\mathcal{A}}_{\mathbb{P}}(uv) - u\bar{\mathcal{A}}_{\mathbb{P}}(v) - v\bar{\mathcal{A}}_{\mathbb{P}}(u).$$

Assume there exists an algebra $\mathcal{U}_{\mathbb{P}} \subset \mathcal{D}(\bar{\mathcal{A}}_{\mathbb{P}}) \cap C_b(X)$. Let

$$\mathcal{U}_{\mathbb{P},2} = \left\{u \in \mathcal{U}_{\mathbb{P}} : \bar{\mathcal{A}}_{\mathbb{P}} u \in \mathrm{L}^2(\mathbb{P}), \bar{\Upsilon}_{\mathbb{P}}(u, u) \in \mathrm{L}^1(\mathbb{P})\right\}.$$

If $\mathbb{P}$ is Markov, it has the usual operator $\mathcal{A}_{\mathbb{P}} u\colon [0, T] \times \mathsf{X} \to \mathbb{R}$. We can see

$$\bar{\mathcal{A}}_{\mathbb{P}} u(t, X_{[0,t]}) = \mathcal{A}_{\mathbb{P}} u(t, X_t)$$

For $\mathbb{P}$, let $R\mathbb{P}$ be the distribution of the inverse process of $X$, i.e., $t \mapsto X_{T-t}$. If $\mathbb{P}$ is Markov, then $R\mathbb{P}$ is also a Markov. In the following, we only consider the Markov case. For $u\colon \mathsf{X} \to \mathbb{R}$ and $u \in \mathcal{D}(\mathcal{A}_{\mathbb{P}})$, we denote

$$\mathcal{A}_{\mathbb{P}} u(t, x) =: \mathcal{A}_{\mathbb{P},t} u(x).$$

For Markov case, we use $\mathcal{U}_{\mathbb{P}}$, $\mathcal{U}_{\mathbb{P},2}$ denotes time-independent $u$ such that the conditions are satisfied for all $t \in [0, T]$.

**Theorem 4.4.1** (Theorem 3.17, Cattiaux et al. 2023)**.** *Let $u, v \in \mathcal{U}_{\mathbb{P},2}$.*

*(a) If the following holds,*

*(i) $(t, x) \mapsto \Upsilon_{\mathbb{P},t}(u, v)(x)$ is continuous.*

*(ii) $\mathcal{U}_{\mathbb{P},2}$ determines the weak convergence of Borel measures on $\mathcal{X}$.*

*(iii) the linear map*

$$v \in \overline{\mathcal{U}}_{2,\mathbb{P}} \to \mathbb{E}\left[\int_0^T \Upsilon_{\mathbb{P},t}(u, v_t)(X_t) dt\right]$$

*determines a finite measure on $[0, T] \times \mathsf{X}$, where*

$$\overline{\mathcal{U}}_{2,\mathbb{P}} = \{v \in C([0, T] \times \mathsf{X}, \mathbb{R})\colon v(t, \cdot) \in \mathcal{U}_{\mathbb{P},2},\ \forall\, t \in [0, T]\}.$$

*then $u \in \mathcal{D}(\mathcal{A}_{R(\mathbb{P})})$ and $\mathcal{A}_{R(\mathbb{P})} u \in \mathrm{L}^1(\mathbb{P})$.*

*(b) If $u \in \mathcal{D}(\mathcal{A}_{R(\mathbb{P})})$ and $\mathcal{A}_{R(\mathbb{P})} u \in \mathrm{L}^1(\mathbb{P})$, then for a.e. $t \in [0, T]$,*

$$\mathbb{E}\left[\left(\mathcal{A}_{\mathbb{P},t} u(X_t) + \mathcal{A}_{R(\mathbb{P}),T-t} u(X_t)\right) v(X_t) + \Upsilon_{\mathbb{P},t}(u, u)(X_t)\right] = 0.$$

**Girsanov Theory on Compact Riemannian Manifolds.** Let $(M, g)$ be a compact Riemannian manifold and isometrically and properly embedding in $\mathbb{R}^p$.

**Theorem 4.4.2.** *Let $\mathbb{Q}$ be the path measure of a Brownian motion on $M$. Let $\mathbb{P}$ be a Markov path measure on $C([0,T], M)$ such that $\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) < \infty$. Then there exists a time-dependent vector field $\beta$ such that $\mathbb{P}$ satisfies the martingale problem for generator $\mathcal{A}$, where for $t \in [0,T]$, $u \in C^2(M)$, and $x \in M$,*

$$\mathcal{A}(t,u)(x) = \langle \beta(t,x), \nabla u(x) \rangle + \frac{1}{2}\Delta_M u(x),$$

*In addition, we have*

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(\mathbb{P}_0 \parallel \mathbb{Q}_0) + \frac{1}{2}\int_0^T \mathbb{E}\left[\|\beta(t, X_t)\|^2\right] dt.$$

*Remark* 4.4.3. For Euclidean case, if $\mathbb{P} \ll \mathbb{Q}$, then for $X \sim \mathbb{P}$, we can find an Euclidean Brownian motion $W$ such that

$$dX_t = v(t, X_t)dt + dW_t,$$

which implies that the generator

$$\mathcal{A}_t = \langle v, \nabla \cdot \rangle + \frac{1}{2}\Delta \cdot .$$

Moreover, by Girsanov's theorem,

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(\mathbb{P}_0 \parallel \mathbb{Q}_0) + \frac{1}{2}\int_0^T \mathbb{E}\left[\|v(t, X_t)\|^2\right] dt.$$

*Proof.* By embedding, if let $B_t^M$ be a Brownian motion on $M$, we have

$$dB_t^M = P_i(B_t^M) \circ dB_t^i = P(B_t^M) \circ dB_t,$$

for the orthogonal projection $P$ and an Euclidean Brownian motion $B$. Note that

$$dB_t^M = \bar{b}(B_t^M)dt + P(B_t^M)dB_t,$$

where $\bar{b}$ satisfies $P(x)\bar{b}(x) = 0$ for all $x \in M$. It implies that

$$\frac{1}{2}\Delta_M u(x) = \left\langle \overline{\nabla}\bar{u}(x), \overline{\nabla}\bar{v}(x) \right\rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x) \right\rangle,$$

for any $u, v \in C_c^2(M)$, where $\bar{u}, \bar{v} \in C_c^2(\mathbb{R}^p)$ be their extensions. Because $\mathbb{Q}$ is the distribution of $B^M$, which can be viewed on $\mathbb{R}^p$, by uniqueness condition and finite entropy, there exists $\bar{\beta}\colon [0,T] \times \mathbb{R}^p \to \mathbb{R}^p$ such that

$$\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}\left(\mathbb{P}_0 \parallel \mathbb{Q}_0\right) + \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|P\left(X_t\right)\bar{\beta}\left(t, X_t\right)\right\|^2\right] dt,$$

and $\mathbb{P}$ satisfies the martingale problem for $\bar{A}\colon [0,T] \times C_c^2(\mathbb{R}^p) \to \mathbb{R}$ with

$$\bar{A}(t, \bar{u})(x) = \langle \bar{b}(x) + P(x)\bar{\beta}(t,x), \overline{\nabla}\bar{u}(x) \rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x) \right\rangle.$$

Let $\beta = P\bar{\beta}$. Note that for any $u \in C_c^2(M)$, if $\bar{u}$ is a extension of $u$ constant-along-normals, then for any $X \in \Gamma(TM)$,

$$\nabla^2 u(X,Y) = \overline{\nabla}^2 \bar{u}(X,Y).$$

By $\Delta_M u = \sum_\alpha \nabla^2 u(P_\alpha, P_\alpha)$,

$$\begin{aligned}
\sum_\alpha \nabla^2 u(P_\alpha, P_\alpha) &= \sum_\alpha \overline{\nabla}^2 \bar{u}(P_\alpha, P_\alpha) \\
&= \sum_\alpha P_\alpha^i P_\alpha^j \frac{\partial^2 \bar{u}}{\partial x^i \partial x^j} \\
&= \sum_{i,j} P_i^j \frac{\partial^2 \bar{u}}{\partial x^i \partial x^j} \\
&= \left\langle P, \overline{\nabla}^2 \bar{u} \right\rangle
\end{aligned}$$

because $P = P^2$. Moreover, $P(x)\bar{b}(x) = 0$ implies that $\bar{b} \in \Gamma(NM)$, which follows that $\left\langle \bar{b}(x), \overline{\nabla}\bar{u} \right\rangle = 0$. So

$$\begin{aligned}
\bar{\mathcal{A}}(t,\bar{u})(x) &= \langle \bar{b}(x) + P(x)\bar{\beta}(t,x), \overline{\nabla}\bar{u}(x) \rangle + \frac{1}{2}\left\langle P(x), \overline{\nabla}^2 \bar{u}(x) \right\rangle \\
&= \langle \beta(t,x), \nabla u(x) \rangle + \frac{1}{2}\Delta_M u.
\end{aligned}$$

□

In particular, when viewing $\mathbb{P}$ as a process on $M$, it satisfies the martingale problem with generator $\mathcal{A}\colon [0,T] \times C_c^2(M) \times M \to \mathbb{R}$ given by

$$\mathcal{A}(t,u)(x) = \langle \beta(t,x), \nabla u(x) \rangle + \frac{1}{2}\Delta_M u(x).$$

**Corollary 4.4.4.** *Let $\mathbb{P}^1, \mathbb{P}^2$ be two Markov path measures on $C([0,T],M)$ with $\mathbb{P}_0^1 = \mathbb{P}_0^2$, associating with SDE*

$$dX_t^i = b_i(t, X_t^i)dt + dB_t^M.$$

*Then we have*

$$\mathrm{KL}(\mathbb{P}^1 \parallel \mathbb{P}^2) = \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|b_1(t,X_t^1) - b_2(t,X_t^2)\right\|^2\right] dt.$$

*Proof.* Let $M$ be a compact manifold isometrically embedded in $\mathbb{R}^p$ such that $M \subset B(0,R)$. Let $\varphi \in C^\infty(\mathbb{R}^p)$ such that $\varphi|_{\overline{B(0,R)}} = 1$ and $\varphi(x) = 0$ for $|x| \geq R+1$. Let $\bar{b}_i \in C_c^2([0,T]\times\mathbb{R}^p, \mathbb{R}^b)$ be the extension of $b_i(t,x)$. Consider

$$\begin{aligned}
d\bar{X}_t^i &= \varphi\left(\bar{X}_t^i\right)\left(\bar{b}^i\left(t, \bar{X}_t^i\right)dt + dB_t^M\right) \\
&= \varphi\left(\bar{X}_t^i\right)\left\{\mathrm{P}\left(\bar{X}_t^i\right)\bar{b}^i\left(t,\bar{X}_t^i\right) + \bar{b}\left(\bar{X}_t\right)\right\}dt + \varphi\left(\bar{X}_t^i\right)\mathrm{P}\left(\bar{X}_t^i\right)\mathrm{dB}_t.
\end{aligned}$$

So for any $x \in M$, $f \in C_c^2(M)$ with extension $\bar{f} \in C_c^2(\mathbb{R}^p)$,

$$\begin{aligned}
\bar{\mathcal{A}}_t^i(\bar{f})(x) &= \left\langle \bar{b}^i(t,x) + \bar{b}(x), \nabla \bar{f}(x) \right\rangle + \frac{1}{2}\left\langle \mathrm{P}(x), \nabla^2 \bar{f}(x) \right\rangle \\
&= \left\langle b^i(t,x), \nabla f(x) \right\rangle + \frac{1}{2}\Delta_M f(x).
\end{aligned}$$

It follows that $\bar{X}^i$ viewing as a process on $M$ has the same generator as $X^i$. So $\bar{X}^i$ has the distribution $\mathbb{P}^i$. Moreover, denote $\mathbb{P}^i$ as $\bar{\mathbb{P}}^i$ when it is viewed as $\mathbb{R}^p$-process. Note that because the

$$\varphi(x) = 0, \ \forall \ \|x\| \geq R+1$$

the global Lipschitz condition of SDE $\bar{X}^i$ is satisfied. Moreover, by the compactness,

$$
\begin{aligned}
\left(d\bar{\mathbb{P}}^1/d\bar{\mathbb{P}}^2\right)\left(\left(\bar{X}_t^1\right)_{t\in[0,T]}\right) = \exp\Bigg[&\int_0^T \left\langle \bar{b}^1\left(t,\bar{X}_t^1\right)-\bar{b}^2\left(t,\bar{X}_t^1\right), \mathrm{P}\left(\bar{X}_t^1\right)d\bar{X}_t^1\right\rangle \\
&-\frac{1}{2}\int_0^T \left\langle \bar{b}^1\left(t,\bar{X}_t^1\right)-\bar{b}^2\left(t,\bar{X}_t^1\right), \mathrm{P}\left(\bar{X}_t^1\right)\left(\bar{b}^1\left(t,\bar{X}_t^1\right)+\bar{b}^2\left(t,\bar{X}_t^1\right)\right)\right\rangle dt\Bigg] \\
&= \exp\left[\frac{1}{2}\int_0^T \left\|\bar{b}^1\left(t,\overline{X}_t^1\right)-\bar{b}^2\left(t,\overline{X}_t^1\right)\right\|^2 \mathrm{d}t + \int_0^T \left\langle \bar{b}^1\left(t,\overline{X}_t^1\right)-\bar{b}^2\left(t,\overline{X}_t^1\right), \mathrm{P}\left(\overline{X}_t^1\right)\mathrm{d}B_t\right\rangle\right].
\end{aligned}
$$

It follows that

$$
\mathrm{KL}\left(\overline{\mathbb{P}}^1 \parallel \overline{\mathbb{P}}^2\right) = \frac{1}{2}\int_0^T \mathbb{E}\left[\left\|b^1\left(t,X_t^1\right)-b^2\left(t,X_t^1\right)\right\|^2\right]dt. \qquad \square
$$

**Proof of Time-reversal Formula.**

**Corollary 4.4.5.** *Let $\mathbb{Q}$ be the path measure of a Brownian motion on $M$ with $\mathbb{Q}_0 = p_{\text{ref}}$ and $\mathbb{P}$ be a path measure on $C([0,T],M)$ with $\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) < \infty$. Then, there exist two time-dependent vector fields $\beta_{\mathbb{P}}$ and $\beta_{R(\mathbb{P})}$ such that $\mathbb{P}$ and $R(\mathbb{P})$ satisfy martingale problem associated with $\mathcal{A}_{\mathbb{P}}$ and $\mathcal{A}_{R(\mathbb{P})}$,*

$$
\begin{aligned}
\mathcal{A}_{\mathbb{P}}(t,u)(x) &= \langle \beta_{\mathbb{P}}(t,x), \nabla u(x)\rangle + \frac{1}{2}\Delta_M u(x), \\
\mathcal{A}_{R(\mathbb{P})}(t,u)(x) &= \left\langle \beta_{R(\mathbb{P})}(t,x), \nabla u(x)\right\rangle + \frac{1}{2}\Delta_M u(x).
\end{aligned}
$$

*Finally, we have that*

$$
\int_0^T \mathbb{E}\left[\left\|\beta_{\mathbb{P}}\left(t,X_t\right)\right\|^2\right]\mathrm{d}t + \int_0^T \mathbb{E}\left[\left\|\beta_{R(\mathbb{P})}\left(t,X_{T-t}\right)\right\|^2\right]\mathrm{d}t < +\infty,
$$

*for $X \sim \mathbb{P}$.*

*Proof.* Just applying above theorem with

$$
\mathrm{KL}(\mathbb{P} \parallel \mathbb{Q}) = \mathrm{KL}(R(\mathbb{P}) \parallel R(\mathbb{Q})) = \mathrm{KL}(R(\mathbb{P}) \parallel \mathbb{Q}) < \infty. \qquad \square
$$

Then by above the Integral by parts, for any $u, v \in C_c^\infty(M)$,

$$
\mathbb{E}\left[v\left(X_t\right)\left(\left\langle \beta_{\mathbb{P}}\left(t,X_t\right)+\beta_{R(\mathbb{P})}\left(T-t,X_t\right), \nabla u\left(X_t\right)\right\rangle + \Delta_M u\left(X_t\right)\right) + \left\langle \nabla u\left(X_t\right), \nabla v\left(X_t\right)\right\rangle\right] = 0.
$$

It follows that

$$
\begin{aligned}
&\int_M \left\{\left\langle \beta_{R(\mathbb{P})}(T-t,x), \nabla u(x)\right\rangle + \langle b(x), \nabla u(x)\rangle\right\} v(x)p_t(x)dp_{\text{ref}}(x) \\
&\quad = \int_M \langle \nabla \log p_t(x), \nabla u(x)\rangle v(x)p_t(x)dp_{\text{ref}}(x).
\end{aligned}
$$

Therefore,

$$
\left\langle \beta_{R(\mathbb{P})}(T-t,x), \nabla u(x)\right\rangle = \langle \nabla \log p_t(x), \nabla u(x)\rangle.
$$

## 4.5 Discretization

Let $(M, g)$ be a complete Riemannian manifold and $B^M$ be a Brownian motion on it. Consider the SDE,

$$dX_t = v(X_t)dt + dB_t^M,\ t \in [0, T], \quad X_0 = Z$$

We always assume $B^M$ is independent of $Z$. Fix $h = T/N$ for some $N \in \mathbb{N}$. Let $t_k = kh$. Constructing process $X_k^h$ by

$$X_{k+1}^h = \exp_{X_k^h}(hv(X_k^h) + \sqrt{h}\xi),$$

where exp is the exponential map, and $\xi$ is a standard Gaussian standard Gaussian with respect to any orthonormal basis of $T_xM$. In the following, we study the convergence of $(X_k^h)_{k=0}^N$.

**Weak convergence.** Assume $M$ has bounded geometry, that is,

$$\operatorname{inj}(M) \geq \iota > 0, \quad \sup_{x \in M} \left|\nabla^k R(x)\right| < \infty,\ \forall\ k,$$

where $R$ is the Riemannian curvature. Let $(R_t)_{t\in[0,T]}$ be the semigroup of $X_t$ and $(R_k^h)_{k=0}^N$ be the semigroup of $X_k^h$. Assume that $v \in C_b^3(TM)$. Then for any $f \in C_b^3(M)$, we have

$$\left\|R_T f - R_N^h f\right\|_\infty \leq Ch.$$

*Proof.* First, for $X_t$, the generator $L$ is

$$Lf = \langle v, \nabla f\rangle + \frac{1}{2}\Delta_M f, \quad \forall\ f \in C_b^3(M),$$

and $R_T = (R_h)^N$. For $X_k^h$, let $R^h = R_1^h$. Then $R_N^h = (R^h)^N$.

**Step 1.** Check: For any $f \in C_b^3(M)$, we have

$$R^h f(x) = f(x) + hLf(x) + \mathcal{O}(h^{3/2}).$$

To prove that, first fix $X_0^h = x$ with $x \in M$ and $f \in C_c^\infty(M)$. We can choose $0 < \rho \leq \iota$ (note that $\rho$ independent of $x$) such that

$$\exp_x\colon B(0, \rho) \subset T_xM \to B(x, \rho) \subset M$$

is a diffeomorphism, moreover, we choose the normal coordinate on $T_xM$. Let

$$Z_h(x) := hv(x) + \sqrt{h}\xi,$$

so that

$$X_1^h = \exp_x(Z_h(x)).$$

Define

$$\varphi_x(z) := f(\exp_x(z)), \quad \forall\ z \in B(0, \rho).$$

Because $f \in C_b^3(M)$ and $M$ has bounded geometry,

$$\sup_{x\in M} \sup_{|z|\leq\rho} \sum_{|\alpha|\leq 3} |\partial_z^\alpha \varphi_x(z)| < \infty$$

Note that

$$R^h f(x) = \mathbb{E}\left[f(X_1^h) \mid X_0^h = x\right] = \mathbb{E}\left[\varphi_x(Z_h(x))\right]$$

I. Fix $h_0 > 0$ sufficiently small such that

$$h_0 \sup_x \|v(x)\| \leq \frac{\rho}{4}.$$

Then for $0 < h < h_0$, if $|\xi| < \rho/(2\sqrt{h})$, then

$$|Z_h(x)| \leq h\|v(x)\| + \sqrt{h}|\xi| \leq \frac{\rho}{4} + \frac{\rho}{2} < \rho.$$

Therefore,

$$A_h := \left\{|\xi| \leq \rho/(2\sqrt{h})\right\} \subset \{Z_h(x) \in B(0,\rho)\}.$$

Moreover, because $\xi$ is Gaussian, then

$$\mathbb{P}(A_h^c) = \mathbb{P}\left(|\xi| > \rho/(2\sqrt{h})\right) \leq c_1 e^{-c_2/h},$$

for some $c_1, c_2 > 0$. Writing

$$R^h f(x) = \mathbb{E}\left[\varphi_x\left(Z_h(x)\right)\mathbb{I}_{A_h}\right] + \mathbb{E}\left[f\left(\exp_x\left(Z_h(x)\right)\right)\mathbb{I}_{A_h^c}\right].$$

Then for the second term,

$$\left|\mathbb{E}\left[f\left(\exp_x\left(Z_h(x)\right)\right)\mathbb{I}_{A_h^c}\right]\right| \leq \|f\|_\infty \mathbb{P}(A_h^c) \leq \|f\|_\infty c_1 e^{-c_2/h} \leq \mathcal{O}(h^2).$$

So the main goal is to bound the first term.

II. For $z \in B(0,\rho)$, by Taylor expansion

$$\varphi_x(z) = \varphi_x(0) + D\varphi_x(0)[z] + \frac{1}{2}D^2\varphi_x(0)[z,z] + R_3(x,z),$$

where

$$R_3(x,z) = \frac{1}{6}\sum_{i,j,k}\partial_{ijk}\varphi_x(\theta z)z_i z_j z_k,$$

for some $\theta = \theta(x,z) \in (0,1)$. For $R_4$, it has

$$|R_3(x,z)| \leq C_f|z|^3, \quad \forall\ |z| \leq \rho.$$

for some $C_f > 0$ (independent of $x$).

For $z = Z_h(x)$, writing

$$\begin{aligned}\mathbb{E}\left[\varphi_x\left(Z_h(x)\right)1_{A_h}\right] =& \varphi_x(0)\mathbb{P}\left(A_h\right) + \mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] \\ &+ \frac{1}{2}\mathbb{E}\left[D^2\varphi_x(0)\left[Z_h, Z_h\right]1_{A_h}\right] \\ &+ \mathbb{E}\left[R_3(x, Z_h(x))1_{A_h}\right]\end{aligned}$$

Next, consider these terms respectively.

(a) First, because

$$\mathbb{P}(A_h) = 1 - \mathbb{P}(A_h^c) = 1 + \mathcal{O}\left(e^{-c_1/h}\right),$$

the zero-order term can be bounded by

$$\varphi_x(0) = \varphi_x(0)\mathbb{P}\left(A_h\right) = f(x) + \mathcal{O}\left(e^{-c_2/h}\right).$$

(b) For the first-order term, we first recall that $Z_h(x) = hv(x) + \sqrt{h}\xi$ and $\mathbb{E}[\xi] = 0$. Because

$$\left\|\mathbb{E}\left[Z_h(x)1_{A_h^c}\right]\right\| \leq \left(\mathbb{E}\left|Z_h(x)\right|^2\right)^{1/2} \left(\mathbb{P}\left(A_h^c\right)\right)^{1/2} \leq \mathcal{O}\left(e^{-c_3/h}\right),$$

we have

$$\begin{aligned}
\mathbb{E}\left[Z_h(x)1_{A_h}\right] &= \mathbb{E}\left[Z_h(x)\right] - \mathbb{E}\left[Z_h(x)1_{A_h^c}\right] \\
&= \mathbb{E}\left[Z_h(x)\right] + \mathcal{O}\left(e^{-c_3/h}\right) \\
&= hv(x) + \mathcal{O}\left(e^{-c_3/h}\right).
\end{aligned}$$

Then we get

$$\mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] = D\varphi_x(0)[hv(x)] + \mathcal{O}\left(e^{-c_2/h}\right).$$

Because $\varphi_x(z) = f(\exp_x(z))$,

$$D\varphi_x(0)[hv(x)] = h\left\langle \nabla f(x), v(x)\right\rangle.$$

So

$$\mathbb{E}\left[D\varphi_x(0)\left[Z_h\right]1_{A_h}\right] = h\left\langle \nabla f(x), v(x)\right\rangle + \mathcal{O}\left(e^{-c_2/h}\right).$$

(c) For the second-order term, first, we have

$$D^2\varphi_x(0)[z,z] = \nabla^2 f(x)(z,z) = z^\top H_x z,$$

where $H_x = \nabla^2 f(x)$ in matrix expression under the normal coordinate. Moreover, because $\xi \in \mathcal{N}(0, I)$,

$$\begin{aligned}
\mathbb{E}\left[Z_h Z_h^\top 1_{A_h}\right] &= \mathbb{E}\left[Z_h Z_h^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\
&= \mathbb{E}\left[(hv + \sqrt{h}\xi)(hv + \sqrt{h}\xi)^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\
&= h^2 vv^\top + h\mathbb{E}\left[\xi\xi^\top\right] + \mathcal{O}\left(e^{-c_2/h}\right) \\
&= h\,\mathrm{Id} + h^2 vv^\top + \mathcal{O}\left(e^{-c_2/h}\right).
\end{aligned}$$

So it implies that

$$\begin{aligned}
\frac{1}{2}\mathbb{E}\left[D^2\varphi_x(0)\left[Z_h, Z_h\right]1_{A_h}\right] &= \frac{1}{2}\mathbb{E}\left[\operatorname{tr}\left(H_x Z_h Z_h^\top\right)1_{A_h}\right] \\
&= \frac{1}{2}\operatorname{tr}\left(H_x \mathbb{E}\left[Z_h Z_h^\top 1_{A_h}\right]\right) \\
&= \frac{1}{2}h\operatorname{tr}\left(H_x\right) + \mathcal{O}\left(h^2\right) + \mathcal{O}\left(e^{-c_2/h}\right) \\
&= \frac{1}{2}h\Delta_M f(x) + \mathcal{O}\left(h^2\right).
\end{aligned}$$

(d) For the remainder term, because $Z_h(x) = hv(x) + \sqrt{h}\xi$ with $\xi \sim \mathcal{N}(0, I)$,

$$\mathbb{E}\left[\left\|Z_h\right\|^3\right] = \mathcal{O}(h^{3/2}).$$

Therefore,

$$\left\|\mathbb{E}\left[R_3\left(x, Z_h(x)\right)1_{A_h}\right]\right\| \leq C_f \mathbb{E}\left[\left\|Z_h\right\|^3\right] = \mathcal{O}\left(h^{3/2}\right).$$

Combining these results, we have

$$R^h f(x) = f(x) + hLf(x) + \mathcal{O}(h^{3/2}),$$

which implies that

$$\left\|R^h f - f - hLf\right\|_\infty \leq C_1 h^{3/2}.$$

**Step 2.** Check:

$$\|R_h f - f - hLf\|_\infty \leq C_2 h^2.$$

Let $u(t,x) = R_t f(x)$. The by the backward Kolmogorov equation,

$$\partial_t u_t = L u_t, \quad u_0(x) = f(x).$$

By the smoothness and boundedness, one can show that $u_t \in C_b^3(M)$ and

$$\sup_{0\leq t\leq T} \|\nabla^\alpha u_t\|_\infty \leq C_3, \quad \forall\ |\alpha| \leq 3. \tag{4.4}$$

In particular, $\|L^2 u_t\|_\infty \leq C_3$.By

$$u_h(x) - f(x) = \int_0^h \partial_t u_s(x) ds = \int_0^h L u_s(x) ds,$$

we get

$$L u_s(x) - Lf(x) = \int_0^s \partial_t L u_r(x) dr = \int_0^s L^2 u_r(x) dr.$$

Therefore,

$$\begin{aligned} u_h(x) &= f(x) + hLf(x) + \int_0^h \int_0^s L^2 u_r(x) dr ds \\ &= f(x) + hLf(x) + \beta_h(x) \end{aligned}$$

with

$$|\beta_h(x)| \leq \int_0^h \int_0^s \left\|L^2 u_r\right\|_\infty dr ds \leq C_3 \int_0^h \int_0^s dr ds = \frac{1}{2} C_3 h^2.$$

It follows that

$$\|R_h f - f - hLf\|_\infty \leq C_2 h^2.$$

**Step 3.** By Step 1 and Step 2, for all $f \in C_c^\infty$,

$$\begin{aligned} \left\|R^h f - R_h f\right\|_\infty &\leq \left\|R^h f - f - hLf\right\|_\infty + \|R_h f - f - hLf\|_\infty \\ &\leq C_1 h^{3/2} + C_2 h^2 \leq C_4 h^{3/2}. \end{aligned}$$

Globally, note that $R_T = (R_h)^N$ and $R_N^h = (R^h)^N$. Writing

$$\left(R^h\right)^N - (R_h)^N = \sum_{k=0}^{N-1} \left(R^h\right)^{N-1-k} \left(R^h - R_h\right) (R_h)^k.$$

Because $R^h$ and $R_h$ are Markov, they are contractions. So

$$\left\|\left(R^h\right)^N f - (R_h)^N f\right\|_\infty \leq \sum_{k=0}^{N-1} \sup_x \left|\left(R^h - R_h\right) g_k(x)\right|,$$

where by (4.4)

$$g_k(x) = \left(R_h^k f\right)(x) = \left(R_{kh} f\right)(x) \in C_b^3(M).$$

Therefore,

$$\begin{aligned}\left\|\left(R^h\right)^N f - \left(R_h\right)^N f\right\|_\infty &\le \sum_{k=0}^{N-1} \left\|R^h g_k - R_h g_k\right\|_\infty \\ &\le \sum_{k=0}^{N-1} C_4 h^{3/2} = C_4 N h^{3/2} = \frac{T}{h} C_4 h^{3/2} = C_T h^{1/2}. \qquad \square\end{aligned}$$

*Remark* 4.5.1. Above proof can be extended to time-inhomogeneous case

$$dX_t = v(t, X_t)dt + g(t)dB_t^M.$$

**Strong convergence.** Assume $M \subset \mathbb{R}^n$ properly and isometrically embedded and assume the second fundamental form is uniformly bounded, i.e.,

$$\sup_{x\in M} \|\mathrm{II}_x\|_{\mathrm{op}}, \ \|\nabla \mathrm{II}_x\|_{\mathrm{op}} < \infty.$$

Assume $v \in C_b^1(M)$ and $X_0 \in L^2$. Let $d$ be the Riemannian distance. We can show that

$$\mathbb{E}\left[\max_{0\le k\le N} d\left(X_{t_k}, X_k^h\right)^2\right] \le Ch$$

*Proof.* Note that

$$\exp_x(v) = x + v + \frac{1}{2}\mathrm{II}_x(v,v) + \mathcal{O}(\|v\|^3),$$

where $\mathrm{II}_x$ is the second form. To prove that, first let $\gamma(t) = \exp_x(tv)$ in $\mathbb{R}^n$, which is also a geodesic in $M$. Then it has

$$\gamma'(0) = \dot\gamma(t) = v$$

and

$$\gamma''(t) = \nabla_{\dot\gamma(t)}\dot\gamma(t) + \mathrm{II}_x(\dot\gamma(t), \dot\gamma(t)) = \mathrm{II}(\dot\gamma(t), \dot\gamma(t)),$$

because $\nabla_{\dot\gamma(t)}\dot\gamma(t)$. Therefore, above can be got by applying the Euclidean Taylor expansion.

**Step 1.** Euclidean EM discretization: Let $\Pi(x)\colon \mathbb{R}^n \to \mathbb{R}^n$ be the orthogonal projection onto $T_xM$. Then we can choose

$$dB_t^M = \Pi \circ dW_t,$$

where $W$ is a standard $\mathbb{R}^n$-Brownian motion that is also independent of $X_0$. Then

$$dX_t = v(X_t)dt + \Pi(X_t)\circ dW_t = \tilde a(X_t)dt + \Pi(X_t)dW_t, \tag{4.5}$$

as an Euclidean SDE, where

$$\tilde a(x) = v(x) + A(x), \quad A^j(x) = \frac{1}{2}\sum_{\alpha=1}^n \Pi_\alpha(x)\left(\Pi_{j\alpha}(x)\right),$$

for $\Pi(x) = (\Pi_{j\alpha}(x))$ when fixing the standard basis $\{e_1, \cdots, e_n\}$ in $\mathbb{R}^n$. Moreover, for

$$\Pi_\alpha(x) = \Pi(x)e_\alpha \in T_xM,$$

we have

$$\Pi_{j\alpha}(x) = \Pi_\alpha(x^j)(x),$$

where $x^j : M \to \mathbb{R}$ is the $j$-th component of the canonical embedding $x : M \hookrightarrow \mathbb{R}^n$.
**Claim:** If $\{E_i(x)\}_{i=1}^d$ is an orthonormal frame of $T_xM \subset \mathbb{R}^n$, then we have

$$A(x) = \frac{1}{2}\sum_{i=1}^{d} \mathrm{II}_x(E_i(x), E_i(x)).$$

To prove that, first, by

$$\Delta_M x^j = \sum_{\alpha=1}^{N} \Pi_\alpha(\Pi_\alpha(x^j)) = \sum_{\alpha=1}^{N} \Pi_\alpha(\Pi_{j\alpha}),$$

we have

$$A^j(x) = \frac{1}{2}\Delta_M x^j(x).$$

On the other hand, we know

$$\Delta_M x^j(x) = \sum_{i=1}^{d} \mathrm{II}_x^j(E_i(x), E_i(x)).$$

For Euclidean SDE (4.5), we consider the Euclidean EM algorithm. Let $t_k = kh, Nh = T, \Delta W_k = W_{t_{k+1}} - W_{t_k}$, and define

$$Y_{k+1}^h = Y_k^h + \tilde{a}\left(Y_k^h\right)h + \Pi\left(Y_k^h\right)\Delta W_k, \quad Y_0^h = X_0.$$

Because $A(x)$ is uniformly bounded and $v \in C_b^1$, $\tilde{a}$ is globally Lipschitz. For $\Pi$, note that $\Pi$ is in fact smooth so it is locally Lipschitz continuous. Moreover, because II is uniformly bounded, by Lemma 4.5.3, $\Pi$ is globally Lipschitz. Combining these with $X_0 \in L^2$, we have

$$\mathbb{E}\left[\max_{0\le k\le N}\left\|X_{t_k} - Y_k^h\right\|^2\right] \le C_T h.$$

Moreover, let

$$\Phi_h^E(x,z) = x + \tilde{a}(x)h + \Pi(x)z.$$

So

$$Y_{k+1}^h = \Phi_h^E(Y_k^h, \Delta W_k)$$

**Step 2.** Geometric EM discretization: For

$$X_{k+1}^h = \exp_{X_k^h}\left(hv\left(X_k^h\right) + \sqrt{h}\xi_{k+1}\right)$$

with $\xi_{k+1}$ a standard Gaussian on $T_xM$. WTLG, assume

$$\xi_{k+1} := \frac{1}{\sqrt{h}}\Pi\left(X_k^h\right)\Delta W_k, \quad \Delta W_k = W_{t_{k+1}} - W_{t_k},$$

where $W$ is the Brownian motion in (4.5). So the geometric EM is

$$X_{k+1}^h = \exp_{X_k^h}\left(hv\left(X_k^h\right) + \Pi\left(X_k^h\right)\Delta W_k\right).$$

Let

$$\Phi_h^G(x,z) = \exp_x(hv(x) + \Pi(x)z),$$

i.e., we have

$$X_{k+1}^h = \Phi_h^G(X_k^h, \Delta W_k).$$

Moreover, by the Taylor expansion of the exponential map, for $u = hv(x) + \Pi(x)z$,

$$\Phi_h^G(x, z) = x + hv(x) + \Pi(x)z + \frac{1}{2}\mathrm{II}_x(u, u) + R_3(x, u).$$

Because $\|\nabla \mathrm{II}\|_{\mathrm{op}} < \infty$ $(\gamma''' = D/dt \nabla \mathrm{II}(\dot{\gamma}, \dot{\gamma}))$,

$$\|R_3(x, u)\| \leq C_1 \|u\|^3,$$

where $C_1$ is independent of $x$.
**Claim:** For $z \sim \mathcal{N}(0, hI_n)$, we have

$$\mathbb{E}\left[\mathrm{II}_x(\Pi(x)z, \Pi(x)z)\right] = 2hA(x).$$

For

$$\Pi(x)z = \sum_{i=1}^{d} \zeta_i E_i(x), \quad \zeta_i = \langle z, E_i(x) \rangle,$$

then $\zeta = (\zeta_1, \cdots, \zeta_d)$ is a Gaussian with

$$\mathbb{E}\left[\zeta_i \zeta_j\right] = \mathbb{E}\left[\langle z, E_i \rangle \langle z, E_j \rangle\right] = E_i^\top \mathbb{E}\left[zz^\top\right] E_j = E_i^\top \left(hI_N\right) E_j = h\delta_{ij},$$

i.e., $\zeta \sim \mathcal{N}(0, hI_d)$. Because $\mathrm{II}_x$ is bilinear,

$$\mathrm{II}_x(\Pi(x)z, \Pi(x)z) = \sum_{i,j=1}^{d} \zeta_i \zeta_j \mathrm{II}_x(E_i, E_j). \tag{4.6}$$

Therefore,

$$\begin{aligned}\mathbb{E}\left[\mathrm{II}_x(\Pi(x)z, \Pi(x)z)\right] &= \sum_{i,j=1}^{d} \mathbb{E}\left[\zeta_i \zeta_j\right] \mathrm{II}_x(E_i, E_j) \\ &= h \sum_{i=1}^{d} \mathrm{II}_x(E_i, E_i) = 2hA(x).\end{aligned}$$

Therefore, for $u = hv(x) + \Pi(x)z$ and $z = \Delta W_k \sim \mathcal{N}(0, hI_n)$,

$$\mathbb{E}\left[\mathrm{II}_x(u, u)\right] = \mathbb{E}\left[\mathrm{II}_x(\Pi(x)z, \Pi(x)z)\right] + h^2 \mathrm{II}_x(v, v) = 2hA(x) + \mathcal{O}(h^2).$$

It follows that

$$\begin{aligned}\mathbb{E}\left[\Phi_h^G(x, \Delta W_k)\right] &= x + hv(x) + hA(x) + \mathcal{O}(h^2) \\ &= x + \tilde{a}(x)h + \mathcal{O}(h^2).\end{aligned}$$

Note that by (4.6), we also have

$$\mathbb{E}\left[\|\mathrm{II}_x(\Pi(x)z, \Pi(x)z)\|^2\right] = \mathcal{O}(h^2),$$

because the second form is uniformly bounded.

□

*Remark* 4.5.2. A direct result of $A(x) = \frac{1}{2}\sum_i \mathrm{II}_x(E_i(x), E_i(x)) \in N_xM$ is when viewing (4.5) as an Euclidean SDE with generator $\tilde{L}$,

$$\tilde{L}|_M = L.$$

For any $f \in C^\infty(M)$, let $\tilde{f} \in C^\infty(\mathbb{R}^n)$ be a normal extension, which means that

$$\overline{\nabla}\tilde{f}(x) = \nabla f(x), \quad \forall\, x \in M,$$

and for all $x \in M$ and $X, Y \in T_xM$,

$$\overline{\nabla}^2\tilde{f}(x)(X,Y) = \nabla f(x)(X,Y).$$

First, we have

$$\tilde{L}\tilde{f}(x) = \left\langle \overline{\nabla}\tilde{f}(x), \tilde{a} \right\rangle + \frac{1}{2}\operatorname{tr}\left(\Pi(x)\Pi(x)^\top \overline{\nabla}^2\tilde{f}(x)\right).$$

For the first term, because $A(x) \in N_xM$,

$$\left\langle \overline{\nabla}\tilde{f}(x), \tilde{a} \right\rangle = \langle \nabla f(x), v \rangle.$$

For the second term, because $\Pi(x)$ is an orthogonal projection, we have

$$\Pi(x)\Pi(x)^\top = \Pi(x) = \sum_{i=1}^d E_i(x) \otimes E_i(x).$$

Therefore,

$$\begin{aligned}
\operatorname{tr}\left(\Pi(x)\Pi(x)^\top \overline{\nabla}^2\tilde{f}(x)\right) &= \sum_{i=1}^d \overline{\nabla}^2\tilde{f}(x)(E_i(x), E_i(x)) \\
&= \sum_{i=1}^d \nabla^2 f(x)(E_i(x), E_i(x)) \\
&= \Delta_M f(x).
\end{aligned}$$

So

$$\tilde{L}\tilde{f}(x) = \langle \nabla f(x), v \rangle + \frac{1}{2}\Delta_M f(x) = Lf(x).$$

**Lemma 4.5.3.**

# Bibliography

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